Supplementary File

Title: Pooled prevalence of adverse pregnancy and neonatal outcomes in Malawi, South Africa, Uganda, and Zimbabwe – Results from a systematic review and meta-analysis to inform trials of novel HIV prevention interventions during pregnancy

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SECTION A: SEARCH TERMS, RESULTS, AND INCLUDED STUDIES

Table A.1. Search Terms, Results, and Date for 10 literature reviews

Type of Outcome	Outcome	Terms*	Results†
Pregnancy outcome	**Pregnancy Loss & Stillbirth**	("pregnancy loss" OR stillbirth* OR "stillborn" OR "still birth" OR "still births" OR "still born" OR stillbirth[mesh] OR "fetal demise" OR "fetal death" OR fetal death [Mesh] OR "spontaneous abortion" OR abortion,spontaneous [mesh] OR "miscarriage") AND (Ethiopia OR Ethiopian OR Kenya OR Kenyan OR Tanzania OR tanzanian Or Uganda OR Ugandan OR Mozambique OR Mozambican OR Malawi OR Malawian OR Zimbabwe OR Zimbabwean OR Zambia OR Zambian OR Botswana OR Botswanan OR "South Africa" OR "South African" OR South Africa [Mesh] OR Lesotho OR Basotho OR Swaziland OR Swazi OR Namibia OR Namibian) AND ((("1998/01/01"[PDat] : "2018/12/31"[PDat]) AND English[lang])	**874**
	Preterm birth	("premature labor"[All Fields] OR "preterm labour"[All Fields] OR "Preterm birth"[All Fields] OR "premature delivery"[All Fields] OR "preterm delivery"[All Fields] OR preterm[All Fields] OR "premature birth"[MESH]) AND (Ethiopia OR Ethiopian OR Kenya OR Kenyan OR Tanzania OR tanzanian Or Uganda OR Ugandan OR Mozambique OR Mozambican OR Malawi OR Malawian OR Zimbabwe OR Zimbabwean OR Zambia OR Zambian OR Botswana OR Botswanan OR "South Africa" OR "South African" OR South Africa [Mesh] OR Lesotho OR Basotho OR Swaziland OR Swazi OR Namibia OR Namibian) AND ((("1998/01/01"[PDat] : "2018/12/31"[PDat]) AND English[lang])	**590**
Infant outcome	**Congenital anomaly**	("Congenital anomaly" OR "congenital abnormalities" OR "congenital abnormality" OR "congenital abnormalities" OR "birth defect" OR "Polydactyly" OR "cranio-facial" OR "cranio facial" OR "craniofacial" OR "neural tube defect" OR "neural tube" OR "spina bifida" OR "anencephaly" OR "heart defect" OR "inguinal hernia" OR "umbilical hernia" OR "micrognathia" OR "epicanthic folds" OR "microcephaly" OR "hydrocephaly" OR "cleft palate" OR "cleft lip") AND (Ethiopia OR Ethiopian OR Kenya OR Kenyan OR Tanzania OR tanzanian Or Uganda OR Ugandan OR Mozambique OR Mozambican OR Malawi OR Malawian OR Zimbabwe OR Zimbabwean OR Zambia OR Zambian OR Botswana OR Botswanan OR "South Africa" OR "South African" OR South Africa [Mesh] OR Lesotho OR Basotho OR Swaziland OR Swazi OR Namibia OR Namibian) AND ((("1998/01/01"[PDat] : "2018/12/31"[PDat]) AND English[lang])	**620**

	Low birth weight	("low birth weight" OR "lbw" OR "low birthweight" OR infant, low birth weight[mesh]) AND (Ethiopia OR Ethiopian OR Kenya OR Kenyan OR Tanzania OR tanzanian Or Uganda OR Ugandan OR Mozambique OR Mozambican OR Malawi OR Malawian OR Zimbabwe OR Zimbabwean OR Zambia OR Zambian OR Botswana OR Botswanan OR "South Africa" OR "South African" OR South Africa [Mesh] OR Lesotho OR Basotho OR Swaziland OR Swazi OR Namibia OR Namibian) AND (("1998/01/01"[PDat] : "2018/12/31"[PDat]) AND English[lang])	**791**
	Neonatal Mortality	("neonatal death" OR perinatal death[mesh] OR perinatal mortality[mesh] OR "neonatal mortality" OR "perinatal mortality") AND (Ethiopia OR Ethiopian OR Kenya OR Kenyan OR Tanzania OR tanzanian Or Uganda OR Ugandan OR Mozambique OR Mozambican OR Malawi OR Malawian OR Zimbabwe OR Zimbabwean OR Zambia OR Zambian OR Botswana OR Botswanan OR "South Africa" OR "South African" OR South Africa [Mesh] OR Lesotho OR Basotho OR Swaziland OR Swazi OR Namibia OR Namibian) AND (("1998/01/01"[PDAT] : "2018/12/31"[PDAT]) AND English[lang])	**834**
Pregnancy complication	**Chorioamnionitis**	(choriomanionitis[mesh] OR "choriomanionitis" OR "intraamniotic infection" OR "intra-amniotic infection" OR "amnionitis" OR "funisitis") AND (Ethiopia OR Ethiopian OR Kenya OR Kenyan OR Tanzania OR tanzanian Or Uganda OR Ugandan OR Mozambique OR Mozambican OR Malawi OR Malawian OR Zimbabwe OR Zimbabwean OR Zambia OR Zambian OR Botswana OR Botswanan OR "South Africa" OR "South African" OR South Africa [Mesh] OR Lesotho OR Basotho OR Swaziland OR Swazi OR Namibia OR Namibian) AND (("1998/01/01"[PDat] : "2018/12/31"[PDat]) AND English[lang])	**6**
	Endometritis	("endometritis"[MeSH Terms] OR "peripartum endometritis" OR "Endometritis"[All Fields] OR "postpartum endometritis"[All Fields] AND (Ethiopia OR Ethiopian OR Kenya OR Kenyan OR Tanzania OR tanzanian Or Uganda OR Ugandan OR Mozambique OR Mozambican OR Malawi OR Malawian OR Zimbabwe OR Zimbabwean OR Zambia OR Zambian OR Botswana OR Botswanan OR "South Africa" OR "South African" OR South Africa [Mesh] OR Lesotho OR Basotho OR Swaziland OR Swazi OR Namibia OR Namibian) AND (("1998/01/01"[PDAT] : "2018/12/31"[PDAT]) AND English[lang])	**43**
	Postpartum Hemorrhage	("peripartum hemorrhage" OR "peripartum haemorrhage" OR postpartum hemorrhage [mesh] OR "postpartum haemorrhage" OR "postpartum hemorrhage") AND (Ethiopia OR Ethiopian OR Kenya OR Kenyan OR Tanzania OR tanzanian Or Uganda OR Ugandan OR Mozambique OR Mozambican OR Malawi OR Malawian OR Zimbabwe OR Zimbabwean OR Zambia OR Zambian OR Botswana OR Botswanan OR "South Africa" OR "South African" OR South Africa	**217**

		[Mesh] OR Lesotho OR Basotho OR Swaziland OR Swazi OR Namibia OR Namibian) AND (("1998/01/01"[PDAT] : "2018/12/31"[PDAT]) AND English[lang])	
	Hypertensive disorders of pregnancy	("preeclampsia" OR preeclampsia [mesh] OR "eclampsia" OR "gestational hypertension" OR "pregnancy induced hypertension" OR "HELLP" OR "preeclampsia-eclampsia" OR "pre-eclampsia") AND (Ethiopia OR Ethiopian OR Kenya OR Kenyan OR Tanzania OR tanzanian Or Uganda OR Ugandan OR Mozambique OR Mozambican OR Malawi OR Malawian OR Zimbabwe OR Zimbabwean OR Zambia OR Zambian OR Botswana OR Botswanan OR "South Africa" OR "South African" OR South Africa [Mesh] OR Lesotho OR Basotho OR Swaziland OR Swazi OR Namibia OR Namibian) AND (("1998/01/01"[PDAT] : "2018/12/31"[PDAT]) AND English[lang])	**549**
	PPROM	("preterm premature rupture membranes" OR "PPROM" OR Fetal Membranes, Premature Rupture [mesh]) AND (Ethiopia OR Ethiopian OR Kenya OR Kenyan OR Tanzania OR tanzanian Or Uganda OR Ugandan OR Mozambique OR Mozambican OR Malawi OR Malawian OR Zimbabwe OR Zimbabwean OR Zambia OR Zambian OR Botswana OR Botswanan OR "South Africa" OR "South African" OR South Africa [Mesh] OR Lesotho OR Basotho OR Swaziland OR Swazi OR Namibia OR Namibian) AND (("1998/01/01"[PDAT] : "2018/12/31"[PDAT]) AND English[lang])	**50**

[*] Swaziland was renamed eSwatini in 2018. Because these literature review searches were conducted in July 2018 and aimed to include data published between 1998 and the time of the search, the search strategy utilized “Swaziland”.
[†] All searches were conducted on 11 July 2018, except the Pregnancy Loss & Stillbirth search which was conducted on 2 July 2018.

Table A.2. Included References by Outcome

Author, Year (Reference)	Country	HIV status*	Outcome											
			Pregnancy Loss	Stillbirth	Congenital Anomaly	LBW	Preterm Birth	Neonatal Mortality	Chorioamnionitis	Endometritis	Postpartum Hemorrhage	Gestational Hypertension	Preeclampsia/ Eclampsia	PPROM
Abdel-Aleem, 2010 (1)	South Africa	NR									X			
Ades, 2013 (2)	Uganda	Positive			X			X						
Ali, 2017 (3)	Malawi	NR	X	X	X			X						
Allanson, 2015 (4)	South Africa	Both		X										
Anderson, 2018 (5)	South Africa	Both				X	X							
Arinaitwe, 2013 (6)	Uganda	Negative		X		X								
Ashorn, 2015 (7)	Malawi	Both	X	X		X	X	X						
Asiki, 2015 (8)	Uganda	Both		X										
Atukunda, 2014 (9)	Uganda	NR									X			
Bailey, 2017 (10)	Malawi	Both		X										
Bamigboye, 1998 (11)	South Africa	NR									X			
Basu, 2010 (12)	South Africa	NR		X			X					X		
Bebell, 2017 (13)	Uganda	Both								X				
Bello, 2010 (14)	South Africa	NR	X	X	X									
Beltman, 2011 (15)	Malawi	NR									X			
Bera, 2010 (16)	South Africa	Positive	X	X	X									
Bloch, 2015 (17)	South Africa	Both					X				X			
Bodkin, 2006 (18)	South Africa	Both					X					X		
Brahmbhatt, 2008 (19)	Uganda	Positive				X	X							
Braun, 2015 (20)	Uganda	Both		X		X	X							
Brittain, 2015 (21)	South Africa	NR	X				X							
Byaruhanga, 2000 (22)	Uganda	NR		X										
Chagomerana, 2017 (23)	Malawi	Positive					X							

Author, Year (Reference)	Country	HIV status*	Outcome											
			Pregnancy Loss	Stillbirth	Congenital Anomaly	LBW	Preterm Birth	Neonatal Mortality	Chorioamnionitis	Endometritis	Postpartum Hemorrhage	Gestational Hypertension	Preeclampsia/ Eclampsia	PPROM
Chetty, 2018 (24)	South Africa	Positive				X	X							
Chevalier, 2017 (25)	Malawi	Positive		X				X		X	X		X	
Chihana, 2015 (26)	Malawi	Both		X				X						
Colbourn, 2013 (27)	Malawi	NR		X				X						
Coutsoudis, 1999 (28)	South Africa	Positive	X	X		X	X	X						
Cutland, 2009 (29)	South Africa	Both		X			X			X				
Dow, 2013 (30)	Malawi	Positive				X	X							
Dreise, 2011 (31)	Uganda	NR			X									
Ebrahim, 2000 (32)	South Africa	NR		X										
Ellison, 2000 (33)	South Africa	NR										X		
Fall, 2015 (34)	South Africa	NR				X	X							
Fatti, 2016 (35)	South Africa	Positive		X										
Feng, 2010 (36)	Malawi	NR				X								
Feresu, 2003 (37)	Zimbabwe	NR				X	X							
Feresu, 2004 (38)	Zimbabwe	NR				X	X							
Feresu, 2005 (39)	Zimbabwe	NR		X										
Feresu, 2015 (40)	Zimbabwe	NR				X	X					X		
Filler, 2006 (41)	Malawi	Both						X						
Frank, 2004 (42)	South Africa	Both										X	X	
Friis, 2004 (43)	Zimbabwe	Both		X		X	X							
Friis, 2009 (44)	Zimbabwe	Both				X	X							
Gebhardt, 2009 (45)	South Africa	Negative	X	X			X					X	X	X
Gessessew, 2007 (46)	Ethiopia	NR												X
Gibb, 2012 (47)	Uganda	Positive		X	X		X							
Govender, 2017 (48)	South Africa	NR		X										
Gray, 2001(49)	Uganda	Both	X	X			X		X		X			

Author, Year (Reference)	Country	HIV status*	Outcome											
			Pregnancy Loss	Stillbirth	Congenital Anomaly	LBW	Preterm Birth	Neonatal Mortality	Chorioamnionitis	Endometritis	Postpartum Hemorrhage	Gestational Hypertension	Preeclampsia/ Eclampsia	PPROM
Gülmezoglu, 2001 (50)	South Africa	NR									X			
Gumede, 2017 (51)	South Africa	Both					X							
Gutman, 2013 (52)	Malawi	Negative		X	X	X	X							
Hall, 2014 (53)	South Africa	Both					X					X		
Harrison, 2007 (54)	South Africa	Both	X		X			X				X		
Hjertholm, 2018 (55)	Malawi	NR				X								
Hofmeyr, 1998 (56)	South Africa	NR									X			
Hussain, 2011 (57)	South Africa	Both		X		X	X							
Immink, 2008 (58)	South Africa	NR											X	
Kalanda, 2006 (59)	Malawi	Both				X	X							
Kalilani, 2010 (60)	Malawi	NR				X								
Kalumbi, 2001 (61)	Malawi	NR		X										
Kananura, 2016 (62)	Uganda	NR		X				X						
Kananura, 2017 (63)	Uganda	NR		X				X						
Kapisi, 2017 (64)	Uganda	Negative				X	X							
Kasumba, 2000 (65)	Uganda	NR		X		X								
Kaye, 2001 (66)	Uganda	NR												X
Kaye, 2006 (67)	Uganda	Both		X		X	X							
Kesande, 2014 (68)	Uganda	NR			X									
Kimani, 2016 (69)	Malawi, Uganda	NR	X	X		X	X							
Kiondo, 2014 (70)	Uganda	Both	X	X		X	X					X	X	
Kujala, 2017 (71)	Uganda	NR		X				X						
Kulmala, 2000 (72)	Malawi	Both	X	X		X		X						
Kumwenda, 2002 (73)	Malawi	Positive	X	X										
Kundodyiwa, 2001 (74)	Zimbabwe	NR									X			

Author, Year (Reference)	Country	HIV status*	Outcome											
			Pregnancy Loss	Stillbirth	Congenital Anomaly	LBW	Preterm Birth	Neonatal Mortality	Chorioamnionitis	Endometritis	Postpartum Hemorrhage	Gestational Hypertension	Preeclampsia/ Eclampsia	PPROM
Lavin, 2016 (75)	South Africa	NR		X										
Lewycka, 2013 (76)	Malawi	NR		X				X						
Liu, 2014 (77)	South Africa	Positive	X	X	X	X	X	X						
Luntamo, 2010 (78)	Malawi	Both				X	X							
Madhi, 2016 (79)	South Africa	Negative		X		X	X							
Majoko, 2002 (80)	Zimbabwe	NR										X		
Majoko, 2007 (81)	Zimbabwe	Both	X	X		X	X	X						
Makanani, 2018 (82)	Malawi, South Africa, Zimbabwe, Uganda	Negative	X	X	X									
Malaba, 2017 (83)	South Africa	Both	X	X		X	X							
McDonald, 2018 (84)	Uganda	Positive		X										
Metaferia, 2009 (85)	Malawi	NR		X										
Moodley, 2016a (86)	South Africa	Both		X		X	X							
Moodley, 2016b (87)	South Africa	Both										X	X	
Moodley, 2017 (88)	South Africa	Both	X	X			X	X						
Moyer, 2016 (89)	Uganda	Both		X		X	X							
Msamati, 2000 (90)	Malawi	NR			X									
Murray, 2018 (91)	South Africa	Both				X	X					X		
Musana, 2009 (92)	Kenya	Both							X					X
Muti, 2015 (93)	Zimbabwe	NR		X		X	X					X	X	
Mwanyumba, 2003 (94)	Kenya	NR							X	X				
Msyamboza, 2009 (95)	Malawi	NR		X		X								
Nabiwemba, 2013 (96)	Uganda	NR				X	X							

Author, Year (Reference)	Country	HIV status*	Outcome											
			Pregnancy Loss	Stillbirth	Congenital Anomaly	LBW	Preterm Birth	Neonatal Mortality	Chorioamnionitis	Endometritis	Postpartum Hemorrhage	Gestational Hypertension	Preeclampsia/ Eclampsia	PPROM
Naidoo, 2011 (97)	South Africa	NR	X	X										
Naidu, 2001 (98)	South Africa	Both		X				X						
Nakubulwa, 2017 (99)	Uganda	Both				X	X			X				X
Nankabirwa, 2011 (100)	Uganda	NR		X										
Nankabirwa, 2015 (101)	Uganda	Both		X				X						
Ndibazza, 2010 (102)	Uganda	Both	X	X		X								
Ndibazza, 2011 (103)	Uganda	Both			X									
Ndirangu, 2012 (104)	South Africa	Both				X	X	X						
Ndyomugyenyi, 2000 (105)	Uganda	NR	X	X		X								
Ndyomugyenyi, 2001 (106)	Uganda	NR		X		X								
Ndyomugyenyi, 2008 (107)	Uganda	NR		X	X	X	X	X						
Ndyomugyenyi, 2011 (108)	Uganda	NR	X	X	X	X	X	X						
Ngwenya, 2016 (109)	Zimbabwe	NR									X			
Ngwenya, 2017 (110)	Zimbabwe	NR											X	
Njiru, 2013 (111)	Uganda	NR				X								
Nkhoma, 2012 (112)	Malawi	Both				X								
Noble, 2005 (113)	Zimbabwe	Both					X							
Ntuli, 2012 (114)	South Africa	NR		X										
Ogbonna, 2007 (115)	Zimbabwe	Both				X								
Ononge, 2016 (116)	Uganda	Both					X				X			
Pattinson, 2003 (117)	South Africa	NR		X				X						
Pattinson, 2010 (118)	South Africa	Both				X		X			X		X	
Rempis, 2017 (119)	Uganda	Both		X										
Roh, 2017 (120)	Uganda	Positive				X	X							
Röllin, 2015 (121)	South Africa	NR				X	X							
Rollins, 2007 (122)	South Africa	Both	X	X		X								

Author, Year (Reference)	Country	HIV status*	Outcome											
			Pregnancy Loss	Stillbirth	Congenital Anomaly	LBW	Preterm Birth	Neonatal Mortality	Chorioamnionitis	Endometritis	Postpartum Hemorrhage	Gestational Hypertension	Preeclampsia/ Eclampsia	PPROM
Sania, 2017 (123)	South Africa	Positive				X								
Sayed, 2008 (124)	South Africa	NR			X									
Schrag, 2012 (125)	South Africa	Both				X		X				X		
Sebitloane, 2008 (126)	South Africa	Positive								X	X			
Sebitloane, 2017 (127)	South Africa	Both		X			X				X			
Shabalala, 2017 (128)	South Africa	Both									X			
Shamu, 2018 (129)	Zimbabwe	Both				X	X							
Sibeko, 2011 (130)	South Africa	Negative	X	X	X		X				X			
Sirak, 2014 (131)	Ethiopia	NR												X
Steyn, 2006 (132)	South Africa	NR				X	X							
Sullivan, 1999 (133)	Malawi	NR					X							
Taha, 2018 (134)	Malawi, South Africa, Zimbabwe, Uganda	Positive	X	X	X	X	X							
Talip, 2010 (135)	South Africa	NR	X	X				X						
Taylor, 2017 (136)	Malawi	NR				X	X							
Theron, 2015 (137)	South Africa	NR			X									
Ticconi, 2003 (138)t	Zimbabwe	Both	X	X										
Ticconi, 2005 (139)	Zimbabwe	Both				X	X							
Turner, 2013 (140)	Malawi	Positive				X	X		X					
Turton, 2017 (141)	South Africa	NR				X	X							
Van Den Broek, 2005 (142)	Malawi	NR	X	X		X	X							
Van Den Broek, 2009 (143)	Malawi	Both					X							
Van Den Broek, 2003 (144)	Malawi	NR		X										

Author, Year (Reference)	Country	HIV status*	Outcome											
			Pregnancy Loss	Stillbirth	Congenital Anomaly	LBW	Preterm Birth	Neonatal Mortality	Chorioamnionitis	Endometritis	Postpartum Hemorrhage	Gestational Hypertension	Preeclampsia/ Eclampsia	PPROM
Van Der Merwe, 2011 (145)	South Africa	Positive				X	X							
Verhoeff, 1999 (146)	Malawi	Both		X		X								
Villar, 2014 (147)	Kenya	NR												X
Wabwire-Mangen, 1999 (148)	Uganda	Positive							X					
Walter, 2006 (149)	Zambia	Positive							X					
Wanyama, 2016 (150)	Uganda	Negative				X								
Worku, 2014 (151)	Ethiopia	Both								X				
Young, 2012 (152)	Uganda	Positive	X	X										

* NR = not reported. Both = Includes both women living with HIV and HIV negative women. In some, but not all of these studies, prevalence was disaggregated by HIV serostatus.

SECTION B: PREGNANCY LOSS & STILLBIRTH

Table B.1. Search Result Flow

	N
Title Review	874
Abstract Review	237
Manuscripts Reviewed*	174
Manuscripts Included From Main Search†	124
Manuscripts Added From Other Searches‡	73
Total Manuscripts Abstracted	197
Total Manuscripts Included For Analysis - 042 Countries Only§	78
Stillbirth**	71
Pregnancy Loss**	29

* Does not include number of references from systematic reviews reviewed.
† Includes systematic review reference reviews.
‡ Some of these include duplicate populations with the studies that were abstracted through the main search. These were removed at analysis.
§ Excludes duplicate populations, studies with incorrect definitions of the outcomes
** Kimani_2016(69) reported prevalence estimates for two study countries and Taha_2018 (134) reported estimates for four countries. These studies were counted as one manuscript each but provided multiple prevalence estimates in the analysis phase.

Table B.2. Pooled Prevalence of Pregnancy Loss – Overall, by MTN-042 Country, and by HIV status

	# of estimates	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95%CI)	I^2 (p-value)
Study Countries	33	49,095	3.3% (0.0, 24.8)	1.6% (0.7, 2.2)	1.9% (1.1, 2.8)	97.2% (<0.01)
Malawi	7	3,956	1.0% (0.2, 1.8)	0.9% (0.2, 1.8)	0.6% (0.2, 1.2)	59.5% (0.02)
South Africa	14	30,080	4.2% (0.1, 20.8)	1.6% (1.2, 6.7)	2.5% (1.1, 4.3)	98.3% (<0.01)
Uganda	8	7,382	1.4% (0.0, 2.5)	1.4% (0.6, 2.2)	1.4% (0.7, 2.1)	75.3% (<0.01)
Zimbabwe	3	7,520	1.8% (0.3, 3.1)	2.2% (0.3, 3.1)	1.0% (0.0, 3.8)	-
Multiple countries(82)	1	157	24.8% -	24.8% -	24.8% (18.3, 32.4)	-
Living with HIV	11	5,255	2.1% (0.0, 9.1)	1.6% (0.7, 2.0)	0.8% (0.3, 1.5)	63.7% (<0.01)
HIV Negative	4	2,161	6.7% (0.3, 20.8)	2.9% (0.9, 12.6)	3.6% (0.5, 9.1)	94.6% (<0.01)

Figure B.1. Forest Plot Summarizing the Pooled Prevalence of Pregnancy Loss – Malawi

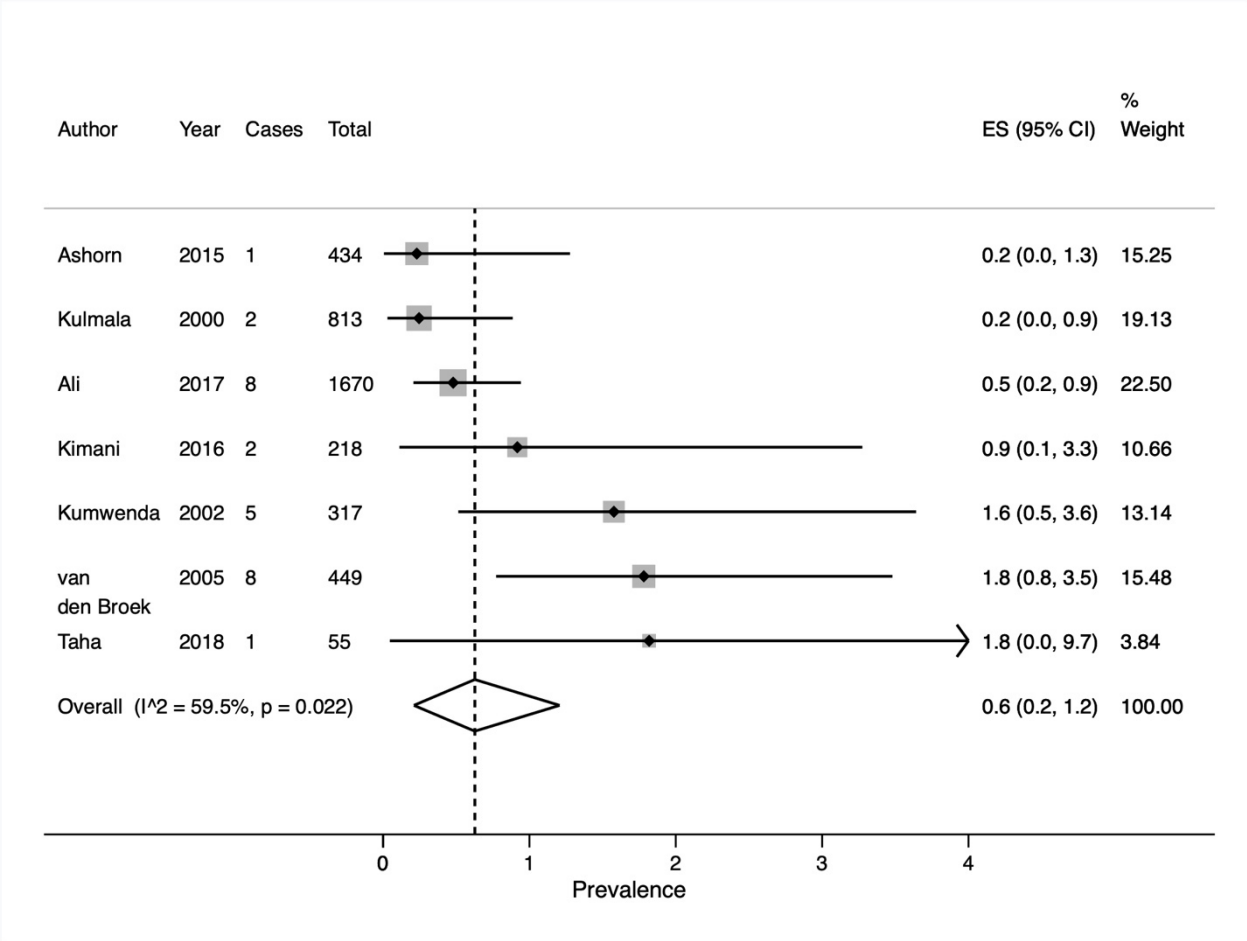


Figure B.2. Forest Plot Summarizing the Pooled Prevalence of Pregnancy Loss - South Africa

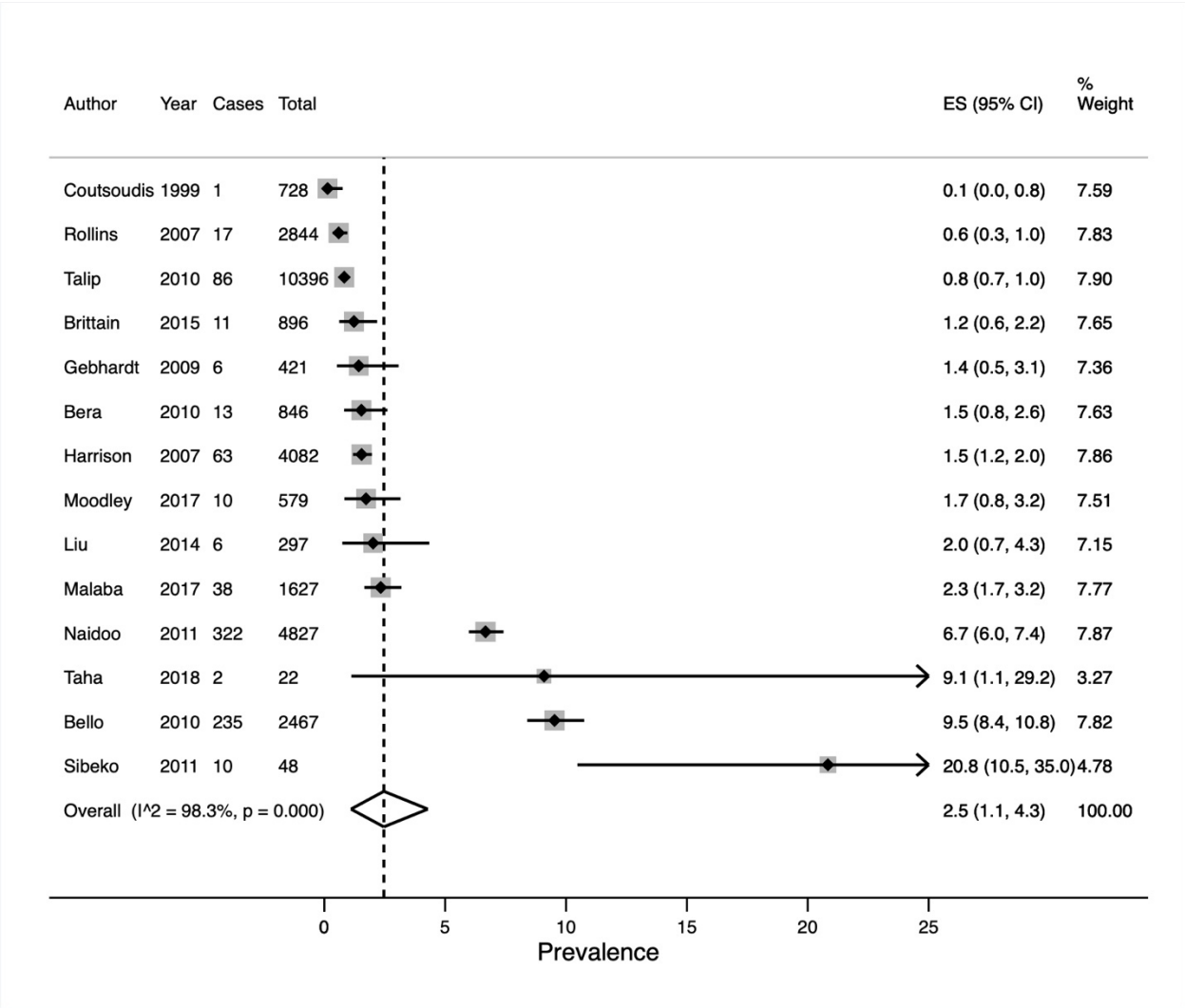


Figure B.3. Forest Plot Summarizing the Pooled Prevalence of Pregnancy Loss – Uganda

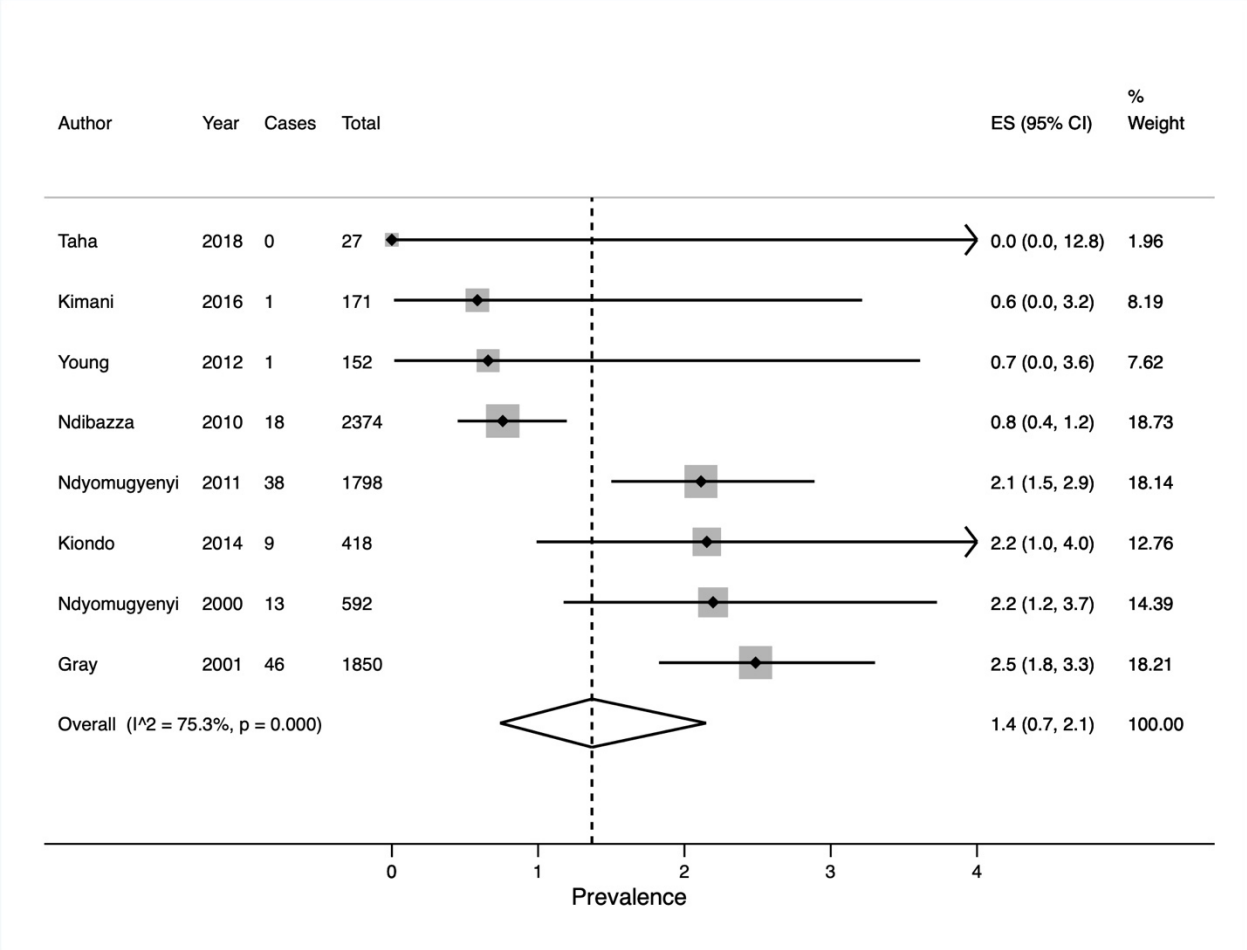


Figure B.4. Forest Plot Summarizing the Pooled Prevalence of Pregnancy Loss - Zimbabwe

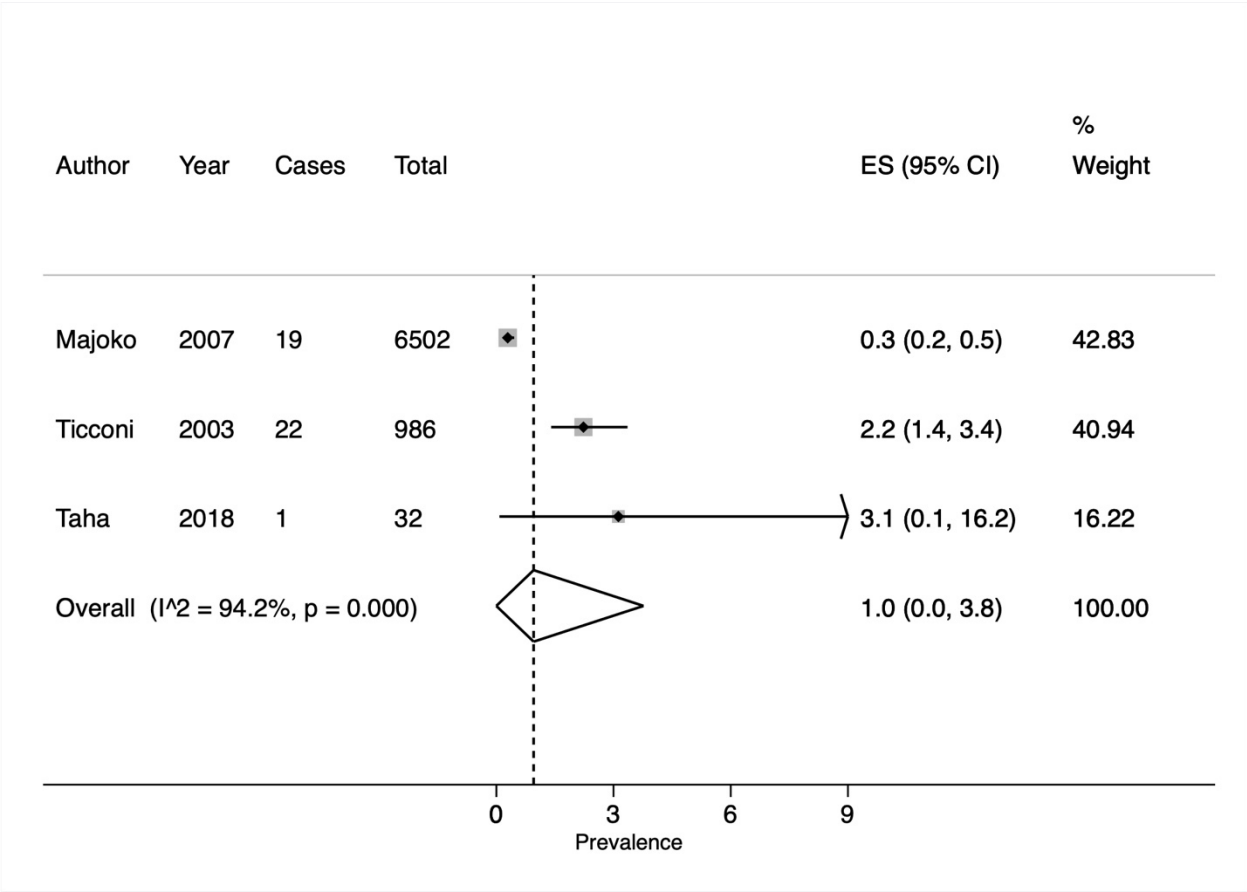


Figure B.5. Forest Plot Summarizing the Pooled Prevalence of Pregnancy Loss Among Women Living with HIV

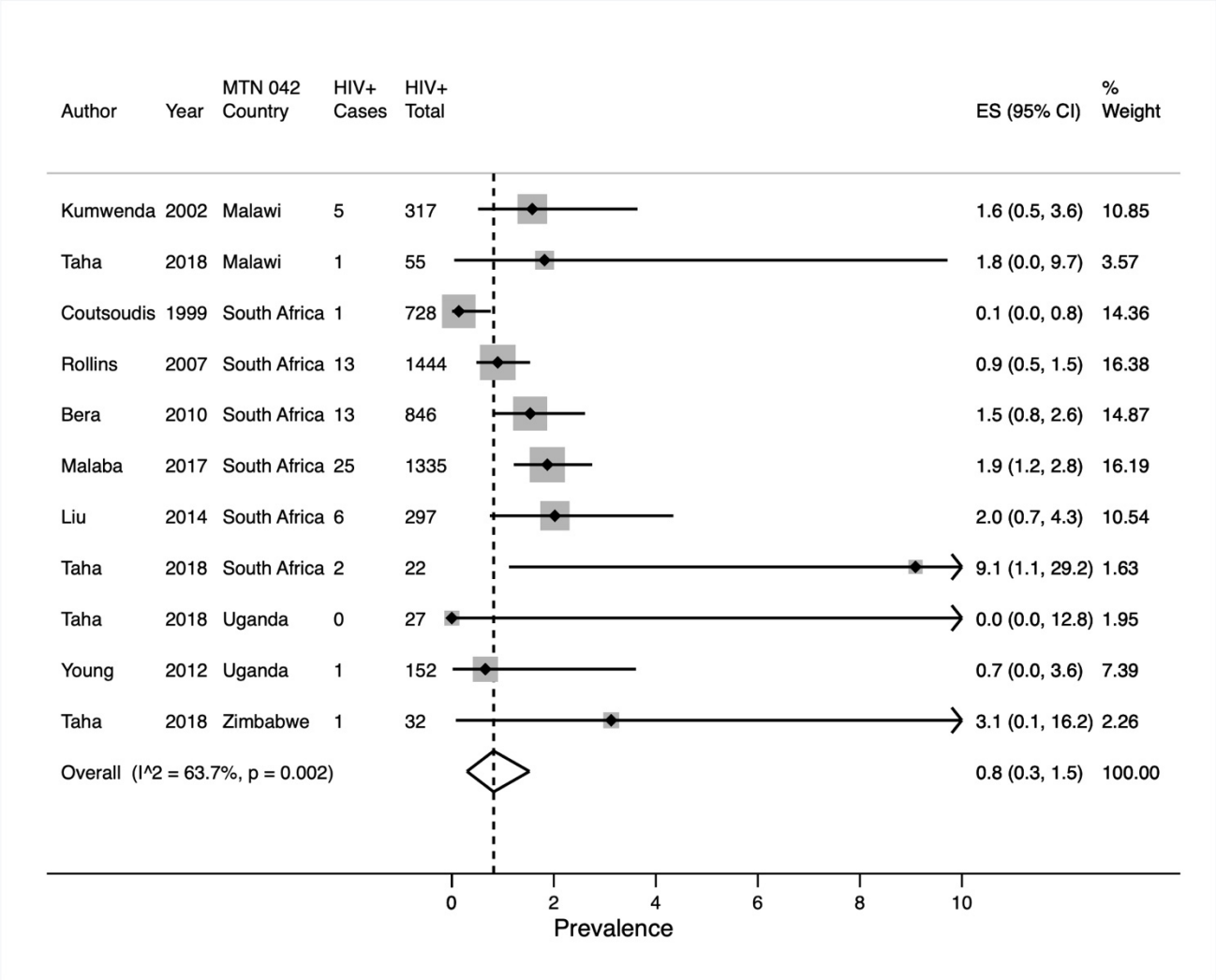


Figure B.6. Forest Plot Summarizing the Pooled Prevalence of Pregnancy Loss Among HIV Negative Women

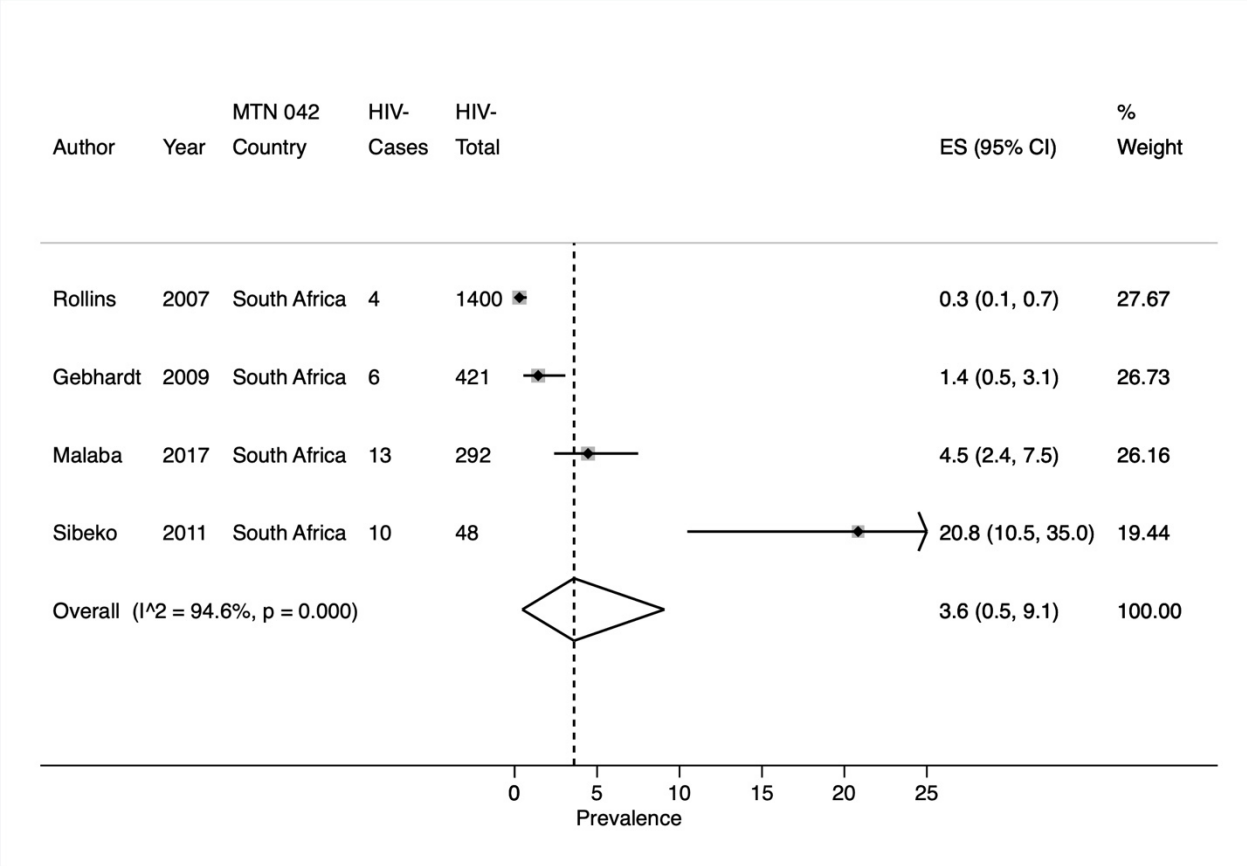


Table B.3. Sensitivity Analysis Excluding Studies with Pregnancy Loss Definition Undefined: Pooled Prevalence of Pregnancy Loss – Overall, by MTN-042 Country, and by HIV status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95%CI)	I^2 (p-value)
Study Countries	25	32,780	3.0% (0.0, 24.8)	1.5% (0.6, 2.2)	1.7% (0.8, 3.0)	97.7% (<0.01)
Malawi	6	3,639	0.9% (0.2, 1.8)	0.7% (0.2, 1.8)	0.5% (0.1, 1.1)	56.2% (<0.01)
South Africa	11	18,740	3.3% (0.1, 9.5)	1.7% (1.4, 6.7)	2.3% (0.8, 4.4)	98.2% (<0.01)
Uganda	4	2,724	0.5% (0.0, 0.8)	0.6% (0.3, 0.7)	0.3% (0.1, 0.6)	0%
Zimbabwe	3	7,520	1.9% (0.3, 3.1)	2.2% (0.3, 3.1)	1.0% (0.0, 3.8)	-
Multiple countries (82)	1	157	24.8% -	24.8% -	24.8% (18.3, 32.4)	-
Living with HIV	10	4,938	2.1% (0.0, 9.1)	1.7% (0.7, 2.0)	0.8% (0.2, 1.5)	66.7% (<0.01)
HIV negative	3	2,113	2.1% (0.3, 4.4)	1.4% (0.3, 4.4%)	1.6% (0.1, 4.7)	92.6% (<0.01)

* Included Studies: Ali_2017 (3), Ashorn_2015 (7), Bello_2010 (14), Bera_2014 (16), Coutsoudis_1999 (8), Gebhardt_2009 (45), Harrison_2007 (54), Kimani_2016 (69), Kulmala_2000 (72), Liu_2014 (77), Majoko_2007 (81), Makanani_2018 (82), Malaba_2017 (83), Moodley_2017 (88), Nadoo_2011 (97), Ndibazza_2010 (102), Rollins_2007 (122), Taha_2018 (134), Ticconi_2003 (138), Young_2012 (152), van den Broek_2005 (142)

Table B.4. Sensitivity Analysis Including Studies Defining Pregnancy Loss ≤20 weeks: Pooled Prevalence of Pregnancy Loss – Overall, by MTN-042 Country, and by HIV status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95% CI)	I^2 (p-value)
Study Countries	8	9,446	2.2% (0.0, 9.1)	1.2% (0.4, 2.7)	0.5% (0.0, 1.6)	83.9% (<0.01)
Malawi	2	1,725	1.1% (0.5, 1.8)	1.1% (0.5, 1.8)	0.2% (0.0, 0.6)	-
South Africa	1	22	9.1% -	9.1% -	9.1% (1.1,29.2)	-
Uganda	2	179	0.3% (0.0, 0.7)	0.3% (0.0, 0.7)	0.2% (0.0,1.9)	-
Zimbabwe	3	7,520	1.9% (0.3, 3.1)	2.2% (0.3, 3.1)	1.0% (0.0, 3.8)	-
Multiple countries	0	-	-	-	-	-
Living with HIV	5	288	2.9% (0.0, 9.1)	1.8% (0.7, 3.1)	1.2% (0.0, 3.9)	27.2% (0.2)
HIV Negative	0	-	-	-	-	-

* Includes studies defining miscarriage as those occurring between 12-20 weeks or ≤20 weeks. Included studies: Ali_2017 (3), Taha_2018 (134), Young_2012 (152), Majoko_2007 (81), Ticconi_2003 (138).

Table B.5. Sensitivity Analysis Including Studies Defining Pregnancy Loss ≤28 weeks: Pooled Prevalence of Pregnancy Loss – Overall, by MTN-042 Country, and by HIV status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95% CI)	I^2 (p-value)
Study Countries	7	11,950	2.0% (0.1, 6.7)	1.5% (0.6, 2.3)	1.7% (0.4, 3.8)	97.6% (<0.01)
Malawi	1	218	0.9% -	-	0.9% (0.1, 3.3)	-
South Africa	5	11,561	2.5% (0.1, 6.7)	2.2% (1.5, 2.3)	2.1% (0.4, 4.9)	98.3% (<0.01)
Uganda	1	171	0.6% -	-	0.6% (0.0, 3.2)	-
Zimbabwe	0	-	-	-	-	-
Living with HIV	3	2,360	1.3% (0.1, 2.0)	1.9% (0.1, 2.0)	1.1% (0.1, 3.0)	89.2% (<0.01)
HIV negative	1	292	4.4% -	-	4.5% (2.4, 7.5)	-

* Does not includes studies defining miscarriage as those occurring between 12-20 weeks or <20 weeks. Included studies: Kimani_2016 (69), Coutsoudis_1999 (28), Harrison_2007 (54), Liu_2014 (77), Malaba_2017 (83), Naidoo_2011(97)

Table B.6. Sensitivity Analysis Excluding Outliers: Pooled Prevalence of Pregnancy Loss – Overall, by MTN-042 Country, and by HIV status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95% CI)	I^2 (p-value)
Study Countries	28	41,574	1.3% (0, 3.1)	1.5% (0.6, 2.1)		86.2% (<0.01)
Malawi	7	3,956	1.0% (0.2, 1.8)	0.9% (0.2, 1.8)	0.6% (0.2, 1.2)	59.5% (<0.01)
South Africa	10	22,716	1.3% (0.1, 2.3)	1.5% (0.8, 1.7)	1.2% (0.8, 1.7)	83.7% (<0.01)
Uganda	8	7,382	1.4% (0.0, 2.5)	1.4% (0.6, 2.2)	1.4% (0.7, 2.1)	75.3% (<0.01)
Zimbabwe	3	7,520	1.9% (0.3, 3.1)	2.2% (0.3, 3.1)	1.0% (0.7, 1.4)	-
Living with HIV	10	5,233	1.4% (0.0, 3.1)	1.6% (0.7, 1.9)	0.9% (0.4, 1.5)	61.0% (<0.01)
HIV negative	3	2,113	2.1% (0.3, 4.4)	1.4% (0.3, 4.4)	1.6% (0.1, 4.7)	92.6% (<0.01)

* N=5 studies had a prevalence estimate that was greater than 1.5 times the interquartile range (IQR) of all included studies. Excluded studies and details are listed in Supplementary Table B.15.

Table B.7. Pooled Prevalence of Stillbirth– Overall, by MTN-042 Country, and by HIV status

	# of estimates	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95% CI)	I^2 (p-value)
Study Countries	75	1,498,361	2.8% (0, 11.0)	2.3% (1.6, 3.8)	2.5% (2.2, 2.7)	98.0% (<0.01)
Malawi	18	536,079	2.6% (0.1, 7.3)	1.9% (1.7, 3.4)	2.3% (1.8, 2.9)	96.9% (<0.01)
South Africa	26	871,383	2.4% (0.0, 5.0)	2.3% (1.5, 3.7)	2.0% (1.7, 2.4)	98.0% (<0.01)
Uganda	23	66,533	3.3% (0.0, 8.8)	2.7% (1.8, 4.5)	3.0% (2.1, 4.1)	97.6% (<0.01)
Zimbabwe	6	24,003	2.4% (1.2, 5.6)	1.7% (1.3, 3.2)	2.0% (0.4, 4.7)	98.4% (<0.01)
Multiple countries	2	363	7.2% (3.4, 11.0)	7.2% (3.4, 11.0)	8.1% (5.5, 11.2)	-
Living with HIV	17	13,377	3.4% (0.0, 11.0)	2.6% (1.9, 4.4)	2.9% (2.0, 3.8)	81.5% (<0.01)
HIV negative	9	9,510	2.2% (0.0, 5.4)	2.2% (1.4, 2.9)	1.9% (1.3, 2.5)	54.0% (<0.01)

Figure B.7. Forest Plot Summarizing Pooled Prevalence of Stillbirth - Malawi

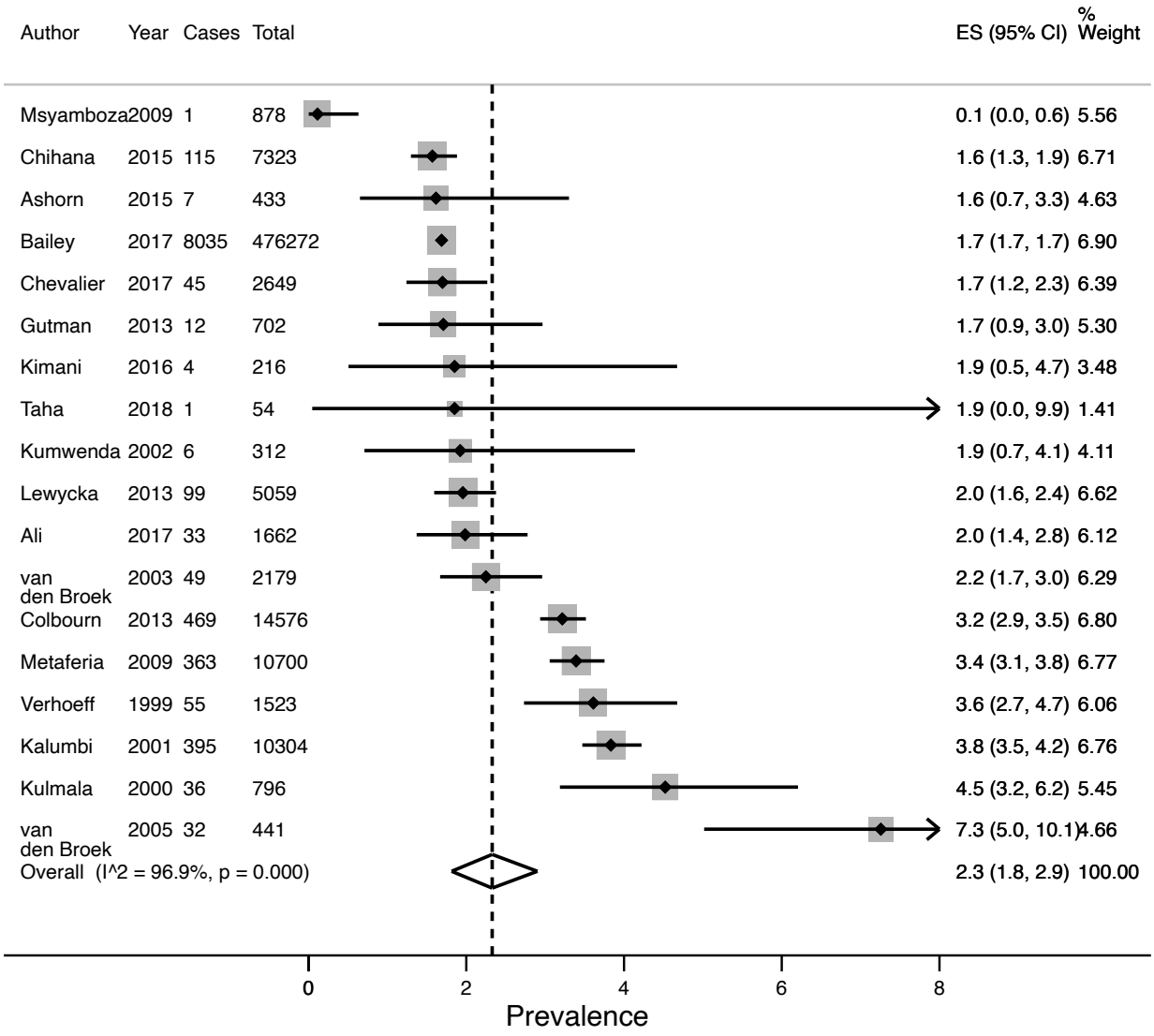


Figure B.8. Forest Plot Summarizing Pooled Prevalence of Stillbirth - South Africa

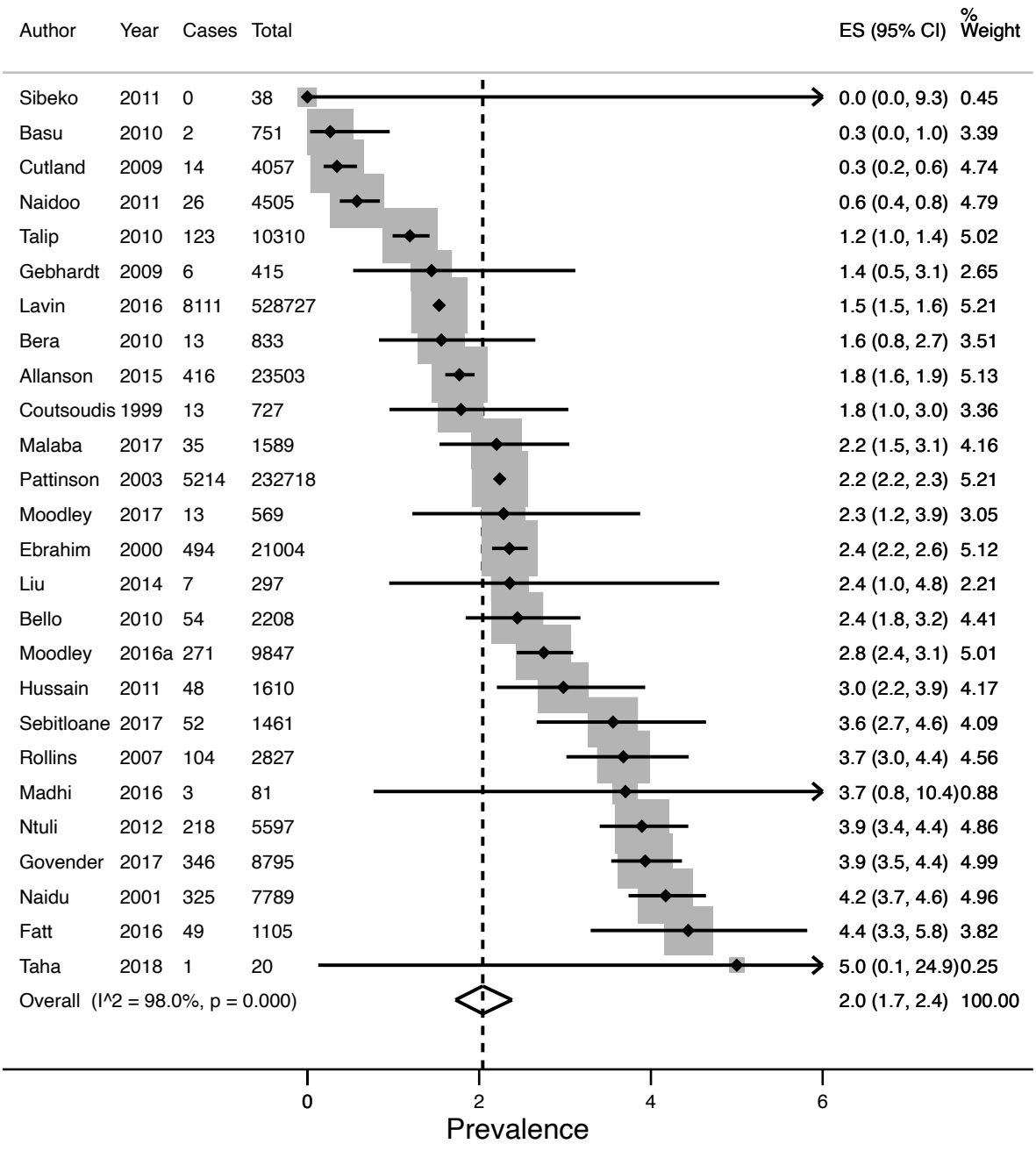


Figure B.9. Forest Plot Summarizing the Pooled Prevalence of Stillbirth - Uganda

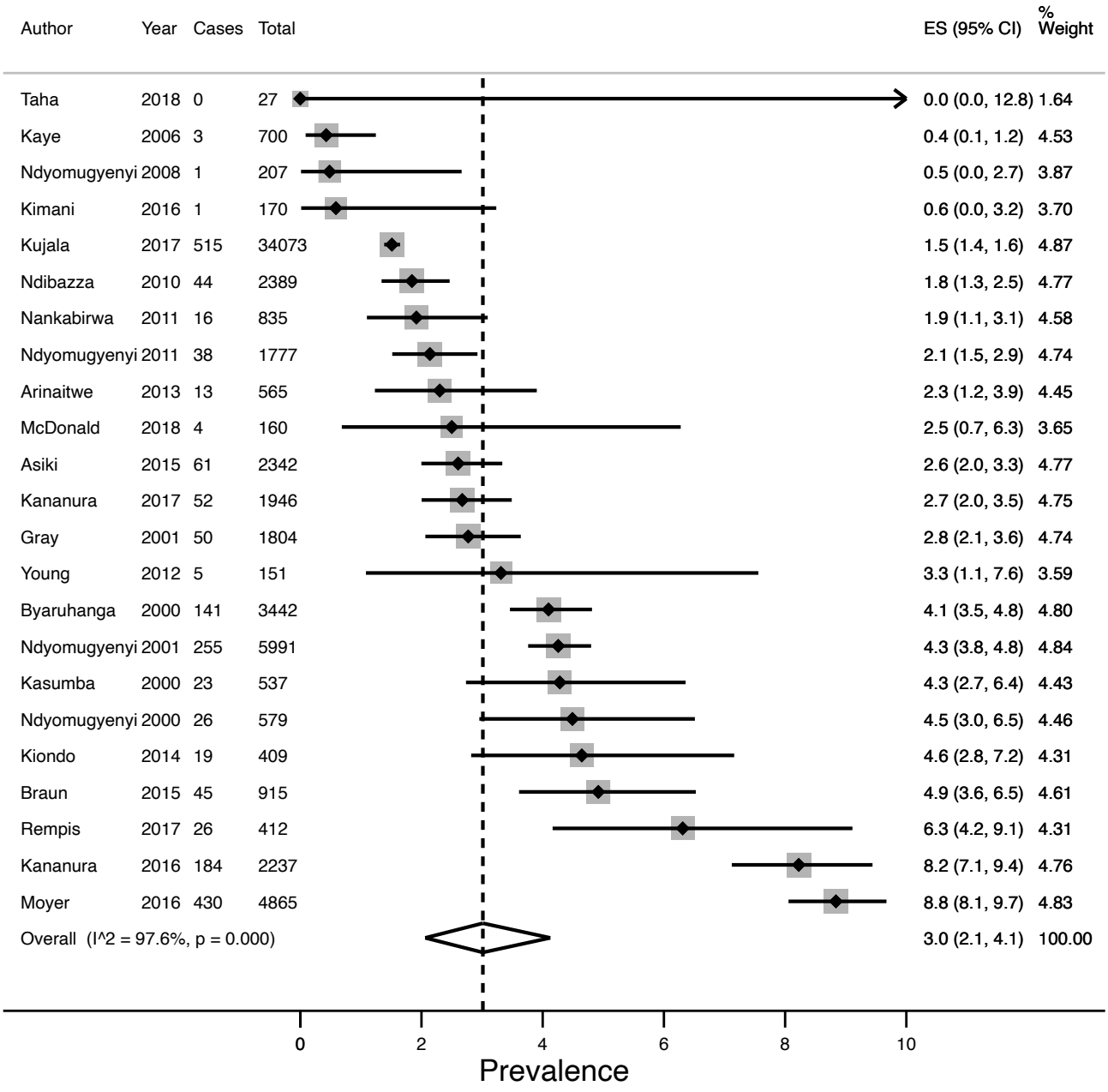


Figure B.10. Forest Plot Summarizing the Pooled Prevalence of Stillbirth - Zimbabwe

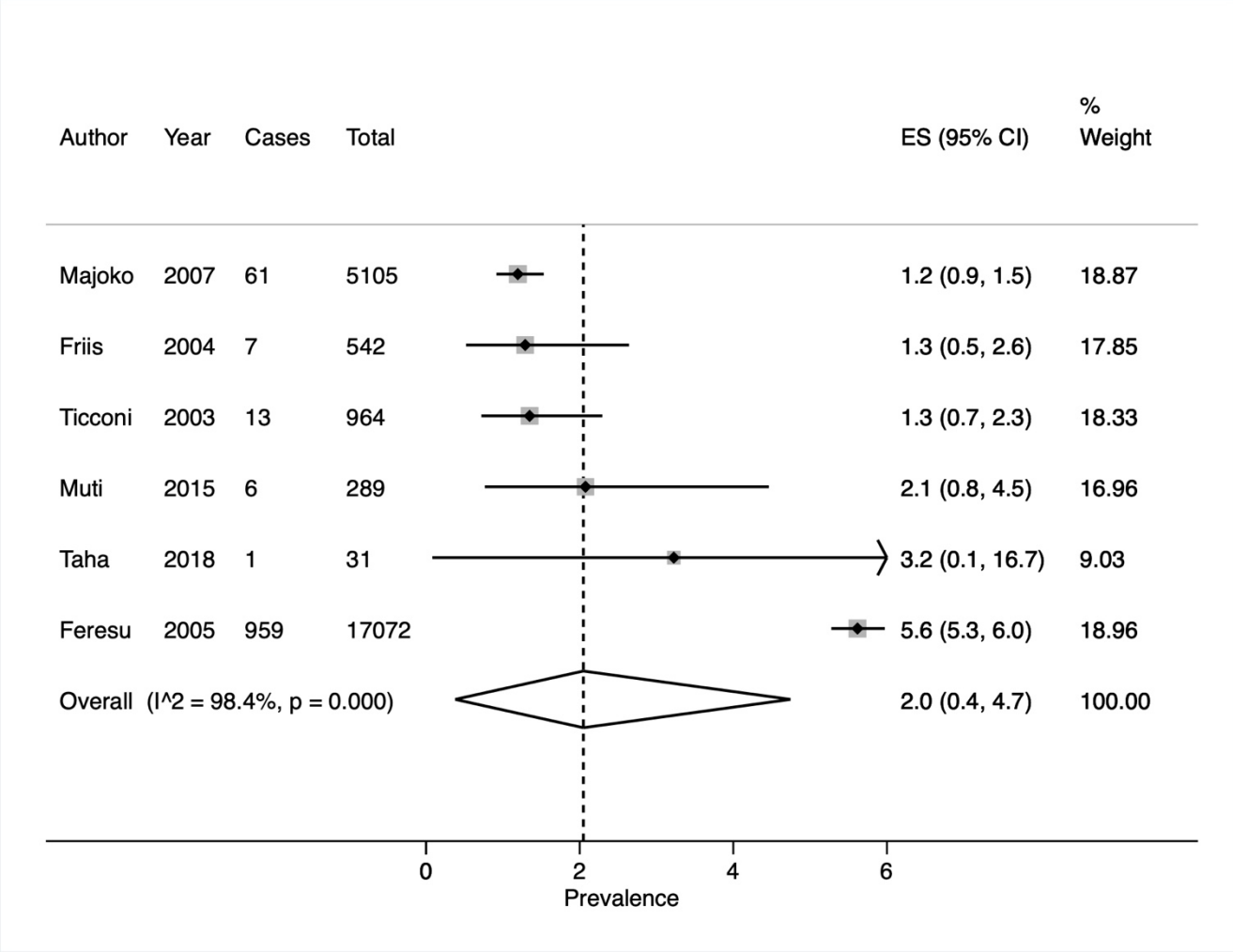


Figure B.11. Forest Plot Summarizing the Pooled Prevalence of Stillbirth among Women Living with HIV

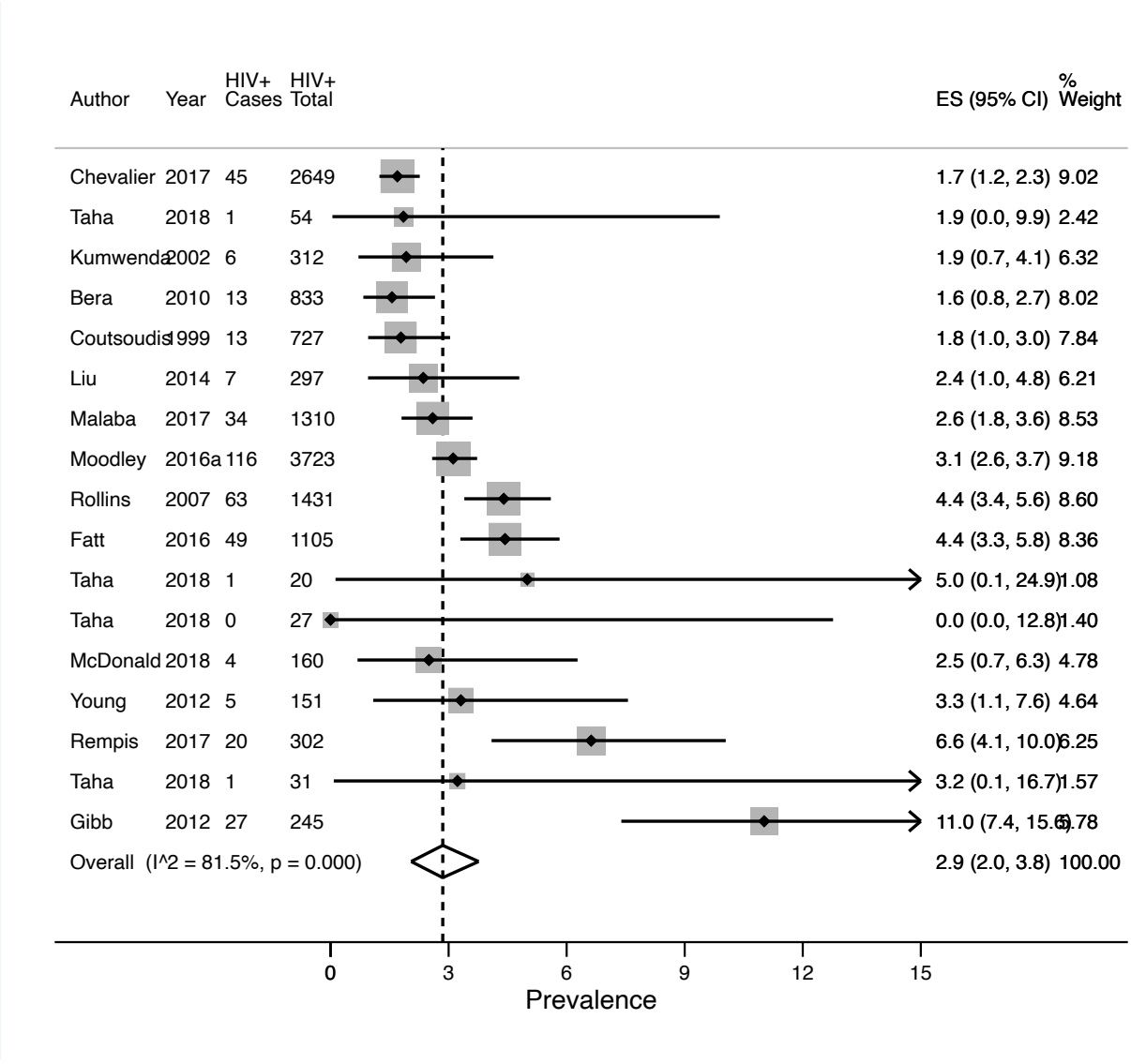


Figure B.12. Forest Plot Summarizing the Pooled Prevalence of Stillbirth among HIV Negative Women

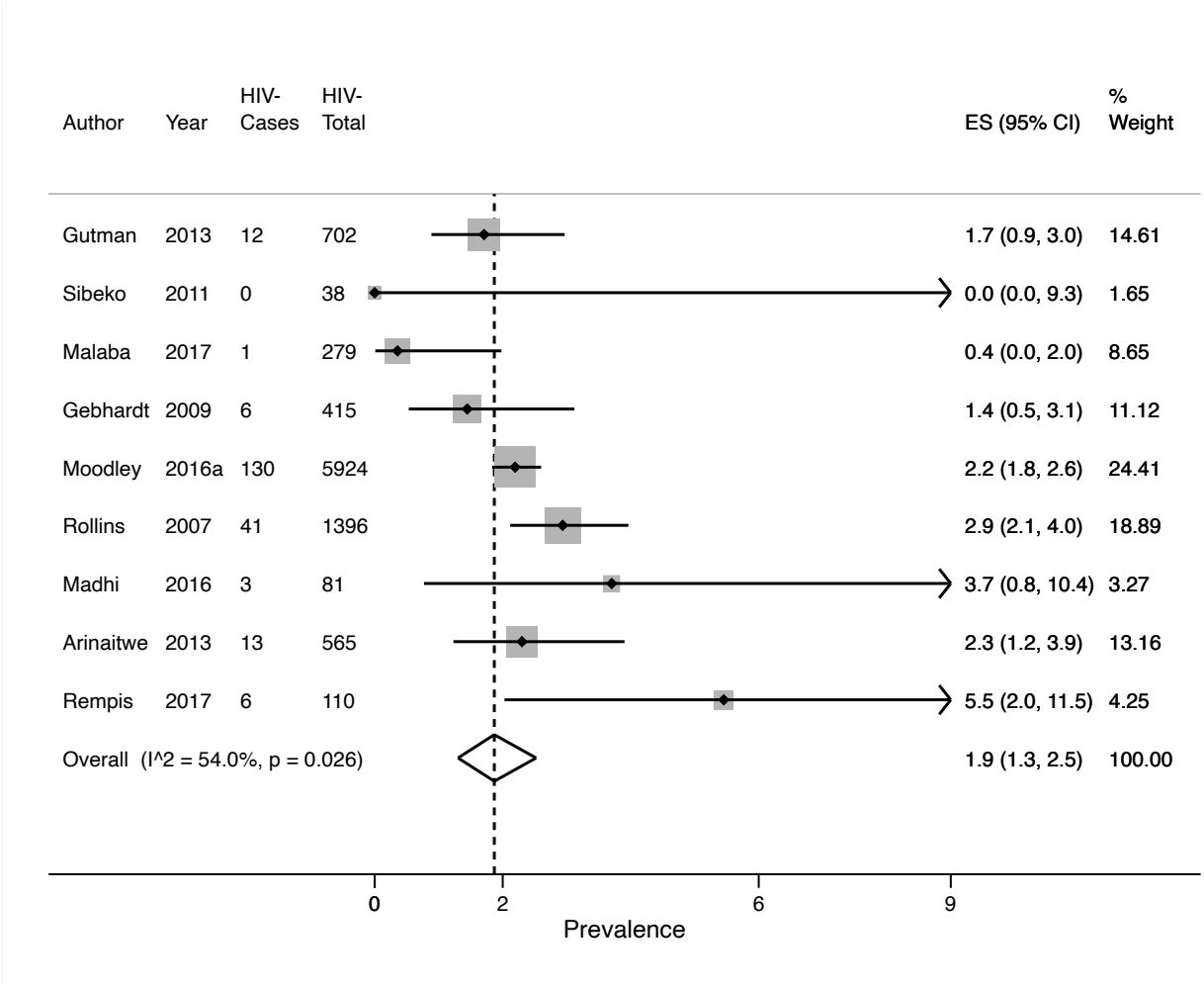


Table B.8. Sensitivity Analysis Excluding Studies with Stillbirth Definition Undefined: Pooled Prevalence of Stillbirth – Overall, by MTN-042 Country, and by HIV status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95% CI)	I^2 (p-value)
Study Countries	37	1,125,008	3.2% (0.0, 11.0)	2.4% (1.8, 3.9)	2.7% (2.3, 3.1)	98.3% (<0.01)
Malawi	9	487,582	2.7% (1.6, 7.3)	1.9% (1.7, 2.0)	2.3% (1.7, 3,0)	87.4% (0.01)
South Africa	10	568,980	2.5% (0.6, 5.0)	2.3% (1.8, 3.7)	1.9% (1.4, 2.5)	96.2% (<0.01)
Uganda	13	50,016	3.4% (0.0, 8.8)	2.6% (1.8, 4.6)	3.2% (1.6, 5.1)	98.4% (<0.01)
Zimbabwe	3	18,067	3.4% (1.3, 5.6)	3.2% (1.3, 5.6)	2.9% (0.2, 7.8)	-
Multiple countries	2	363	7.2% (3.4, 11.0)	7.2% (3.4, 11.0)	8.1% (5.5, 11.2)	-
Living with HIV	12	6,094	3.6% (0.0, 11.0)	2.9% (1.8, 4.7)	3.1% (1.8, 4.9)	84.4% (<0.01)
HIV negative	2	1,506	4.2% (2.9, 5.4)	4.2% (2.9, 5.4)	2.9% (2.1, 3.9)	-

* Included studies: Ali_2017(3), Allanson_2015 (4), Ashorn_2015 (7), Asiki_2015 (8), Bailey_2017 (10), Bello_2010 (14), Chevalier_2017 (25), Coutsoudis_1999 (28), Feresu_2005(39), Gibb_2012 (47), Kujala_2017 (71), Kananura_2016 (62), Kananura_2017(63), Kimani_2016 (69), Kiondo_2014 (70), Kulmala_2000 (72), Lavin_2016 (75), Lewycka_2013 (76), Liu_2014 (77), Makanani_2018 (82), McDonald_2018 (84),

Moodley_2017 (88), Moyer_2016 (89), Nankabirwa_2011 (100), Naidoo_2011(97), Ndibazza_2010 (102), Ntuli_2012(114
Rollins_2007 (122), Taha_2018 (134), Ticconi_2003 (138), Young_2012 (152), van den Broek_2005 (142)

Table B.9. Pooled Prevalence of Antepartum Stillbirth – Overall and by MTN-042 Country

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95% CI)	I^2 (p-value)
Study Countries	9	563,797	1.9% (0.4, 3.5)	1.6% (1.1, 2.7)	1.7% (1.1, 2.4)	99.2% (<0.01)
Malawi	3	501,539	0.9% (0.4, 1.6)	0.8% (0.4, 1.6)	0.8% (0.4, 1.4)	-
South Africa	3	36,889	2.0% (1.1, 2.8)	2.2% (1.1, 2.8)	2.0% (1.0, 3.2)	-
Uganda	2	8,297	2.5% (1.6, 3.5)	2.5% (1.6, 3.5)	2.6% (2.3, 3.0)	-
Zimbabwe	1	17,072	2.7% -	-	2.7% (2.4, 2.9)	-

* Included studies: Allanson_2015 (4), Bailey_2017 (10), Byaruhanga_2000 (22), Colbourn_2013 (27), Feresu_2005 (39), Metaferia_2009 (85), Moyer_2016 (89), Naidu_2001 (98), Ntuli_2012 (114)

Table B.10. Pooled Prevalence of Intrapartum Stillbirth – Overall and by MTN-042 Country

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95% CI)	I^2 (p-value)
Study Countries	10	564,042	2,2% (0.6, 5.1)	1.7% (1.1, 2.5)	1.8% (1.3, 2.4)	98.5% (<0.01)
Malawi	3	501,539	1.4% (0.9, 1.8)	1.6% (0.9, 1.8)	1.4% (0.8, 2.1)	-
South Africa	3	36,889	1.2% (0.6, 1.9)	1.1% (0.6, 1.9)	1.2% (0.5, 2.1)	-
Uganda	2	8,297	3.8% (2.5, 5.1)	3.8% (2.5, 5.1)	3.9% (3.5, 4.4)	-
Zimbabwe	1	17,072	1.2% -	-	1.2% (1.0, 1.4)	-
Multiple Countries	1	245	4.9% -	-	4.9% (2.6, 8.4)	-

* Included studies: Allanson_2015 (4), Bailey_2017 (10), Byaruhanga_2000 (22), Colbourn_2013 (27), Feresu_2005 (39), Gibb_2012 (47), Metaferia_2009 (85), Moyer_2016 (89), Naidu_2001 (98), Ntuli_2012 (114)

Table B.11. Sensitivity Analysis Including Studies Defining Stillbirth ≥20 Weeks Gestation: Pooled Prevalence of Stillbirth – Overall and by MTN-042 Country

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95% CI)	I^2 (p-value)
Study Countries	12	35,853	3.2% (0.0, 8.2)	2.9% (1.5, 4.4)	3.7% (1.4, 4.3)	97.5% (<0.01)
Malawi	2	2,703	1.8% (1.7, 1.9)	1.8% (1.7, 1.9)	1.3% (0.9, 1.8)	-
South Africa	2	5,617	4.4% (3.9, 5.0)	4.4% (3.9, 5.0)	2.9% (2.4, 3.4)	-
Uganda	4	4,361	3.6% (0.0, 8.2)	3.0% (1.3, 5.8)	3.5% (0.7, 7.9)	95.7% (<0.01)
Zimbabwe	4	23,172	2.8% (1.2, 5.6)	2.3% (1.3, 4.4)	2.3% (0.2, 6.1)	99.0% (<0.01)

* Included studies: Chevalier_2017 (25), Feresu_2005 (39), Kananura_2016 (62), Kananura_2017 (63), Majoko_2007 (81), Ntuli_2012 (114), Taha_2018 (134), Ticconi_2003 (138), Young_2012 (152)

Table B.12. Sensitivity Analysis Including Studies Defining Stillbirth ≥28 weeks Gestation: Pooled Prevalence of Stillbirth – Overall and by MTN-042 Country

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95% CI)	I^2 (p-value)
Study Countries	16	1,084,552	2.5% (0.6, 8.8)	1.8% (1.6, 2.4)	2.1% (1.8, 2.4)	97.9% (<0.01)
Malawi	3	481,547	1.8% (1.7, 2.0)	1.9% (1.7, 2.0)	1.6% (1.5, 1.8)	-
South Africa	5	557,759	1.6% (0.6, 2.4)	1.8% (1.5, 1.8)	1.4% (1.0, 1.8)	92.1% (<0.01)
Uganda	8	45,246	3.3% (0.6, 8.8)	2.2% (1.7, 4.4)	2.9% (1.2, 5.4)	98.8% (<0.01)
Zimbabwe	0	-	-	-	-	-
Living with HIV	4	1486	3.3% (1.8, 6.,6)	2.4% (2.1, 4.6)	3.1% (1.3, 5.5)	78.4% (<0.01)
HIV negative	1	110	5.5% -	-	5.5% (2.0, 11.5)	-

* Included studies: Allanson_2015 (4), Asiki_2015 (8), Bailey_2017 (10), Coutsoudis_1999 (28), Kujala_2017 (71), Kimani_2016 (69), Lavin_2016 (75), Lewycka_2013 (76), Liu_2014 (77), McDonald_2018 (84), Moyer_2016 (89), Nankabirwa_2011 (100), Naidoo_2011(97), Ndibazza_2010 (102), Rempis_2017 (119)

Table B.13. Sensitivity Analysis Excluding Outliers: Pooled Prevalence of Stillbirth – Overall, by MTN-042 Country, and by HIV status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95% CI)	I^2 (p-value)
Study Countries	71	1,490,573	2.5% (0, 6.3)	2.2% (1.6, 3.6)	2.2% (2.0, 2.4)	97.5% (<0.01)
Malawi	17	535,638	2.3% (0.1, 4.5)	1.9% (1.7, 3.2)	2.2% (1.7, 2.7)	96.9% (<0.01)
South Africa	26	871,383	2.4% (0.0, 5.0)	2.3% (1.5, 3.7)	2.0% (1.7, 2.4)	98.0% (<0.01)
Uganda	21	59,431	2.7% (0.0, 6.3)	2.6% (1.8, 4.3)	2.6% (0.9, 1.5)	93.6% (<0.01)
Zimbabwe	6	24,003	2.4% (1.2, 5.6)	1.7% (1.3, 3.2)	2.0% (0.4, 4.7)	98.4% (<0.01)
Multiple countries	1	118	3.4% -	-	3.4% (0.9, 8.5)	-
Living with HIV	16	13,132	2.9% (0.0, 6.6)	2.2% (0, 5.4)	2.5% (1.8, 3.2)	74.1% (<0.01)
HIV negative	9	9,510	2.2% (0.0, 5.4)	2.2% (1.4, 2.9)	1.9% (1.3, 2.5)	54.0% (<0.01)

* N=4 studies had a prevalence estimate that was greater than 1.5 times the interquartile range (IQR) of all included studies. Excluded studies and details are listed in Supplementary Table B.14.

Notes B.1. Pregnancy Loss and Stillbirth Review and Meta-Analyses: Study Specific Notes

- Basu_2010 (12) & Ebrahim_2000(32) - Some overlap in study years at same study hospital
    - Potential duplicate population, but included both
- Lewycka_2013 (76)
    - Had a 6 month inception period, and an intervention period. Abstracted the intervention period because sample size was bigger
- Colbourn_2013 (27)
    - Presents data for a baseline period, then an intervention period that had a control and 3 intervention arms. Included the baseline period only because it contributed the largest number of infants
- Harrison_2007 (54)
    - Could not stratify pregnancy loss/stillbirth estimates by trial arm, included overall estimate

Table B.14. Pregnancy Loss and Stillbirth Meta-Analyses Outliers

Outcome	Manuscript	Country	Frequency	Definition	Outcome Ascertainment	Notes
Pregnancy Loss	Sibeko_2011(130)	South Africa	20.8%	Miscarriage – weeks not defined, does not include terminated pregnancies	Monthly pregnancy tests	CAPRISA 004 trial. Miscarriage rate includes early preclinical pregnancy losses.
	Naidoo_2011 (97)	South Africa	6.7%	Spontaneous miscarriage ≤28 weeks gestation	Interviews with the women	
	Bello_2010 (14)	South Africa	9.5%	Spontaneous abortion – weeks not defined	Retrospective self-report	
	Makanani_2018 (82)	Malawi, South Africa, Uganda, Zimbabwe	24.8%	Spontaneous abortion, weeks not defined; estimate does not include therapeutic/elective abortion or ectopic pregnancies	Prospective follow-up of women who became pregnant until pregnancy outcome occurred	ASPIRE trial
	Taha_2018 (134)	South Africa	9.1%	Abortion/miscarriage <20 weeks	Prospective follow-up of women who became pregnant for pregnancy outcome	PROMOTE study incident pregnancies. Small sample size (2/22 pregnancies)
Stillbirth	Moyer_2016 (89)	Uganda	8.8%	Stillbirth - Not defined but text is suggestive of ≥28 weeks	Records review	
	Van den Broek_2003 (142)	Malawi	7.3%	Not born alive – no weeks specified	Interviews with the women	

	Kananura_2017 (63)	Uganda	8.2%	≥20 weeks gestation	Face to face interview within 12 months of delivery	
	Gibb_2012 (47)	Uganda, Zimbabwe	11.0%	≥22 weeks gestation	Prospective follow-up	

Table B.15. Pregnancy Loss Study Definition – Included Studies

B.15A. Individual study characteristics

Author, Year	Study Design	Pregnancy Loss Definition										Gestational Age*	
		<20 weeks	≤20 weeks	<21 weeks	<22 weeks	<24 weeks	≤24 weeks	<28 weeks	≤28 weeks	2nd trimester	Weeks Not Reported	Inclusion/ Exclusion Criteria	Determination Method
Majoko, 2007 (81)	RCT	X										Enrolled at ANC booking	Fundal heights
Ticconi, 2003 (138)	Prospective Cohort	X										Admitted pregnant women. Median GA at enrollment was in 3rd trimester	Not reported
Taha, 2018 (134)	Prospective cohort	X										Incident pregnancies, age at confirmation not reported	Not reported
Ali, 2017(3)	Prospective cohort		X†									Mean GA at enrollment=34.7 and 33.9 weeks	Not reported
Young, 2012 (152)	Prospective cohort		X									12-28 weeks	LMP and ultrasounds
Moodley, 2017 (88)	Retrospective Cohort			X†								Enrolled at ANC registration; median GA at enrollment=24 weeks	<24 weeks=obstetric ultrasounds; ≥24 weeks=fundal heights, LMP, and palpation
Kulmala, 2000 (72)	Prospective cohort				X							Enrollment at ANC registration	Fundal height
Ashorn, 2015 (7)	RCT				X							<20 weeks	Ultrasound

Rollins, 2007 (122)	Prospective cohort						X					Enrolled at ANC	LMP
Kiondo, 2014 (70)	RCT						X					12-22 weeks	Not reported
Van den Broek, 2005 (142)	Prospective cohort					X						<24 weeks	Ultrasound
Bera, 2010 (16)	Prospective cohort					X						≥14 weeks	Ultrasound
Liu, 2014 (77)	Prospective cohort							X				Enrolled at ANC or HIV care clinic	LMP and ultrasound
Kimani, 2016 (69)	RCT							X†				14-26 weeks	Ultrasound
Ndibazza, 2010 (102)	RCT							X				In 2nd or 3rd trimester per midwife assessment	Not reported
Harrison, 2007 (54)	RCT								X			Enrolled at first ANC visit	LMP and fundal height
Malaba, 2017 (83)	Prospective cohort								X			Women seeking ANC	LMP and fundal height
Lewycka, 2013 (76)	RCT								X			Incident pregnancies, participants visited monthly	Not reported
Naidoo, 2011(97)	Cross sectional								X†			Self-reported pregnancies	Self-report
Gebhardt, 2009 (45)	Prospective cohort									X		Enrollment at ANC booking	Ultrasounds
Sibeko, 2011 (130)	RCT										X	Incident pregnancies, monthly urine pregnancy testing	Not reported
Brittain, 2015 (21)	Prospective cohort										X	20-28 weeks	Fundal height
Talip, 2010 (135)	Cross sectional										X	n/a (delivery records review)	Ultrasound
Ndyomugyenyi, 2000 (105)	RCT										X	In 1st or 2nd trimester	Hackett's method
Kumwenda, 2002 (73)	RCT										X	18-28 weeks	LMP
Gray, 2001(49)	RCT										X†	Various. Approximately 1/3	Ballard score

												enrolled in each trimester.	
Bello, 2010(14)	Cross sectional										X†	Self-report on most recent pregnancy	Self-report
Ndyomugyenyi, 2011(108)	RCT										X†	≥16 weeks	Fundal height
Makanani, 2018 (82)	RCT										X†	Incident pregnancy, monthly urine pregnancy testing	LMP, ultrasound, and physical exam

B.15B. Summary of included studies by gestational age at enrollment details and likelihood of including first trimester pregnancy losses. Gray represents studies that did not include or included few first trimester pregnancies.

Gestational age at enrollment (or under assessment)	# of studies	Likelihood of including 1st trimester pregnancy losses
At the time of ANC booking or at ANC (no mean/median GA)	7	**Possible but few (**ANC typically initiated later in pregnancy)
Median or mean GA = 2nd or 3rd trimester	3	**Possible but few**
Incident pregnancy (with regular pregnancy testing)	4	**Likely**
≥12 weeks	2	**Possible but few**
<20 weeks, <24 weeks	2	**Not included**
≥14 weeks	6	**Not included**
Self-reported	2	**Possible**
Based on medical records after delivery	1	**Possible but few** since data collection from delivery records
1st or 2nd trimester	1	**Likely**
Any trimester	1	**Likely**

* ANC=antenatal care/clinic; GA=gestational age; LMP=last menstrual period
† Specifically defined as spontaneous abortions only.

Table B.16. Stillbirth Study Definition – Included Studies

Author, Year	Study Design	Stillbirth/IUFD Definition								Gestational Age*	
		>20 weeks	≤20 weeks	≥21 weeks	≥22 weeks	>24 weeks	>28 weeks	≥28 weeks	Weeks Not Reported	Inclusion/ Exclusion Criteria	Determination Method
Ntuli, 2012 (114)	Cross sectional	X								n/a	Not reported
Feresu, 2005 (39)	Cross sectional	X								>20 weeks	LMP
Young, 2012 (152)	Prospective Cohort	X								12-28 weeks	LMP and ultrasounds
Majoko, 2007 (81)	RCT	X								Enrolled at ANC booking	Fundal heights
Ticconi, 2003 (138)	Prospective Cohort		X							Admitted pregnant women. Median GA at enrollment was in 3rd trimester	Not reported
Kananura, 2017 (63)	Cross sectional		X							n/a	Self report
Kananura, 2016 (62)	Cross sectional		X							n/a	Self report
Chevalier, 2017 (25)	Prospective cohort		X							<30 weeks	Not reported
Taha, 2018 (134)	Prospective cohort		X							Incident pregnancies, age at confirmation not reported	Not reported
Moodley, 2017 (88)	Retrospective cohort			X						Enrolled at ANC registration; median GA at enrollment=24 weeks	<24 weeks=obstetric ultrasounds; ≥24 weeks=fundal heights, LMP, and palpation

Ali, 2017 (3)	Prospective cohort			X						Mean GA at enrollment=34.7 and 33.9 weeks	Not reported
Ashorn, 2015 (7)	RCT				X					<20 weeks	Ultrasound
Rollins, 2007 (122)	Prospective cohort					X				Enrolled at ANC	LMP
Kiondo, 2014 (70)	RCT					X				12-22 weeks	Not reported
Van den Broek, 2005 (142)	Prospective cohort					X				<24 weeks	Ultrasound
Bera, 2010 (16)	Prospective cohort					X				≥14 weeks	Ultrasound
Bailey, 2017 (10)	Cross sectional							X		n/a	Not reported
Allanson, 2015 (4)	Cross sectional							X		n/a	Fundal height
Rempis, 2017 (119)	Cross sectional							X		n/a	Finnstrom scoring
Moyer, 2016 (89)	Cross sectional							X		n/a	Not reported
Kimani, 2016 (69)	RCT							X		14-26 weeks	Ultrasound
Kujala, 2017 (71)	Cross sectional							X		n/a	Self-report
Liu, 2014 (77)	Prospective cohort							X		Enrolled at ANC or HIV care clinic	LMP and ultrasound
McDonald, 2018 (84)	Prospective cohort							X		12-28 weeks gestation	Not reported
Naidoo, 2011 (97)	Cross sectional							X		Self-reported pregnancies	Self-report
Ndibazza, 2010 (102)	RCT							X		In 2nd or 3rd trimester per midwife assessment	Not reported
Harrison, 2007 (54)	RCT						X			Enrolled at first ANC visit	LMP and fundal height
Malaba, 2017 (83)	Prospective cohort						X			Women seeking ANC	LMP and fundal height

Asiki, 2015 (8)	Cross sectional						X			n/a	Not reported
Lewycka, 2013 (76)	RCT						X			Incident pregnancies, participants visited monthly	Not reported
Lavin, 2016 (75)	Cross sectional						X			n/a	LMP, ultrasound, clinical examination
Nankabirwa, 2011 (100)	Prospective cohort						X			>7 months	Not reported
Coutsoudis, 1999 (28)	RCT						X			Recruited from ANC, mean age at enrollment 28 weeks	Not reported
Byaruhanga, 2000 (22)	Cross sectional								X	n/a	Not reported
Verhoeff, 1999 (146)	Prospective cohort								X	Enrolled at first ANC	Not reported
Makanani, 2018 (82)	RCT								X	Incident pregnancy, monthly urine pregnancy testing	LMP, ultrasound, and physical exam
Kaye_2006 (67)	Prospective cohort								X	>13 weeks	LMP, ultrasound, and clinical exam
Fatti_2016 (35)	Prospective cohort								X	Enrolled at maternal faciltiies	Not reported
Kumwenda_2002 (73)	RCT								X	18-28 weeks	LMP
Muti_2015 (93)	Cross sectional								X	n/a	Not reported
Ndyomugyenyi_2008 (107)	RCT								X	≥16 weeks	Fundal height and LMP
Kasumba_2000 (65)	Cross sectional								X	n/a	Physical examination
Ndyomugyenyi_2011 (108)	RCT								X	≥16 weeks	Fundal height
Ndyomugyenyi_2000 (105)	RCT								X	In 1st or 2nd trimester	Hackett's method
Ndyomugyenyi_2001 (106)	Cross sectional								X	n/a	Not reported
Pattinson_2003 (117)	Cross sectional								X	n/a	Not reported

Talip_2010 (135)	Cross sectional								X	n/a (delivery records review)	Ultrasound
Bello_2010 (14)	Cross sectional								X	Self-report on most recent pregnancy	Self-report
Govender_2017 (48)	Cross sectional								X	n/a	Not reported
Ebrahim_2000 (32)	Cross sectional								X	n/a	Not reported
Gebhardt, 2009 (45)	Prospective cohort								X	Enrollment at ANC booking	Ultrasounds
Basu_2010 (12)	Cross sectional								X	n/a	Ultrasound
Colbourn_2013 (27)	RCT								X	n/a	Not reported
van den Broek_2003 (144)	Cross sectional								X	n/a	Not reported
Msyamboza_2009 (95)	Prospective cohort								X	n/a	Not reported
Metaferia_2009 (85)	Cross sectional								X	n/a	LMP
Kalumbi_2001 (61)	Cross sectional								X	n/a	Not reported
Arinaitwe_2013 (6)	Cross sectional								X	n/a	Not reported
Madhi_2016 (79)	RCT								X	28-35 weeks	Ultrasound, LMP, and fundal height
Sibeko_2011 (130)	RCT								X	Incident pregnancies, monthly urine pregnancy testing	Not reported
Gutman_2013 (52)	Cross sectional								X	n/a	Ballard score
Friis_2004 (43)	RCT								X	22-36 weeks	LMP and fundal height
Gray_2001 (49)	RCT								X	Various. Approximately 1/3 enrolled in each trimester.	Ballard score
Braun_2015 (20)	Cross sectional								X	n/a	Not reported
Hussain_2011 (57)	Cross sectional								X	n/a	Ballard score
Moodley_2016 (86)	Cross sectional								X	n/a	Not reported

Naidu_2001 (98)	Cross sectional								X	n/a	Not reported
Sebitloane_2017 (127)	Cross sectional								X	n/a	Not reported
Kulmala_2000 (72)	Prospective cohort								X[†]	Enrollment at ANC registration	Fundal height
Chihana_2015 (26)	Prospective cohort								X	n/a	Not reported

[*] ANC=antenatal clinic/care; GA=gestational age; LMP=last menstrual period
[†] However, based on definition of abortion, it is likely the stillbirth definition are losses at ≥22 weeks.

SECTION C: PRETERM BIRTH

Table C.1. Search Result Flow

	N
Title Review	590
Abstract Review	265
Manuscripts Reviewed*	189
Manuscripts Included From Main Search†	109
Manuscripts Added From Other Searches‡	48
Total Manuscripts Abstracted	157
Total Manuscripts Included For Analysis – MTN 042 Countries Only§	63

* Does not include number of references from systematic reviews reviewed.
† Includes systematic review reference reviews.
‡ Some of these include duplicate populations with the studies that were abstracted through the main search. These were removed at analysis.
§ Kimani_2016 (69) reported prevalence estimates for two study countries and Taha_2018 (134) reported estimates for four countries. These studies were counted as one manuscript each but provided multiple prevalence estimates in the analysis phase.

Table C.2. Pooled Prevalence of Preterm Birth – Overall, by MTN-042 Country, and by HIV status

	# of estimates *	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95%CI)	I^2 (p-value)
Study Countries	67	134,763	13.4% (0.0, 32.1)	14.4% (8.6, 18.4)	12.7% (11.2, 14.3)	98.4% (<0.01)
Malawi	13	9,850	14.7% (0.6, 23.8)	17.4% (11.3, 19.7)	13.5% (9.1, 18.5)	97.7% (<0.01)
South Africa	28	85,559	13.6% (0.0, 32.1)	13.7% (6.8, 19.5)	12.6% (10.0, 15.5)	99.2% (<0.01)
Uganda	15	11,066	12.0% (3.7, 23.5)	11.9% (4.8, 16.6)	11.4% (9.0, 14.1)	93.1% (<0.01)
Zimbabwe	10	28,063	13.9% (0.0, 23.9)	15.1% (10.7, 17.4)	14.6% (12.4, 16.9)	93.7% (<0.01)
Multiple Countries†	1	225	9.8% -	-	9.8% (6.2, 14.4)	-
Living with HIV	21	18,592	14.3% (0.0, 25.1)	16.1% (9.8, 21.8)	14.1% (11.0, 17.6)	97.2% (<0.01)
HIV negative	13	11,108	11.0% (0.6, 20.9)	9.7% (7.5, 16.4)	10.0% (5.7, 15.4)	98.1% (<0.01)

* No outliers were identified so no sensitivity analysis excluding outliers was conducted.
† Uganda+ Zimbabwe, Gibb_2012 (47)

Figure C.1. Forest Plot Summarizing the Pooled Prevalence of Preterm Birth – Malawi

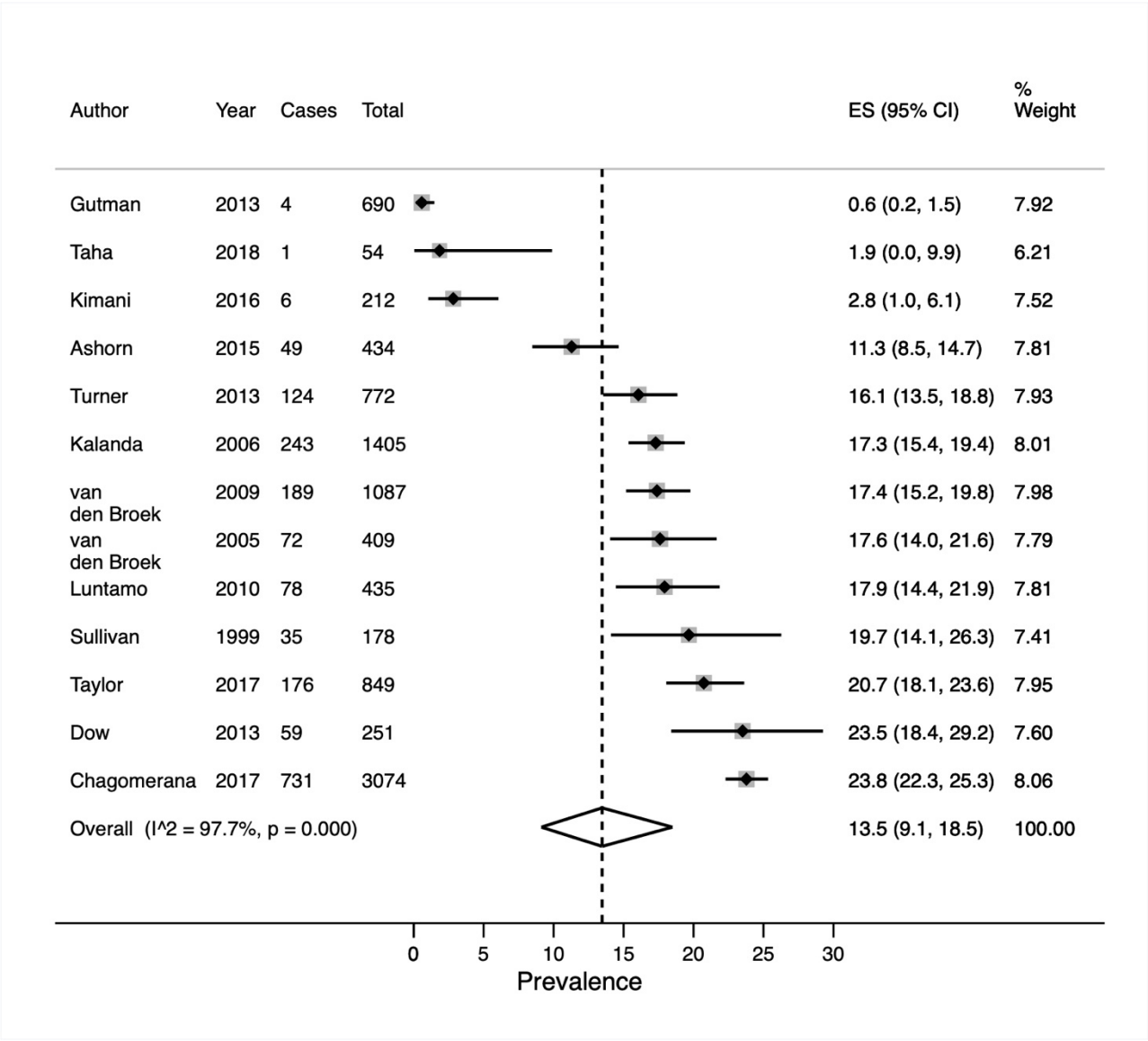


Figure C.2. Forest Plot Summarizing the Pooled Prevalence of Preterm Birth – South Africa

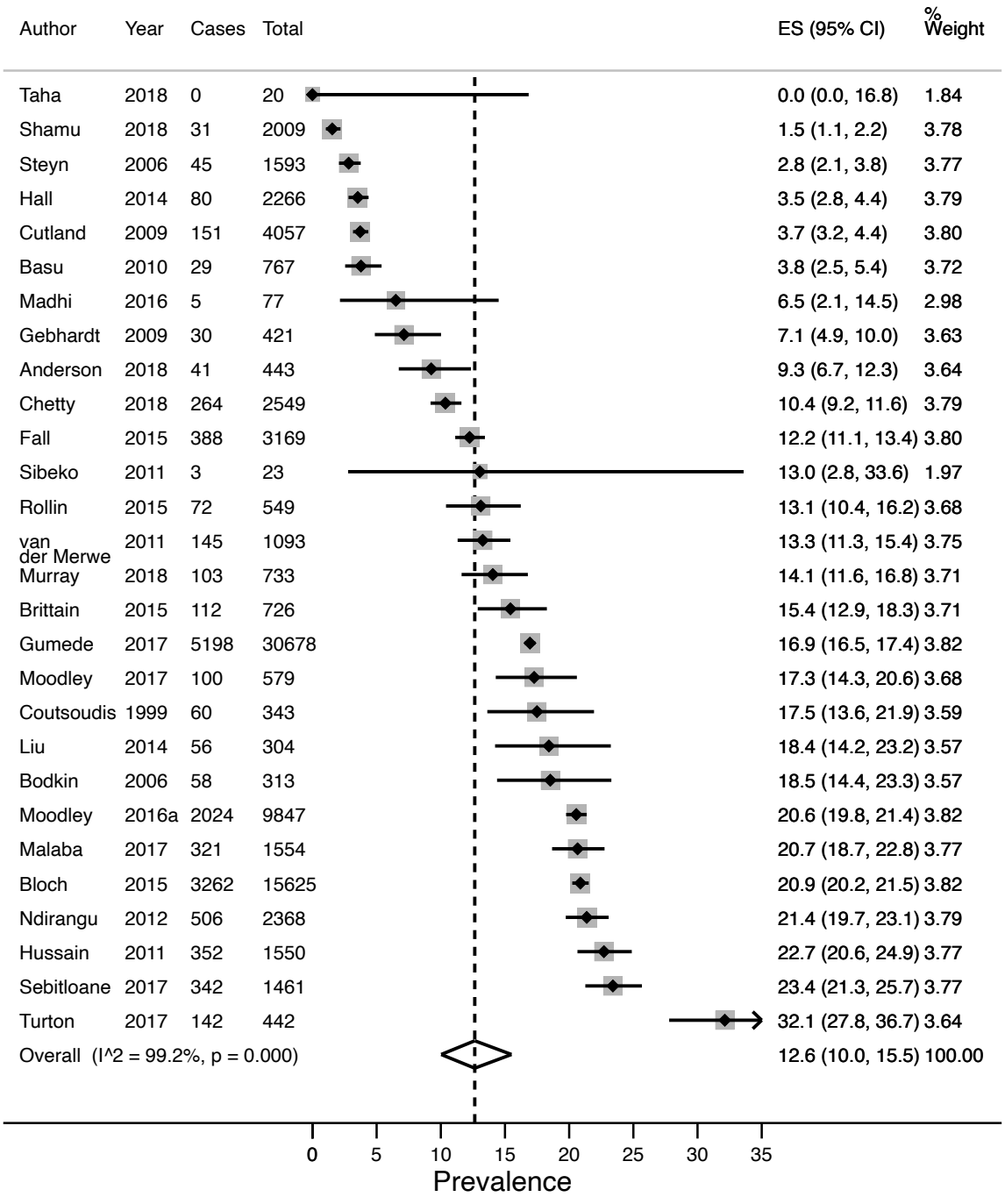


Figure C.3. Forest Plot Summarizing the Pooled Prevalence of Preterm Birth – Uganda

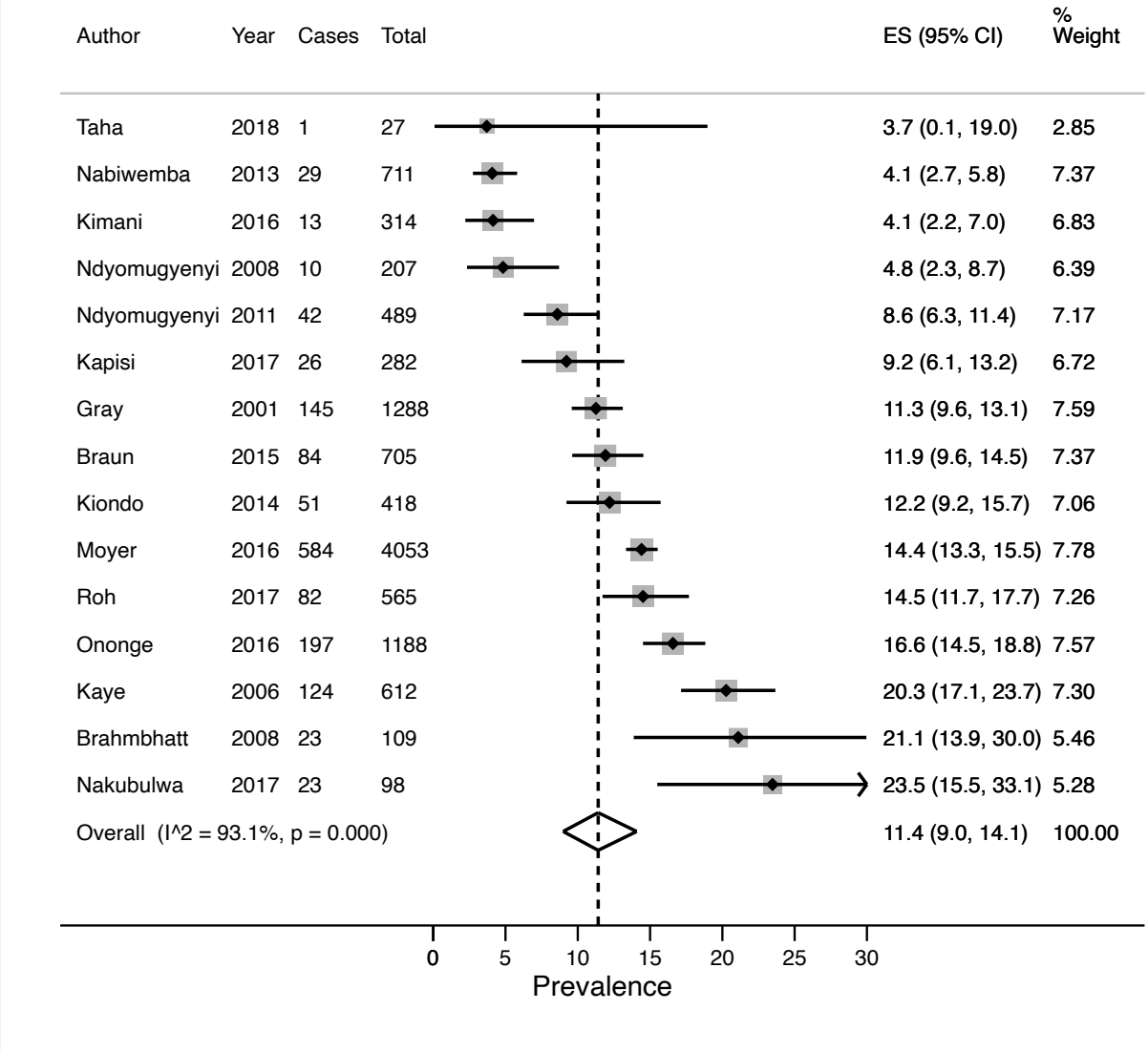


Figure C.4. Forest Plot Summarizing the Pooled Prevalence of Preterm Birth – Zimbabwe

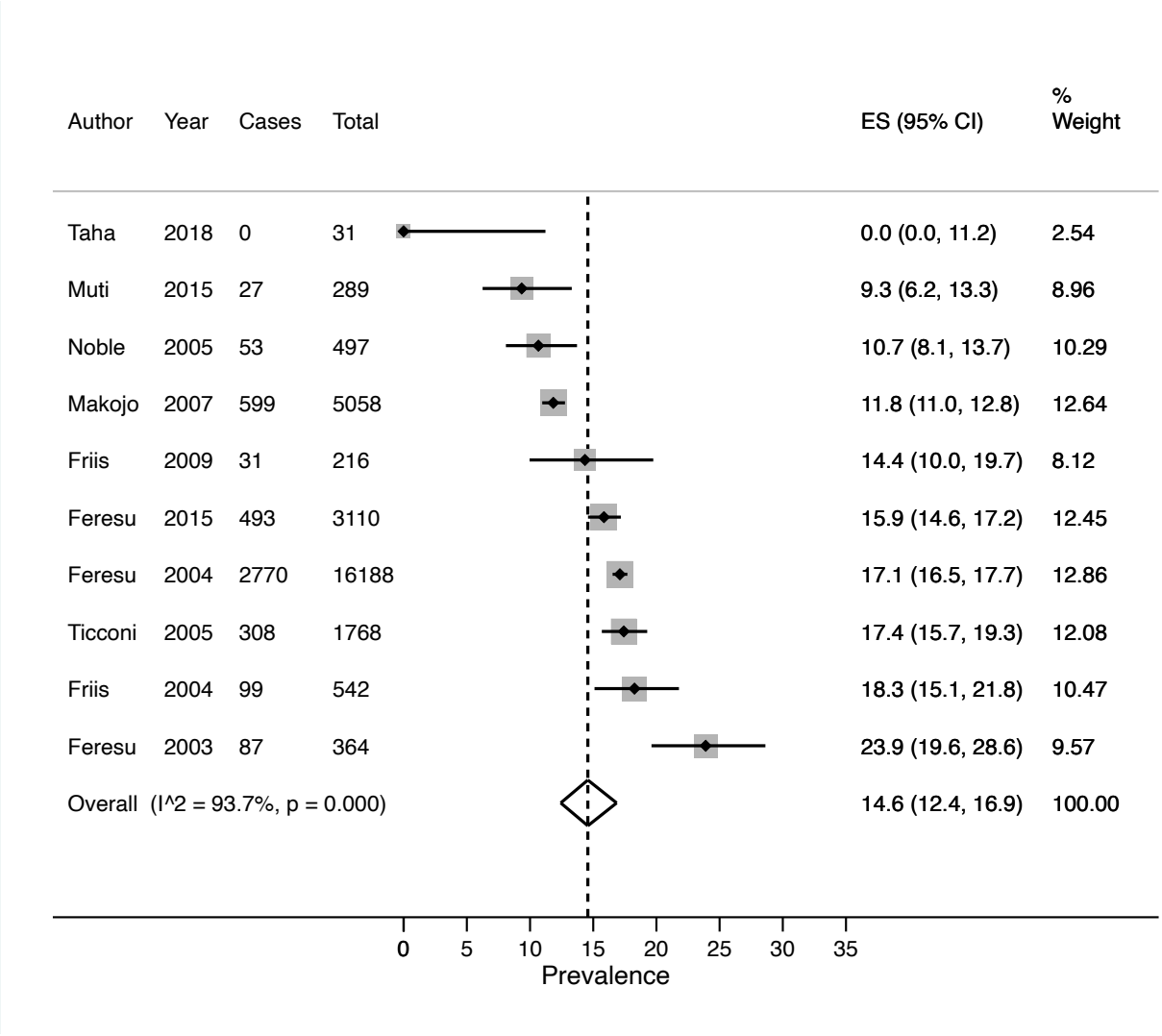


Figure C.5. Forest Plot Summarizing the Pooled Prevalence of Preterm Birth Among Women Living with HIV

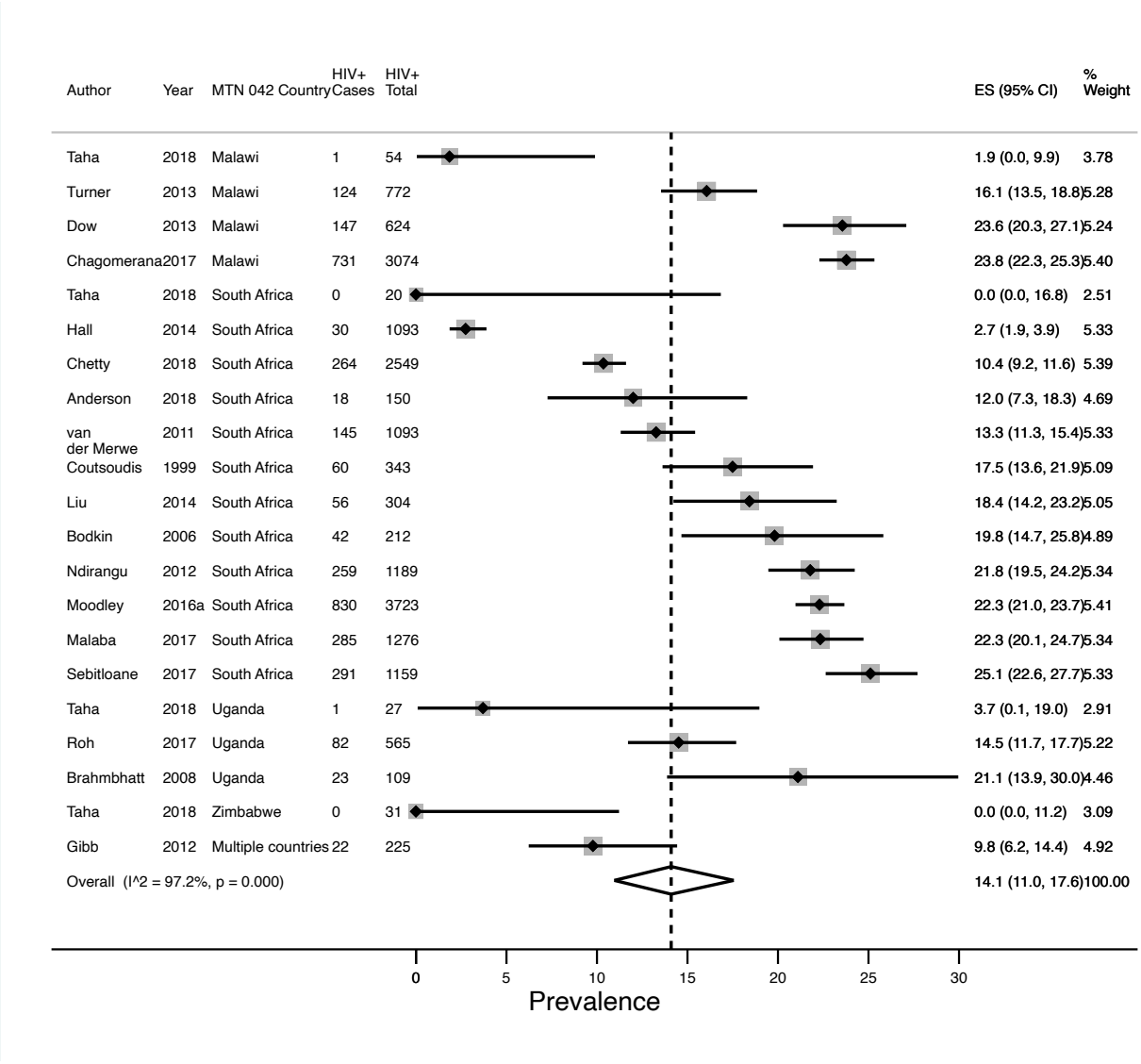


Figure C.6. Forest Plot Summarizing the Pooled Prevalence of Preterm Birth Among HIV Negative Women

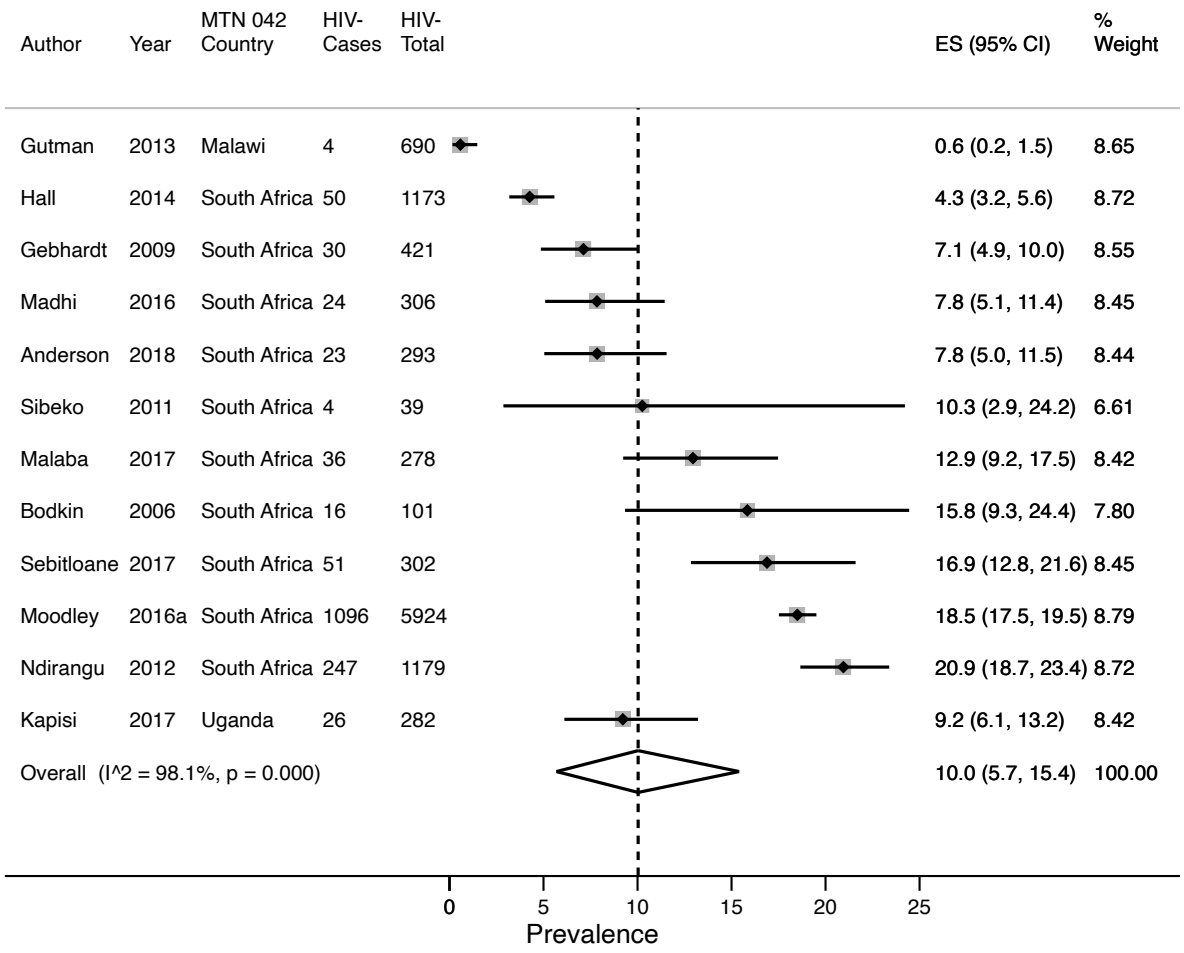


Table C.3. Sensitivity Analysis Live Birth Only: Pooled Prevalence of Preterm Birth – Overall, by MTN-042 Country, and by HIV Status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95% CI)	I^2 (p-value)
Study Countries	30	47,187	14.0% (0.6, 32.1)	15.6% (8.6, 19.7)	12.7% (10.1, 15.6)	98.5% (<0.01)
Malawi	7	6,740	14.0% (0.6, 23.8)	17.3% (2.8, 19.7)	12.2% (5.7, 20.8)	98.7% (<0.01)
South Africa	13	16,707	15.0% (1.5, 32.1)	15.4% (10.4, 20.7)	13.8% (8.8, 19.6)	99.0% (<0.01)
Uganda	7	4,217	12.0% (4.1, 23.5)	11.3% (4.1, 20.3)	10.8% (6.7, 15.7)	95.2% (<0.01)
Zimbabwe	2	19,298	16.5% (15.9, 17.1)	16.5% (15.9, 17.1)	16.9% (16.4, 17.4)	-
Multiple Countries†	1	225	9.8% -	-	9.8% (10.6, 16.1)	-
Living with HIV	8	9,732	17.5% (9.8, 23.8)	18.0% (13.2, 22.1)	17.3% (13.0, 22.1)	96.9% (<0.01)
HIV negative	5	2,492	10.5% (0.6, 20.9)	10.3% (7.8, 12.9)	9.0% (1.4, 22.0)	98.8% (<0.01)

* Only studies where the preterm birth estimate was clearly defined as prevalence among live-births only were included. Included studies: Braun_2015 (20), Brittain_2015 (21), Chagomerana_2017 (23), Chetty_2018 (24), Coutsoudis_1999 (28), Fall_2015 (34), Feresu_2004 (38), Feresu_2015 (40), Gibb_2012 (47), Gray_2001 (49), Gutman_2013 (52), Hussain_2011 (57), Kalanda_2006 (59), Kaye_2006 (67), Kimani_2016 (69), Liu_2014 (77), Madhi_2016 (79), Malaba_2017 (83), Nabiwemba_2013 (96), Nakubulwa_2017 (99), Ndirangu_2012 (104), Ndyomugyenyi_2011 (108), Shamu_2018 (129), Sibeko_2011(130), Steyn_2006 (132), Sullivan_1999 (133), Turner_2013 (140), Turton_2013 (141), van den Broek_2005 (142)

† Uganda+ Zimbabwe, Gibb_2012 (47)

Table C.4. Sensitivity Analysis Excluding Studies with Preterm Birth Definition Undefined: Pooled Prevalence of Preterm Birth – Overall, by MTN-042 Country, and by HIV Status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95%CI)	I^2 (p-value)
Study Countries	59	123,610	13.4% (0.0, 32.1)	14.1% (8.6, 18.3)	12.9% (11.4, 14.4)	98.1% (<0.01)
Malawi	13	9.850	14.7% (0.6, 23.8)	17.4% (11.3, 19.7)	13.5% (9.1, 18.5)	97.7% (<0.01)
South Africa	24	80,948	14.3% (0.0, 32.1)	14.7% (8,2, 20.6)	13.8% (11.3, 16.4)	98.9% (<0.01)
Uganda	12	6,292	10.4% (3.7, 23.5)	10.2% (4.4, 13.4)	9.9% (7.2, 12.9)	92.0% (<0.01)
Zimbabwe	9	26,295	13.5% (0.0, 23.9)	14.4% (10.7, 17.1)	14.1% (11.8, 16.7)	94.3% (<0.01)
Multiple Countries†	1	225	9.8% -	-	9.8% (6.2, 14.4)	-
Living with HIV	18	17,178	14.2% (0.0, 25.1)	15.3% (9.8, 22.3)	15.0% (12.3, 18.0)	95.5% (<0.01)
HIV negative	10	9,795	11.3% (0.6, 20.9)	9.2% (7.8, 16.9)	10.3% (5.5, 16.4)	99.3% (<0.01)

* Excluded studies: Bodkin_2006, Brahmbhatt_2008, Hall_2014, Kaye_2006, Moyer_2016, Shamu_2018, Sibeko_2011, Ticconi_2005
† Uganda+ Zimbabwe, Gibb_2012 (47)

SECTION D: CONGENITAL ANOMALIES

Table D.1. Search Result Flow

	N
Title Review	620
Abstract Review	75
Manuscripts Reviewed*	40
Manuscripts Included From Main Search†	15
Manuscripts Added From Other Searches	25
Total Manuscripts Abstracted	40
Total Manuscripts Included For MTN-042 Analysis‡	19

* Does not include # of references from systematic reviews reviewed
† Including systematic review reference reviews
‡ Included all prevalence estimates regardless of whether it was reported for live births only or included stillbirths, and independent of HIV status. Some studies reported all congenital abnormalities, or all on a long list of possible abnormalities of interest. Other studies only reported no prevalence of specific anomalies (ex: NTDs, cleft lip) and do not report on other abnormalities that may have occurred in the same study population.

Table D.2. Pooled Prevalence of Congenital Anomalies – Overall, by MTN-042 Country, and by HIV status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95%CI)	I^2 (p-value)
Study Countries	22	402,215	1.5% (0.0, 7.5)	0.3% (0.02, 2.7)	0.4% (0.2,0.7)	97.9% (<0.01)
Malawi	4	27,951	0.4% (0.0, 1.2)	0.2% (0.1,0.7)	0.2% (0.0, 0.6)	78.8% (<0.01)
South Africa	8	312,903	1.3% (0.0, 6.7)	0.1% (0.0, 1.6)	0.2% (0.0, 0.6)	98.5% (<0.01)
Uganda	7	60,997	1.6% (0.1, 6.0)	0.3% (0.1, 3.7)	0.6% (0.0, 1.5)	98.7% (<0.01)
Zimbabwe	1	31	0.0% -	-	0.0% (0.0, 11.2)	-
Multiple Countries	2	333	5.3% (3.1, 7.5)	5.3% (3.1, 7.5)	4.3% (2.3, 6.8)	-
Living with HIV	8	1,846	2.1% (0.0, 6.7)	1.8% (0.0, 3.4)	1.8% (0.5, 3.6)	67.8% (<0.01)
HIV negative	2	739	0.6% (0.0, 1.2)	0.6% (0.0, 1.2)	0.7% (0.1, 1.6)	-

* Naevus/birthmarks were excluded when possible. They were retained in the overall estimates when unable to confirm # of infants versus # anomalies.

Figure D.1. Forest Plot Summarizing the Pooled Prevalence of Congenital Anomalies

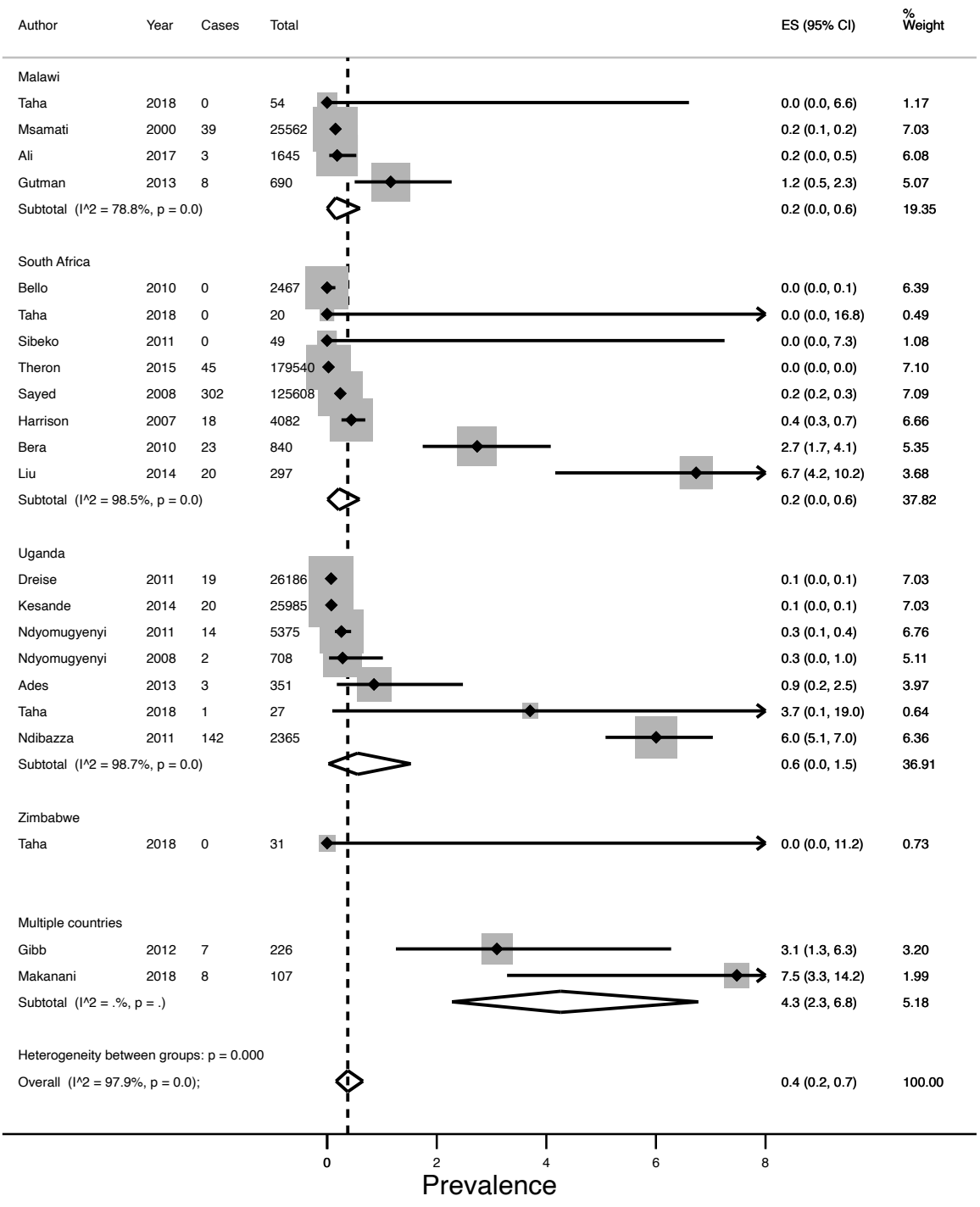
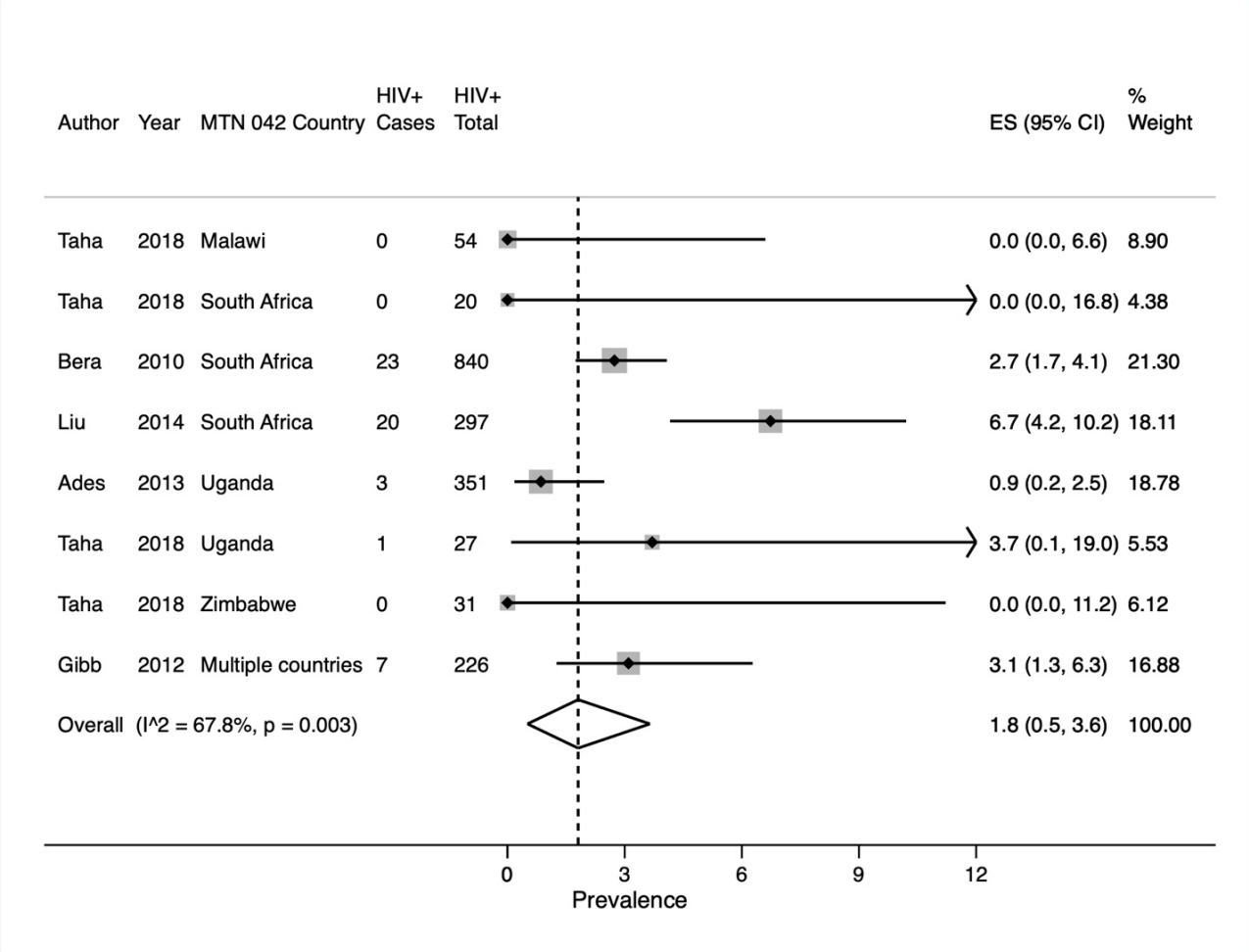


Figure D.2. Forest Plot Summarizing the Pooled Prevalence of Congenital Anomalies Among Women Living with HIV



Note: Only two studies reported congenital anomaly prevalence among HIV negative women so no forest plot was generated.

Table D.3. Sensitivity Analysis Excluding Outliers: Pooled Prevalence of Congenital Anomalies – Overall, by MTN-042 Country, and by HIV status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95%CI)	I^2 (p-value)
MTN 042 Countries	21	402,108	1.2% (0.0, 6.7)	0.2% (0.03, 1.2)	0.3% (0.1,0.6)	98.0% (<0.01)
Malawi	4	27,951	0.4% (0.0, 1.2)	0.2% (0.1,0.7)	0.2% (0.0, 0.6)	78.8% (<0.01)
South Africa	8	312,903	1.3% (0.0, 6.7)	0.1% (0.0, 1.6)	0.2% (0.0, 0.6)	98.5% (<0.01)
Uganda	7	60,997	1.6% (0.1, 6.0)	0.3% (0.1, 3.7)	0.6% (0.0, 1.5)	98.7% (<0.01)
Zimbabwe	1	31	0.0% -	-	0.0% (0.0, 11.2)	-
Multiple Countries	1	226	3.1% -	-	3.1% (1.3, 6.3)	-
Living with HIV	8	1,846	2.1% (0.0, 6.7)	1.8% (0.0, 3.4)	1.8% (0.5, 3.6)	67.8% (<0.01)
HIV negative	2	739	0.6% (0.0, 1.2)	0.6% (0.0, 1.2)	0.7% (0.1, 1.6)	-

* N=1 study had a prevalence estimate that was greater than 1.5 times the interquartile range (IQR) of all included studies. The excluded study was Makanani_2018 (82). Details in Supplemental Table D.7.

Table D.4. Sensitivity Analysis Excluding Non-Randomized Trial Data: Pooled Prevalence of Congenital Anomalies – Overall, by MTN-042 Country, and by HIV status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95%CI)	I^2 (p-value)
MTN 042 Countries	8	13,263	2.3% (0.0, 7.5)	0.6% (0.3, 4.6)	1.5% (0.2, 3.6)	97.7%
Malawi	0	-	-	-	-	-
South Africa	2	4,131	0.2% (0, 0.4)	0.2% (0, 0.4)	0.1% (0.0, 0.3)	-
Uganda	4	8,799	1.9% (0.3, 6.0)	0.6% (0.3, 3.4)	1.3% (0.0, 5.1)	98.8%
Zimbabwe	0	-	-	-	-	-
Multiple Countries	2	333	5.3% (3.1, 7.5)	5.3% (3.1, 7.5)	4.3% (2.3, 6.8)	-
Living with HIV	2	577	2.0% (0.9, 3.1)	2.0% (0.9, 3.1)	1.6% (0.7, 2.8)	-
HIV negative	1	49	0%	0%	0.0% (0.0, 7.3)	-

* Includes studies using randomized trial data as a proxy for closer surveillance for anomalies. Included studies = Ades_2013 (2), Gibb_2012 (47), Harrison_2007 (54), Makanani_2018 (82), Ndibazza_2011 (103), Ndyomugyenyi_2008 (107), Ndyomugyenyi_2011 (108), Sibeko_2011 (130)

Figure D.3. Forest Plot Summarizing the Pooled Prevalence of Congenital Anomalies Among Women Living with HIV – Restricted to Studies Using Randomized Trial Data

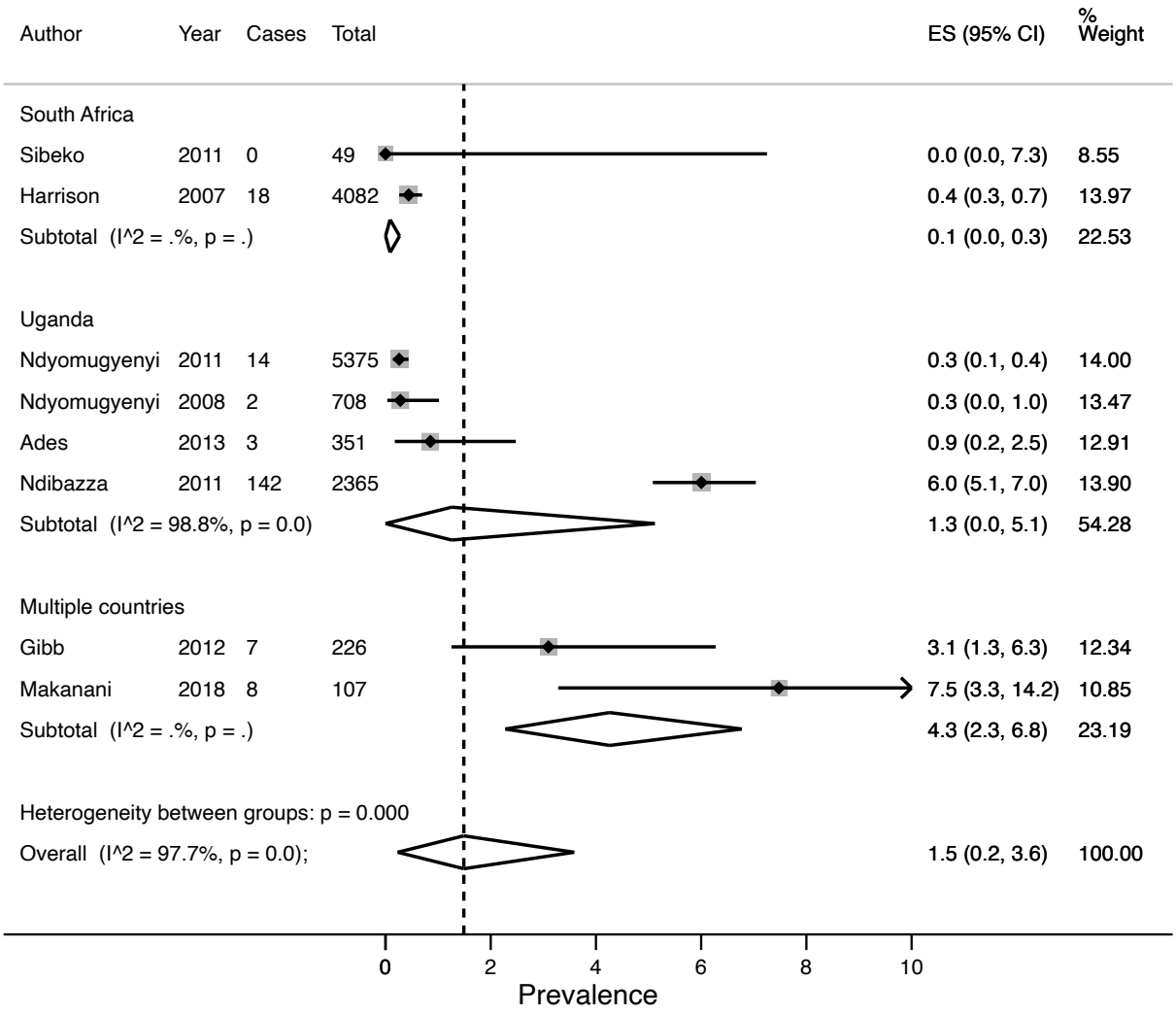


Table D.5. Overall pooled prevalence of specific and system-specific congenital anomalies

Anomaly Type*	# manuscripts	# cases	# of pregnancies/ infants at risk	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95%CI)	Per 1000	I^2 (p-value)
Not defined	8	49	13,372	0.75% (0.0, 3.10)	0.36% (0.13, 0.96)	0.34% (0.04, 0.82)	3.4	85.3% (<0.01)
Cleft Lip and/or Palate	8	111	205,537	0.14% (0.04, 0.34)	0.10% (0.07, 0.21)	0.03% (0.01, 0.05)	0.3	58.9% (0.02)
NTDs and/or Hydrocephalus	3	182	153,535	0.11% (0.09, 0.13)	0.12% (0.09, 0.13)	0.11% (0.08, 0.14)	1.1	32.2% (0.23)
Cardiovascular	3	8	3,502	0.43% (0.12, 1.01)	0.17% (0.12, 1.01)	0.23% (0.01, 0.65)	2.3	57.0% (0.10)
Polydactyly & Syndactyly	6	43	4,668	0.82% (0.14, 1.35)	1.00% (0.28, 1.14)	0.70% (0.30, 1.22)	7	56.4% (0.04)
Musculoskeletal including clubfoot	5	14	5,855	0.31% (0.06, 0.67)	0.28% (0.24, 0.30)	0.20% (0.07, 0.38)	2	25.7% (0.25)
Umbilical & Inguinal Hernia	4	79	3,609	2.25% (0.48, 4.67)	1.92% (0.74, 3.75)	1.73% (0.40, 3.81)	17.3	89.2% (<0.01)
Esophageal, gastrointestinal, or anorectal	3	47	182,745	0.06% (0.03, 0.12)	0.04% (0.03, 0.12)	0.02% (0.0, 0.07)	0.2	31.8% (0.23)
Genitourinary	2	8	2,662	0.46% (0.25, 0.67)	0.46% (0.25, 0.67)	0.23% (0.06, 0.48)	2.3	-
Trisomy	3	3	4,850	0.07% (0.04, 0.12)	0.07% (0.04, 0.12)	0.05% (0.05, 0.15)	0.5	0% (0.7)
Multiple Systems	1	10	2,365	0.42%	0.42%	0.42% (0.20, 0.78)	4.2	-
Other†	6	123	130,916	0.82% (0.06, 1.87)	0.58% (0.07, 1.71)	0.40% (0.06, 0.98)	4	91.5% (<0.01)

[*] Congenital anomalies were grouped into subtypes by common types (ex: NTDs) and then by system (ex: musculoskeletal). a subgroup was created when there was more than one reported case or study reporting the type/system of the anomaly. When multiple anomalies were listed per infant, the infant was included as one overall infant but was included as a case in each of the anomalies sub-types. If specific anomalies were not specified, such infants were included in the "multiple systems" sub-group. Naevus/birthmarks were excluded when possible for the overall and type specific analyses.

[†] Includes singular, or infrequent, reports of rare or non-specific anomalies that did not fit well into other defined sub-groups. Examples include natal tooth, anophthalmia, facial asymmetry, arachnoid cyst, hypopigmented skin, macrocephaly with brain defect, subtle dysmorphism, and plagiocephaly.

Table D.6. Congenital Anomaly: Study Definition

Author, Year	Study Design	Definition of Congenital Anomaly	Method for Ascertainment	Timing of Ascertainment	Anomalies Reported (diagnoses)	Notes
Msamati, 2000 (90)	Cross sectional	Cleft lip, cleft palate, spina bifida, hydrocephalus	Delivery and nursery records in obstetrics & gynecology and pediatrics departments	At (or near) time of delivery	Clefts (lip or palate), spina bifida, and hydrocephalus	Only assessed for specific set of anomalies
Gutman, 2013 (52)	Cross sectional	Physical abnormality	Not reported	At (or near) time of delivery	Details not reported.	
Ali, 2017 (3)	Cohort	Malformations = physical defect in a live infant that was identified by a clinician at designated health facilities thorough a clinical examination form that included a screening process to detect abnormalities	Physical examination ("Newborn babies presenting with health problems, per mothers' reports, were referred to the designated health facility where a qualified medical professional did a medical examination to detect and manage any malformations or other health issues."	Near time of delivery (exact timing not specified, visited at home by village health volunteer after delivery)	Congenital limb defect, Down's syndrome, and "not growing normally"	Children with abnormalities were referred to government health-care system for final diagnosis and care
Bera, 2010 (16)	Cohort	Birth defects	Physical examination by a pediatrician or senior obstetrician	At delivery (and potentially up to 6 weeks when mother/baby follow-up ended)	Arachnoid cyst, pulmonary stenosis, postaxial polydactyly, facial asymmetry and overlapping fingers, bilateral clubfeet, congenital naevus, and umbilical hernia, Trisomy 19, arthrogryposis multiplex congenita, oesophageal atresia with tracheo-oesophageal fistura, postaxpostaxial polydactyly of fingers and preaxial polydactyly with syndactyly of toes, and lower central incisor tooth	Pregnancy registry for EFV-based ART exposures
Sayed, 2008 (124)	Before/After	Neural Tube Defects (NTDs) and Non NTDs	Health care workers were trained on clinical recognition of the priority birth defects by	At or near time of delivery, or hospitalization	Anencephaly, spina bifida, encephalocoele, cleft lip, cleft	Assessment of NTDs before and

			a medical geneticist. Information on cases was extracted from hospital records.		palate, Down syndrome, and oculo-cutaneous albinism	after folic acid fortification in South Africa; primarily assessed for a specific set of anomalies
Theron, 2015 (137)	Cross sectional	Anorectal malformations	Reviewing the admission and theater registries and patient's medical records	Not clear; retrospective assessment but unclear ages of included patients	Anorectal malformation	Only included anorectal malformations
Bello, 2010 (14)	Cross sectional	Malformation at birth	Retrospective, self report	At birth	No anomalies reported	
Liu, 2014 (77)	Cohort	A major congenital anomaly has an adverse outcome on either the function or the social acceptability of the individual, and a minor congenital anomaly has no medical or cosmetic importance	South Africa: examination by specialists and infants with suspected congenital anomalies by the clinical geneticist. Zambia: study- midwives trained to assess for congenital anomalies by physical exam.	Birth, 6 weeks, 3 months, 6 months, 1 year	Major: Ambiguous genitalia, bilateal club foot, BPES, cleft lip, genu recurvartum, hypospadias, optic nerve atrophy, persistent foramen ovale, plagiocephaly, secundum atrial septal defect, syndactyly, ventricular septal defect, macrocephaly with brain atrophy Minor: hypopigmented skin, polydactyly, umbilical hernia, preauricular sinus and cyst, preauricular tag, strabismus, subtle dysmorphism	ART registry
Sibeko, 2011 (130)	RCT	Not defined	On pregnancy cessation, the clinical staff collected data on pregnancy complications and documented pregnancy outcomes. Pregnant women presented their babies once in the early postpartum period to the study clinician for	Early postpartum	No anomalies reported for early neonatal period; live born only (term and preterm)	CAPRISA 004; zero anomalies reported so used the overall population total in the analysis

			assessment for abnormalities.			(versus placebo arm only)
Gibb, 2012 (47)	Cohort (secondary analysis of an RCT)	Not defined	Not specified	Near to delivery, length of potential follow-up unclear	Talipes, cardiac (atrial septal defect and patent ductus arteriousus), undescended testes, hydrochephalus, and skin tag	DART trial
Ndibazza, 2011(103)	Cohort (secondary analysis of an RCT)	Any structural defect present at birth	"Midwives carried out an external examination for overt anomalies, and no specific examination routine was followed. Anomalies detected by the physicians followed a thorough physical examination, supported when necessary by referral to specialists. Special investigations such as an echocardiogram were only done if there were clinical indication. Infants who had congenital anomalies involving more than one system were recorded once as having a multiple congenital anomaly. When no definite diagnosis was made, the infant was classified and coded as having an unspecified congenital anomaly." Anomalies were classified by organ system according to the ICD10.	At delivery and "accumulated to 1 year of age (per original trial methods, Ndibazza, 2010, CID)	GIT, Genitourinary, Cardiovascular, Dysmorphic, Head and neck, Multiple systems, Musculoskeletal, Central nervous system, Skin (birthmarks and poliosis), and unspecific cardiorespiratory	38 skin related abnormalities were reported. We excluded these cases from the prevalence analysis.
Ndyomugyenyi, 2011(108)	RCT	Not defined	Physical examination	At delivery, potentially through 28 days	Not reported	Used the overall population total in the analysis (versus

						placebo arm only)
Dreise, 2011 (31)	Cross sectional	All clefts	"The coordinator made the final diagnosis of the cleft, while a diagnosis of other defects was made by the plastic surgeons during the first visit of the baby to the cleft clinic."	At birth	Isolated cleft lip, cleft lip and palate, and isolated palate; three had other anomalies (constriction ring syndrome, trisomy 13 or 18, Pierre Robin syndrome)	Only assessed for specific set of anomalies; live birth only
Kesande, 2014 (68)	Cross sectional	Cleft lip and cleft palate	Medical records	At birth	Cleft lip and cleft palate	Only assessed for specific set of anomalies; live birth only
Ades, 2013 (2)	Cohort (secondary analysis of RCT data)	Not defined	Delivery data were collected by study physicians who reviewed the medical record	At birth, and up to 1 month	Cleft lip and palate, Polydactyly, and Other	
Ndyomugyenyi, 2008 (107)	RCT	Not defined	Physical examination	At birth, and up to 1 month	talipes equinovarus of the right foot and talipes equinovarus of the left foot, cleft palate, and multiple fingers	Used the overall population total in the analysis (versus placebo arm only)
Harrison, 2007 (54)	RCT	Not defined	Records review?	At delivery	Details not reported.	
Makanani, 2018 (82)	Cohort (secondary analysis of RCT data)	Structural anomalies only	Infant congenital anomalies identified at the time of delivery were determined by participant report and medical record review, when available.	At or around delivery	Umbilical hernia that was reducible, micrognathia and epicanthic folds, inguinal hernia that was repaired, craniofacial (structural) right frontal skull depression positional plagiocephaly, reducible umbilical hernia, umbilical hernia, and polydactyly	ASPIRE trial; While anomalies were reported by study arm, we included all congenital anomalies (overall) in our analysis. Prevalence was similar in both arms.

Taha, 2018 (134)	Cohort	Not defined	Physical examination	At delivery, potentially later follow-up	Not reported	PROMOTE Study

Table D.7. Congenital Anomaly Meta-Analyses Outliers

Manuscript	Country	Frequency	Definition	Outcome Ascertainment	Notes
Makanani_2018 (82)	Malawi, South Africa, Uganda, Zimbabwe	7.5%	All were structural anomalies (no further definition)	Congenital anomalies were identified at the time of delivery by participant report and medical records review, when available	ASPIRE study. Similar prevalence in placebo and dapivirine vaginal ring arm

SECTION E: LOW BIRTH WEIGHT

Table E.1. Search Result Flow

	N
Title Review	791
Abstract Review	213
Manuscripts Reviewed*	140
Manuscripts Included From Main Search†	115
Manuscripts Added From Other Searches‡	60
Total Manuscripts Abstracted	175
Total Manuscripts Included For Analysis - 042 Countries Only§	64

* Does not include number of references from systematic reviews reviewed.
† Systematic review reference review was not completed.
‡ Some of these include duplicate populations with the studies that were abstracted through the main search. There will be removed at analysis.
§ Kimani_2016 (69) reported prevalence estimates for two study countries and Taha_2018 (134) reported estimates for four countries. These studies were counted as one manuscript each, but provided multiple prevalence estimates in the analysis phase.

Table E.2. Pooled Prevalence of Low Birth Weight – Overall, by MTN-042 Country, and by HIV status

	# of estimates	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95%CI)	I^2 (p-value)
Study Countries	68	117,578	12.0% (2.8, 37.1)	11.3% (8.7, 14.4)	11.7% (10.6, 12.9)	97.1% (<0.01)
Malawi	17	14,827	10.6% (2.8, 21.2)	11.8% (7.1, 13.3)	10.4% (8.5, 12.5)	93.0% (<0.01)
South Africa	20	54,144	12.8% (5.0, 24.8)	12.6% (9.2, 15.1)	12.7% (10.9, 14.5)	96.8% (<0.01)
Uganda	20	19,760	12.3% (3.7, 37.1)	10.1% (9.2, 13.5)	11.9% (9.8, 14.2)	94.4% (<0.01)
Zimbabwe	11	28,847	12.0% (3.2, 22.5)	11.4% (8.0, 16.7)	11.8% (8.0, 16.2)	98.7% (<0.01)
Living with HIV	18	17,181	13.1% (3.2, 22.4)	14.1% (7.4, 17.8)	13.7% (11.2, 16.3)	94.5% (<0.01)
HIV negative	10	20,529	10.4% (3.0, 14.3)	11.1% (9.4, 13.5)	10.0% (7.7, 12.5)	95.1% (<0.01)

Figure E.1. Forest Plot Summarizing the Pooled Prevalence of Low Birth Weight– Malawi

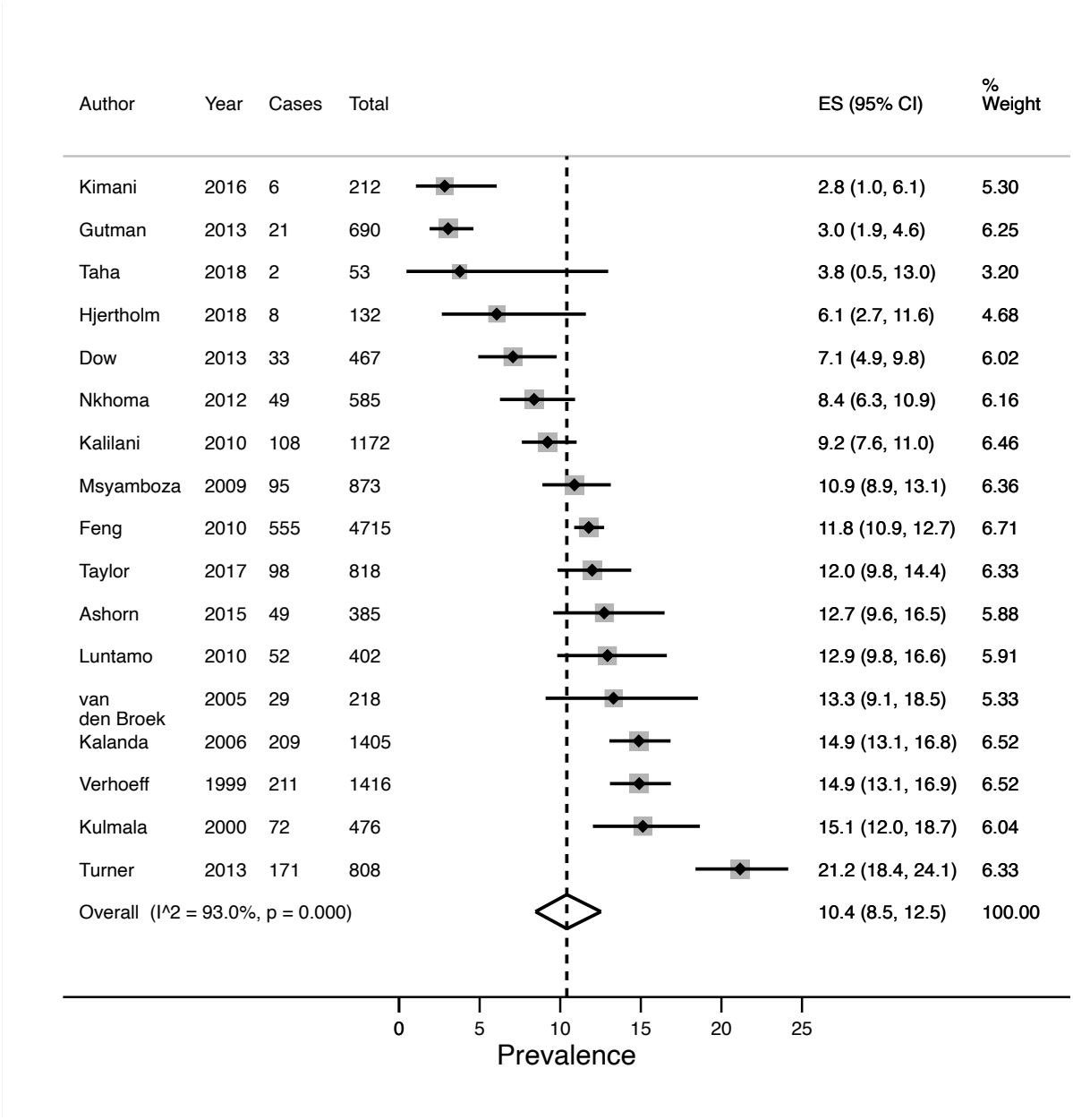


Figure E.2. Forest Plot Summarizing the Pooled Prevalence of Low Birth Weight - South Africa

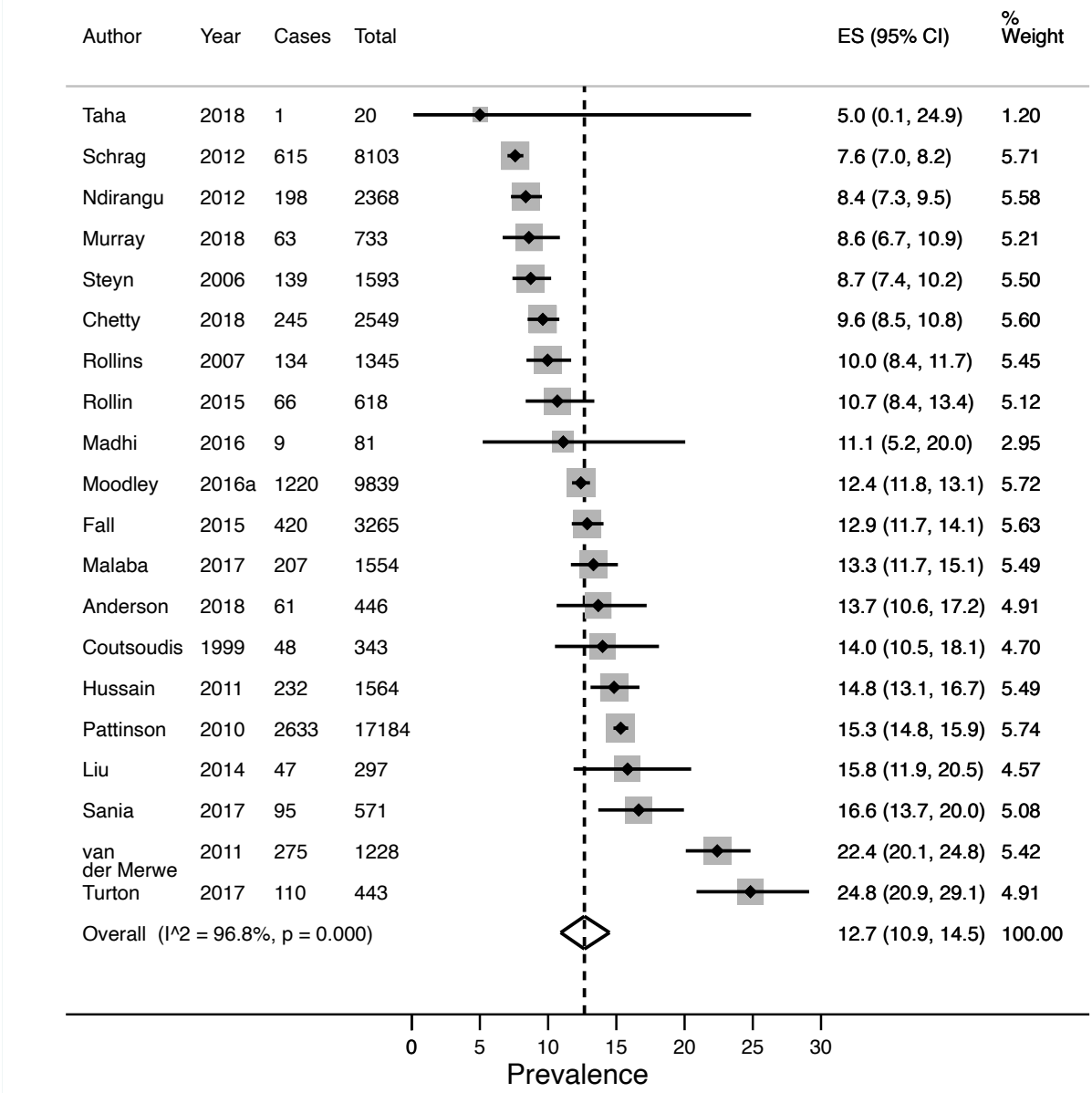


Figure E.3. Forest Plot Summarizing the Pooled Prevalence of Low Birth Weight – Uganda

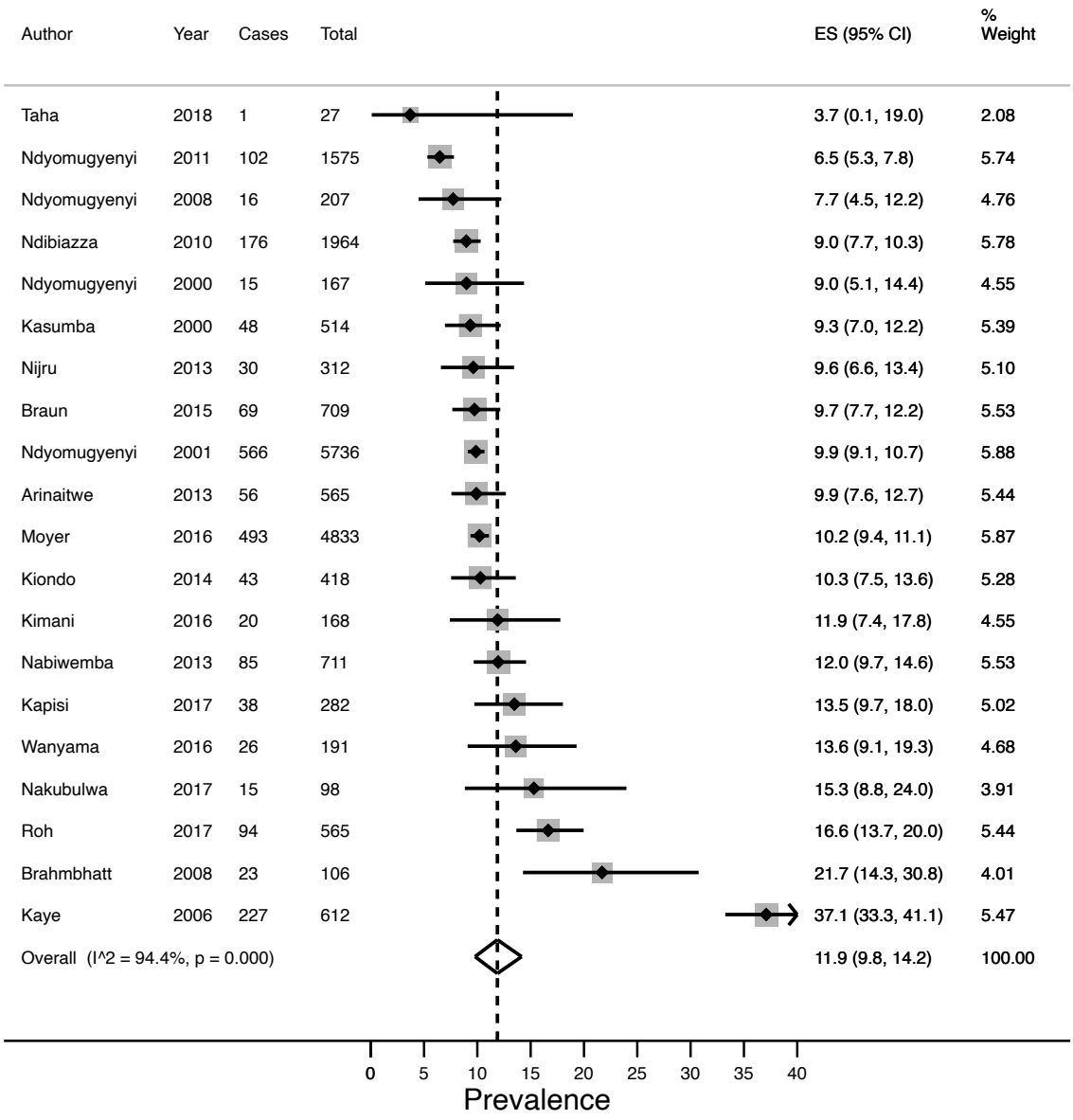


Figure E.4. Forest Plot Summarizing the Pooled Prevalence of Low Birth Weight – Zimbabwe

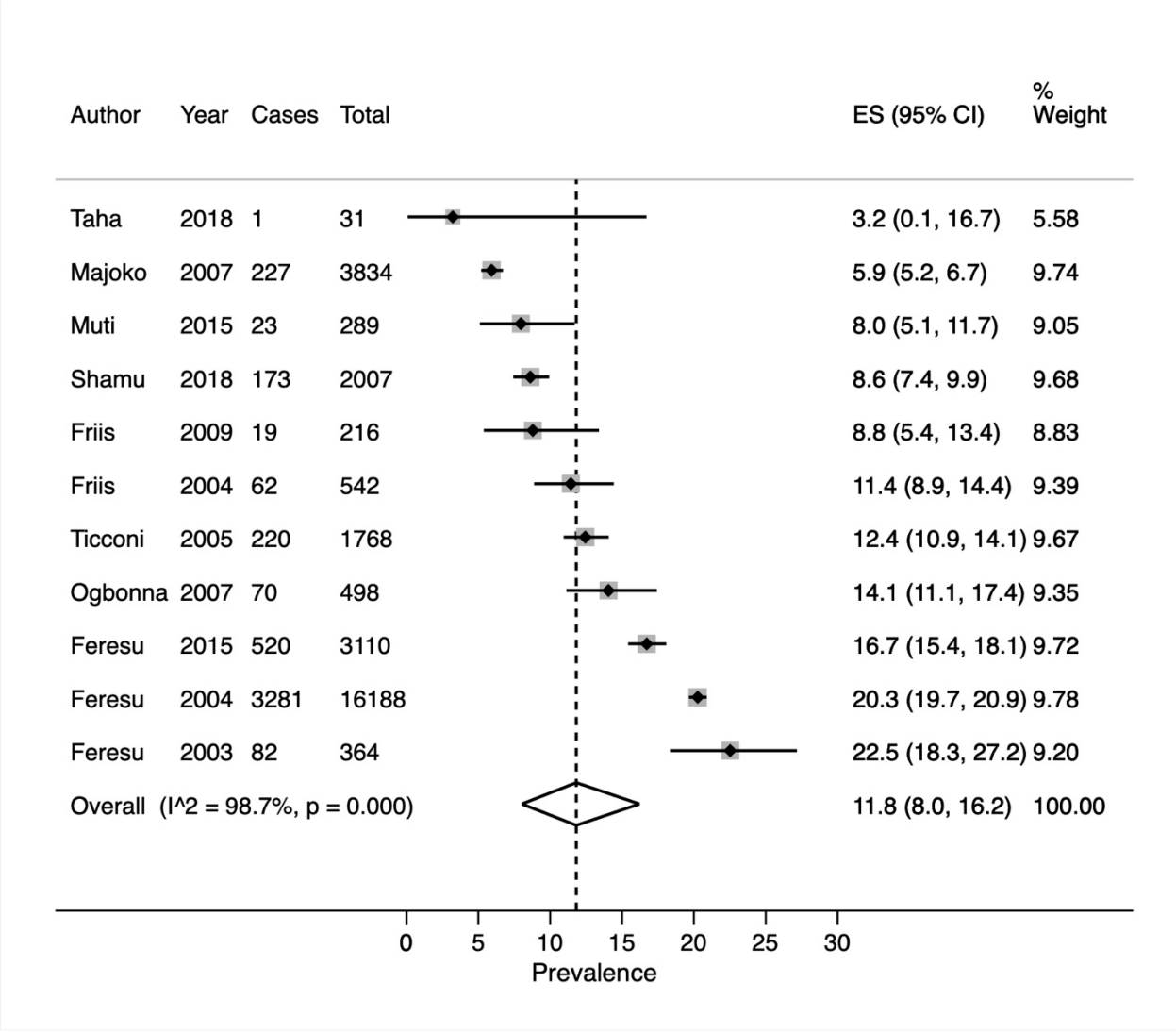


Figure E.5. Forest Plot Summarizing the Pooled Prevalence of Low Birth Weight - Among Women Living with HIV

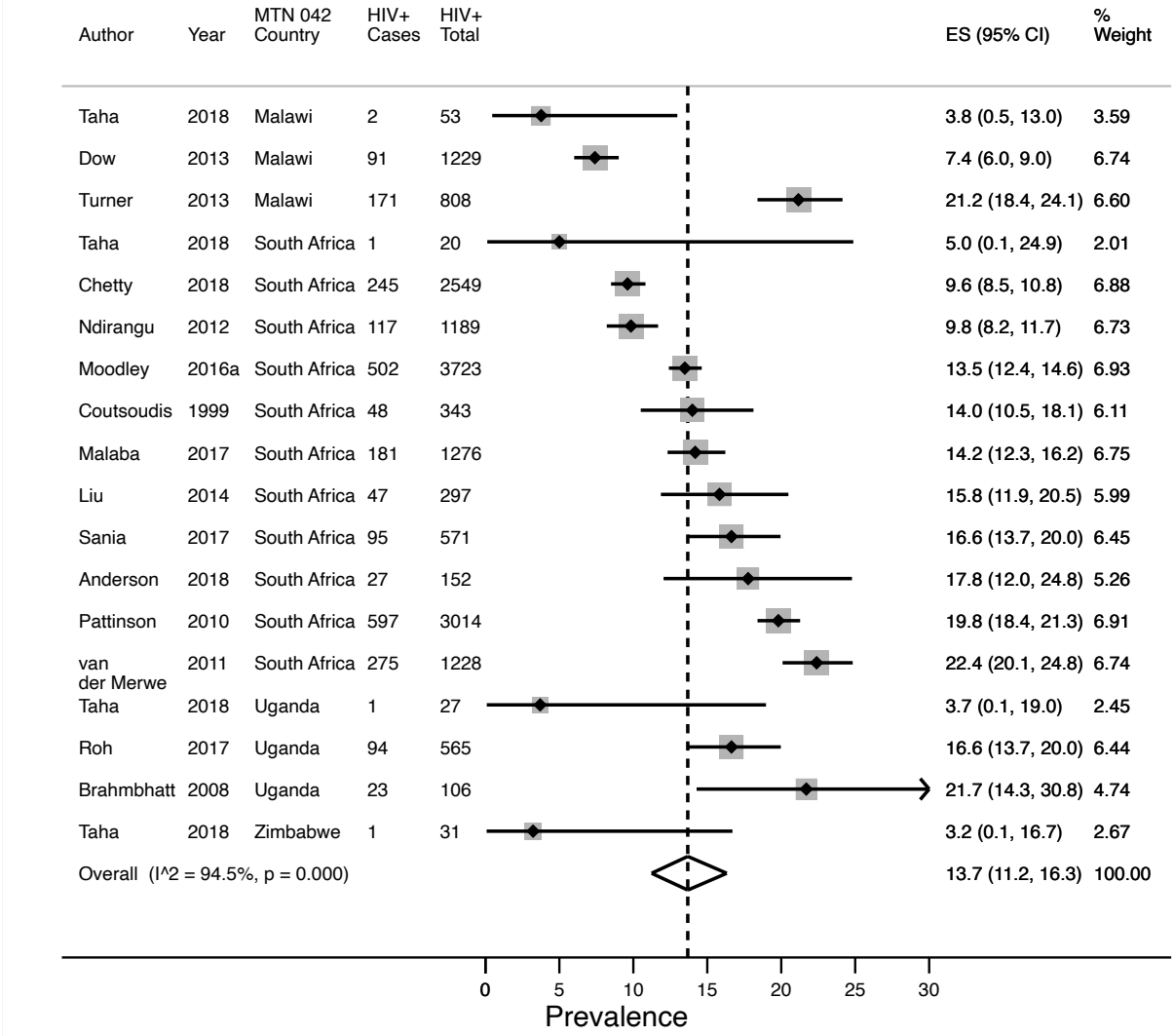


Figure E.6. Forest Plot Summarizing the Pooled Prevalence of Low Birth Weight Among HIV Negative Women

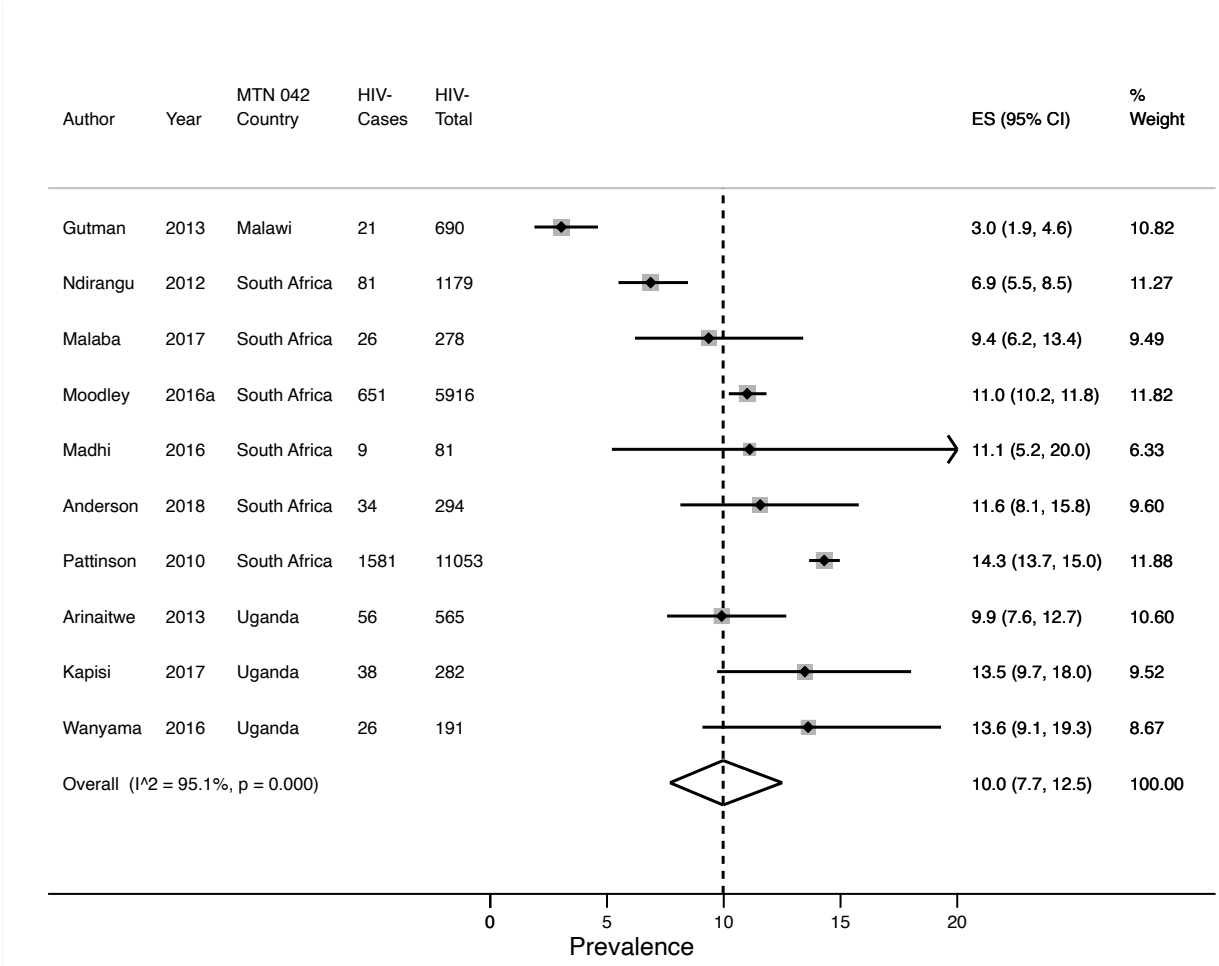


Table E.3. Sensitivity Analysis Including Studies with Live Birth Only: Pooled Prevalence of Low Birth Weight – Overall, by MTN 042 Country, and by HIV Status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95%CI)	I^2 (p-value)
Study Countries	34	53,021	13.4% (2.8, 37.1)	13.1% (9.3, 15.3)	12.9% (10.9, 14.9)	97.6% (<0.01)
Malawi	7	5,392	12.1% (2.8, 21.2)	14.9% (3.0, 15.1)	11.3% (6.9, 16.5)	96.7% (<0.01)
South Africa	11	14,774	13.4% (8.4, 24.8)	13.3% (9.6, 15.8)	12.9% (10.8, 15.1)	92.8% (<0.01)
Uganda	11	10,688	12.9% (6.5, 37.1)	9.9% (9.0, 13.6)	12.2% (8.4, 16.6)	96.7% (<0.01)
Zimbabwe	5	22,167	16.4% (8.6, 22.5)	16.7% (14.1, 20.3)	16.0% (11.4, 21.3)	98.2% (<0.01)
Living with HIV	8	7,185	14.9% (9.6, 21.2)	15.0% (11.9, 17.2)	14.5% (11.5, 17.8)	92.4% (<0.1)
HIV negative	5	2,632	8.9% (3.0, 13.6)	9.4% (6.9, 11.6)	8.2% (4.9, 12.3)	90.6% (<0.01)

*Included studies: Anderson_2018(5), Ashorn_2015 (7), Braun_2015 (20), Chetty_2018 (24), Coutsoudis_1999 (28), Fall_2015 (34), Feresu_2003 (37), Feresu_2004 (38), Feresu_2015 (40), Gutman_2013 (52), Kalanda_2006 (59), Kasumba_2000 (65), Kaye_2006(67), Kimani_2016 (69), Kulmala_2000 (72), Liu_2014 (77), Malaba_2017 (83), Nabiwemba_2013 (96), Nakubulwa_2017 (99), Ndirangu_2012 (104), Ndyomugyenyi_2000 (105), Ndyomugyenyi_2001 (106), Ndyomugyenyi_2008 (107), Ndyomugyenyi_2011 (108), Ogbonna_2007 (115), Rollins_2007 (122), Sania_2017 (123), Shamu_2018 (129), Steyn_2006 (132), Turner_2013 (140), Turton_2017 (141), Verhoeff_1999 (146), Wanyama_2016 (150)

Table E.4. Sensitivity Analysis Excluding Outliers: Pooled Prevalence of Low Birthweight – Overall, by MTN-042 Country, and by HIV status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95%CI)	I^2 (p-value)
Study Countries	66	116,523	11.4% (2.8, 22.5)	11.0% (8.6, 14.0)	11.3% (10.2, 12.4)	96.8% (<0.01)
Malawi	17	14,827	10.6% (2.8, 21.2)	11.8% (7.1, 13.3)	10.4% (8.5, 12.5)	93.0% (<0.01)
South Africa	19	53,701	12.2% (5.0, 22.4)	12.4% (8.7, 14.8)	12.1% (10.4, 13.9)	96.7% (<0.01)
Uganda	19	19,148	11.0% (3.7, 21.7)	9.9% (9.0, 13.5)	10.5% (9.4, 11.6)	76.8% (<0.01)
Zimbabwe	11	28,847	12.0% (3.2, 22.5)	11.4% (8.0, 16.7)	11.8% (8.0, 16.2)	98.7% (<0.01)
Living with HIV	18	17,181	13.1% (3.2, 22.4)	14.1% (7.4, 17.8)	13.7% (11.2, 16.3)	94.5% (<0.01)
HIV negative	10	20,529	10.4% (3.0, 14.3)	11.1% (9.4, 13.5)	10.0% (7.7, 12.5)	95.1% (<0.01)

*N=2 studies had a prevalence estimate that was greater than 1.5 times the interquartile range (IQR) of all included studies. The excluded studies are described in Supplemental Table E.5

Table E.5. Low Birthweight Meta-Analysis Outliers

Manuscript	Country	Frequency	Definition	Outcome Ascertainment	Notes
Turton_2017(141)	South Africa	24.8%	<2500g	Abstraction from maternal hospital record	The aim of this study was to investigate the oral health status and pregnancy outcomes of mothers attending an antenatal clinic in South Africa in order to assess periodontal disease as an independent risk factor for low birthweight. “The exclusion criteria were a history of medical problems that may have affected the study outcome, such as current use of systemic corticosteroids and/or antibiotics, existing heart disease, hypertension, diabetes, asthma, blood-cell disorders and chronic renal disease. Also excluded were mothers with induced labour, multiple pregnancies, smokers and those who used or abused alcohol.”
Kaye_2006 (67)	Uganda	37.1%	<2500g	At time of delivery, participant interview by research assistant	Enrolled women in second trimester (excluded women with chronic disease associated with frequent hospitalization or likely to cause LBW)

SECTION F: NEONATAL MORTALITY

Table F.1. Search Result Flow

	N
Title Review	834
Abstract Review	379
Manuscripts Reviewed*	211
Manuscripts Included From Main Search†	100
Manuscripts Added From Other Searches‡	41
Total Manuscripts Abstracted	141
Total Manuscripts Included For Neonatal Mortality Analysis – MTN-042 Countries Only§	26

* Does not include number of references from systematic reviews reviewed.
† Includes systematic review reference reviews.
‡ Some of these include duplicate populations with the studies that were abstracted through the main search. These were removed at analysis.
§ Excludes duplicate populations, studies with incorrect definitions of the outcomes, studies that only reported early neonatal mortality. Definitions accepted for neonatal mortality meta-analysis includes deaths occurring within 28 days, before 29 days, within one month, and before 30 days.

Table F.2. Pooled Prevalence of Neonatal Mortality – Overall, by MTN-042 Country, and by HIV status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95% CI)	I^2 (p-value)
Study Countries	26	342,853	2.0% (0.3, 4.2)	1.9% (0.7, 3.0)	1.7% (1.4, 2.1)	97.2% (<0.01)
Malawi	8	22,030	2.7% (0.5, 4.2)	3.2% (1.3, 3.9)	2.4% (1.4, 3.5)	94.8% (<0.01)
South Africa	10	276,251	1.0% (0.3, 2.5)	0.7% (0.6, 1.3)	0.9% (0.6, 1.2)	95.4% (<0.01)
Uganda	7	40,616	2.7% (2.0, 3.7)	2.7% (2.2, 3.4)	2.6% (2.3, 3.0)	38.7% (0.1)
Zimbabwe	1	3,956	1.3% -	-	1.3% (0.9, 1.7)	-
Living with HIV	5	6,713	1.3% (0.3, 3.7)	1.0% (0.5,1.1)	1.0% (0.5, 1.8)	81.2% (<0.01)
HIV negative	1	11,053	0.5% -	-	0.5% (0.4, 0.6)	-

* No outliers were identified so no sensitivity analysis excluding outliers was conducted.

Figure F.1. Forest Plot Summarizing the Pooled Prevalence of Neonatal Mortality – Overall and by Country

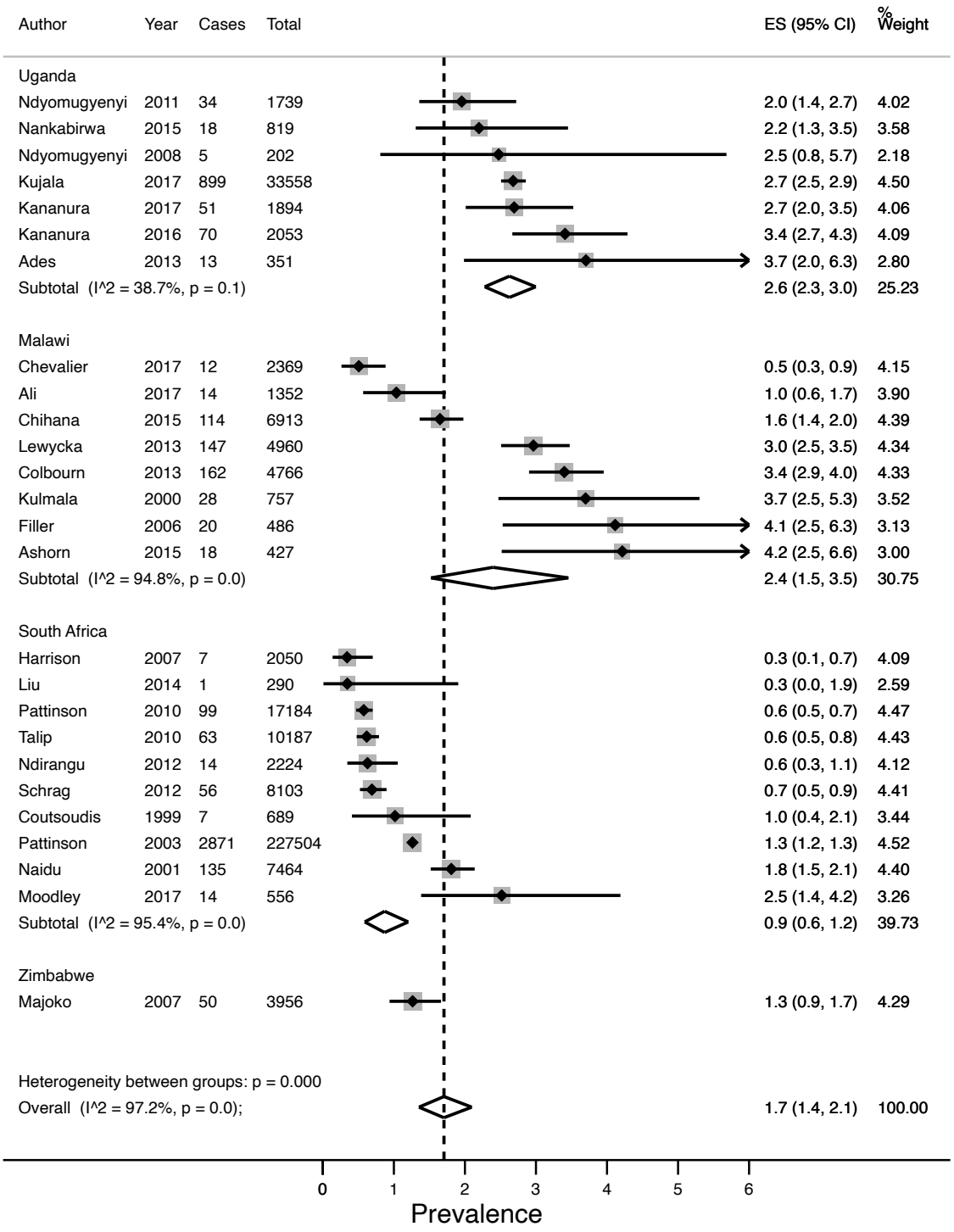
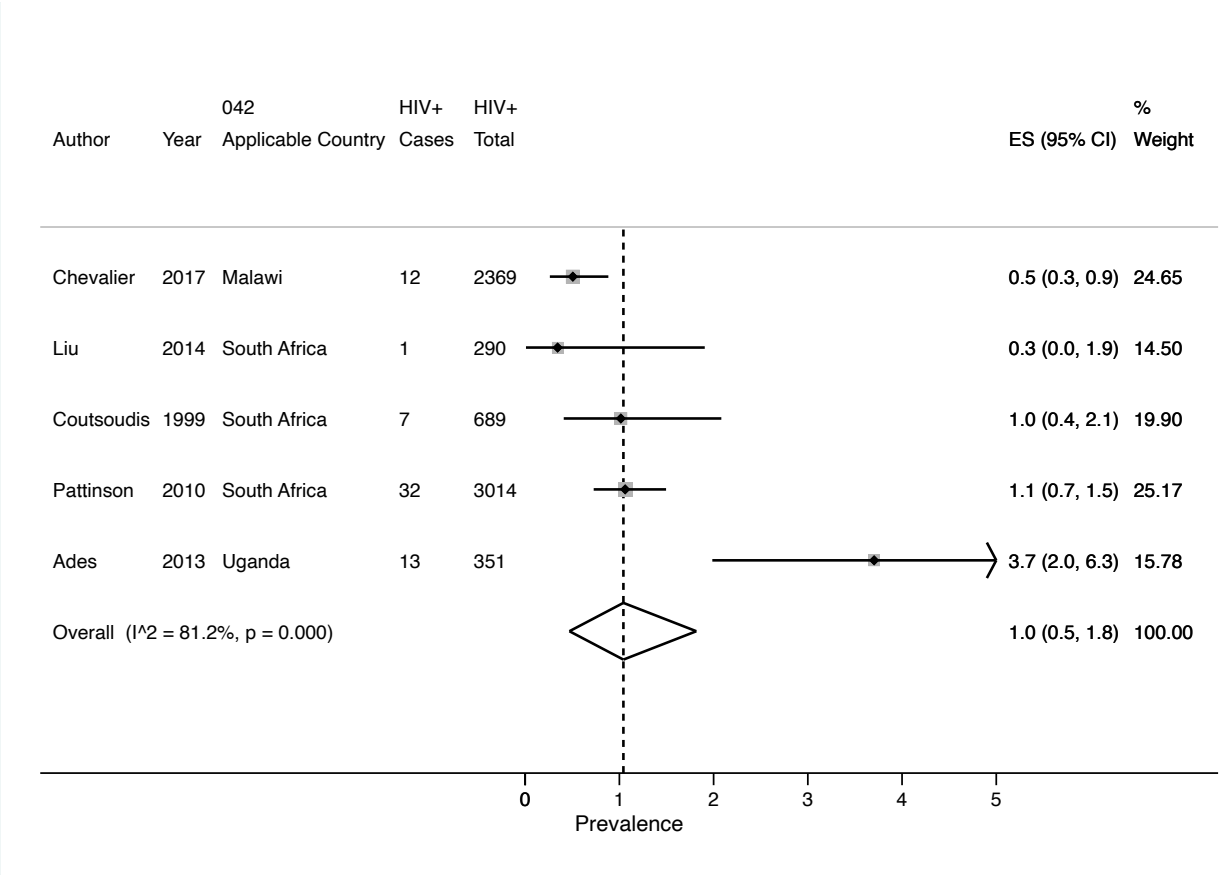


Figure F.2. Forest Plot Summarizing the Pooled Prevalence of Neonatal Mortality Among Women Living with HIV



Note: No forest plot was generated for HIV negative women as there was only one study reporting a prevalence estimate.

Table F.3. Neonatal Mortality (<28 days) Summary - Sensitivity analysis excluding studies that did not define "neonatal mortality"

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95%CI)	I^2 (p-value)
Study Countries	19	73,952	2.3% (0.3, 4.2)	2.5% (1.0, 3.4)	2.0% (1.5, 2.6)	95.2% (<0.01)
Malawi	8	22,030	2.7% (0.5, 4.2)	3.2% (1.3, 3.9)	2.4% (1.5, 3.5)	94.8% (<0.01)
South Africa	4	11,306	0.7% (0.3, 1.0)	0.7% (0.5, 0.9)	0.6% (0.5, 0.8)	0.0% (0.7)
Uganda	7	40,616	2.7% (2.0, 3.7)	2.7% (2.2, 3.4)	2.6% (2.3, 3.0)	38.7% (0.1)
Zimbabwe	0	-	-	-	-	-
Living with HIV	4	3,699	1.4% (0.3, 3.7)	0.8% (0.4, 2.4)	1.1% (0.3, 2.4)	75.0% (<0.01)
HIV negative	0	-	-	-	-	-

* Included studies: Ades_2013 (2), Ali_2017 (3), Ashorn_2015 (7), Chevalier_2017 (25), Chihanna_2015 (26), Colbourn_2013 (27), Coutsoudis_1999 (28), Filler_2006 (41), Kananura_2016 (62), Kananura_2017 (63), Kujala_2017 (71), Kulmala_2000 (72), Lewycka_2013 (76), Liu_2014 (77), Nankabirwa_2015 (101), Ndirangu_2012 (104), Ndyomugyenyi_2008 (107), Ndyomugyenyi_2011 (108), Schrag_2012 (125)

Notes F.1. Neonatal Mortality Study Specific Notes:

- RCTs where overall proportion (not control prevalence) was used:
    - Filler 2006 was an RCT testing monthly SP vs 2 dose SP during pregnancy in women living with HIV and HIV negative women. The n was not presented by arm or HIV status, just proportions. The analysis therefore included the combined (intervention + control) prevalence estimate.
- Pattinson_2010
    - Used neonatal mortality rates and N for HIV subgroups to calculate group specific prevalence; the total number of deaths by calculation (n=99) is smaller than a number reported in the paper (103). Used 99 as the numerator and included the paper since the prevalence difference using 99 vs 103 was insignificant given the total denominator at risk n=17184).

SECTION G: CHORIOAMNIONITIS

Table G.1. Search Result Flow

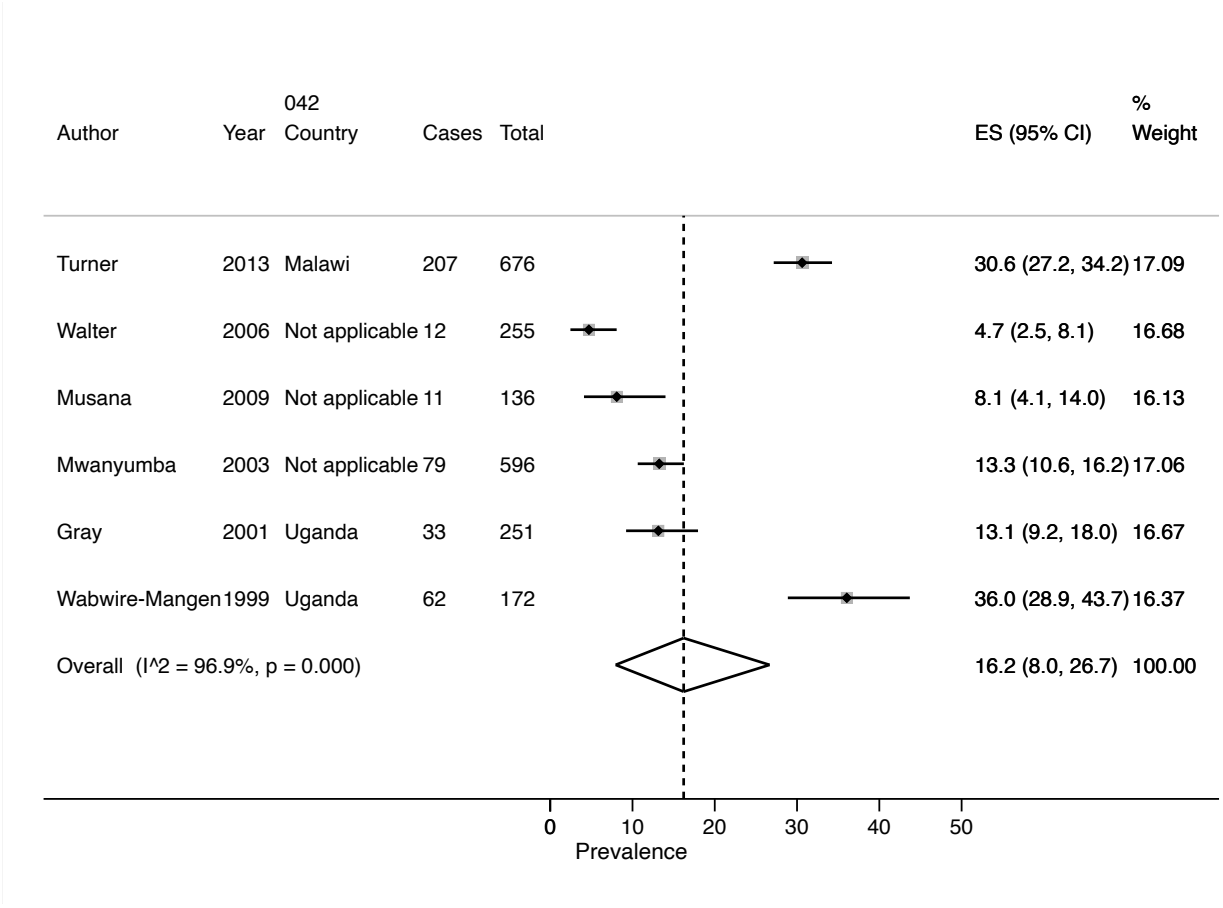
	N
Title Review	6
Abstract Review	1
Manuscripts Reviewed	1
Manuscripts Included From Main Search*	1
Manuscripts Added From Other Searches	5
Total Manuscripts Abstracted†	6
Total Manuscripts Included For Analysis‡	6

* Does not include number of references from systematic reviews reviewed
† Includes systematic review reference reviews
‡ Includes studies from non-MTN042 countries

Table G.2. Pooled Prevalence of Chorioamnionitis – Overall, by MTN-042 Country, and by HIV status

	# of estimates	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95% CI)	I^2 (p-value)
Total	6	2,086	17.6% (4.7, 36.0)	13.2% (8.1, 30.6)	16.2% (8.0, 26.7)	96.9% (<0.01)
Study Countries	3	1,099	26.6% (13.1, 36.0)	30.6% (13.1, 36.0)	25.9% (14.0, 39.8)	-
Malawi	1	676	30.6% -	-	30.6% (27.2, 34.2)	-
South Africa	0	-	-	-	-	-
Uganda	2	4,23	24.6% (13.1, 36.0)	24.6% (13.1, 36.0)	21.5% (17.7, 25.6)	-
Zimbabwe	0	-	-	-	-	-
Total - Living with HIV	4	1,171	20.4% (4.7, 36.0)	20.4% (7.5, 33.3)	18.7% (5.6, 36.9)	97.5% (<0.01)
Total - HIV Negative	1	68	5.9% -	-	5.9% (1.3, 13.0)	-

Figure G.1. Forest Plot Summarizing the Pooled Prevalence of Chorioamnionitis



Notes G.1. Chorioamnionitis Study Specific Notes

- Definitions of Chorioamnionitis
  - Clinical symptom: Walter_2006 (149)
  - Histopathology: Turner_2013 (140), Gray_2001 (49), Mwanyumba_2003 (94), Wabwire-Mangen_1999 (148)
  - Undefined: Musana_2009 (92)
- Studies that included women living with HIV with low CD4 count or advanced AIDS (WHO 3 or 4)
    - Turner_2013 (140)
      - Malawi. All were ART naive at enrollment, but 95% received single dose nevirapine at delivery.
    - Walter_2006 (149)
      - Zambia. All had low CD4 and excluded women who began taking ART while pregnant.
    - Musana_2009 (92)
      - Kenya. Included women Living with HIV with advanced disease.
    - Wabwire-Mangen_1999 (148)
      - Uganda. Study was conducted in 1990-1992, pre-ART.

SECTION H: ENDOMETRITIS

Table H.1. Search Result Flow

	N
Title Review	43
Abstract Review	12
Manuscripts Reviewed*	4
Manuscripts Included From Main Search†	4
Manuscripts Added From Other Searches	4
Total Manuscripts Abstracted	8
Total Manuscripts Included For Analysis	7

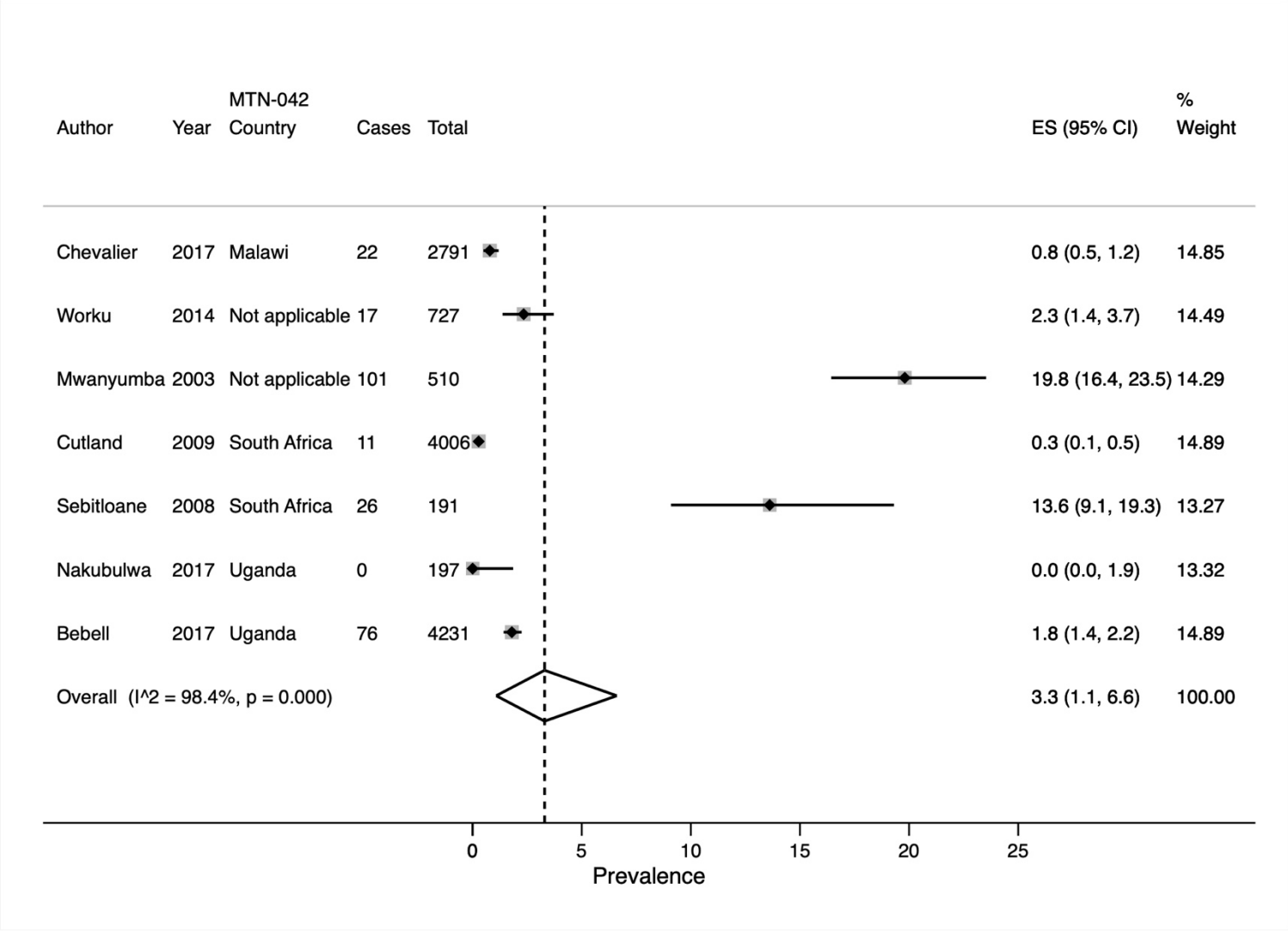
* Does not include number of references from systematic reviews reviewed.
† Includes systematic review reference reviews.

Table H.2. Pooled Prevalence of Endometritis – Overall, by MTN-042 Country, and by HIV status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95% CI)	I^2 (p-value)
Total	7	12,653	5.5% (0.0, 19.8)	1.8% (0.3, 13.6)	3.3% (1.1, 6.6)	98.4% (p<0.01)
Study Countries	5	11,416	3.3% (0.0, 13.6)	0.8% (0.3, 1.8)	1.6% (0.4, 3.4)	96.7% (p<0.01)
Malawi	1	2,791	0.8% -	-	0.8% (0.5, 1.2)	-
South Africa	2	4,197	6.9% (0.3, 13.6)	6.9% (0.3, 13.6	0.3% (0.2, 0.6)	-
Uganda	2	4,428	0.9% (0.0, 1.8)	0.9% (0, 1.8)	1.6% (0.4, 3.4)	-
Zimbabwe	0	-	-	-	-	-
Total - Living with HIV	3	4,022	5.0% (0.5, 13.6)	0.8% (0.5, 13.6)	2.9% (0.3, 7.6)	96.9% (p<0.01)
Total - HIV Negative	1	2, 916	0.2% -	-	0.2% (0.1, 0.4)	-

* No outliers were identified so no sensitivity analysis excluding outliers was conducted.

Figure H.1. Forest Plot Summarizing the Pooled Prevalence of Endometritis



SECTION I: POSTPARTUM HEMORRHAGE

Table I.1. Search Result Flow

	N
Title Review	217
Abstract Review	68
Manuscripts Reviewed*	32
Manuscripts Included From Main Search†	16
Manuscripts Added From Other Searches	17
Total Manuscripts Abstracted for Antepartum and Postpartum Hemorrhage	33
Total Manuscripts Abstracted for Postpartum Hemorrhage only	28
Total Manuscripts Included For Postpartum Hemorrhage Analysis- MTN 042 Only‡	17

* Does not include number of references from systematic reviews reviewed.
† Includes systematic review reference reviews

Table I.2. Pooled Prevalence of Postpartum Hemorrhage – Overall, by MTN-042 country, and by HIV status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95%CI)	I^2 (p-value)
Study Countries	17	71,308	5.6% (0.6, 17.4)	3.1% (1.6, 9.0)	4.4% (3.0, 6.0)	98.7% (<0.01)
Malawi	2	5,875	2.1% (1.1, 3.1)	2.1% (1.1, 3.1)	2.0% (1.7, 2.4)	-
South Africa	10	57,046	4.7% (0.6, 15.8)	3.1% (0.7, 7.0)	3.6% (2.0, 5.6)	98.9% (<0.01)
Uganda	3	3,564	9.7% (2.8, 17.4)	9.0% (2.8, 17.4)	8.8% (2.5, 18.3)	-
Zimbabwe	2	4,823	7.4% (1.6, 13.3)	7.4% (1.6, 13.3)	1.9% (1.5, 2.3)	-
Living with HIV	5	7,541	8.2% (0.7, 33.3)	1.1% (0.7, 4.9)	4.5% (1.3, 9.4)	98.3% (<0.01)
HIV negative	4	11,650	7.1% (0.3, 21.4)	3.4% (1.5, 12.7)	5.2% (0.4, 14.0)	97.0% (<0.01)
≥500mL lost†	7	25,720	8.9% (1.6,17.4)	9.0% (2.5, 15.8)	7.5% (4.6, 11.1)	98.5% (<0.01)

* No outliers were identified so no sensitivity analysis excluding outliers was conducted.
† For the purpose of this sub-analysis analysis, any study reporting ≥500mL of blood lost independent of length of time or delivery type (vaginal, c-section) was included. Studies included: Beltman_2011 (15), Bloch_2015 (17), Abdel-Aleem_2010 (1), Ononge_2016 (116), Atukunda_2014 (9), Ngwenya_2016 (109), Kundodyiwa_2001 (74)

Figure I.1. Forest Plot Summarizing the Pooled Prevalence of Postpartum Hemorrhage

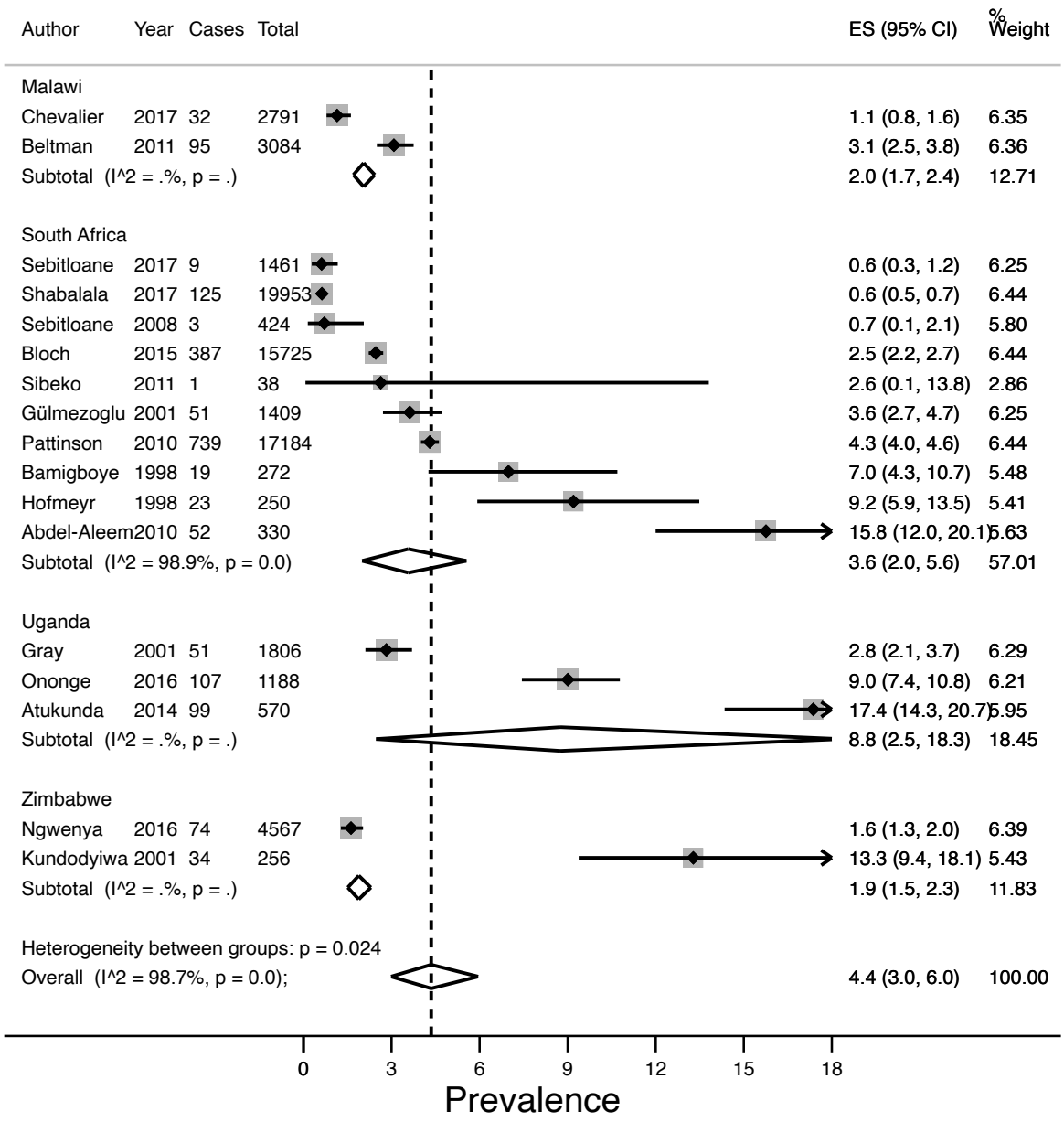


Figure I.2. Forest Plot Summarizing the Pooled Prevalence of Postpartum Hemorrhage Among Women Living with HIV

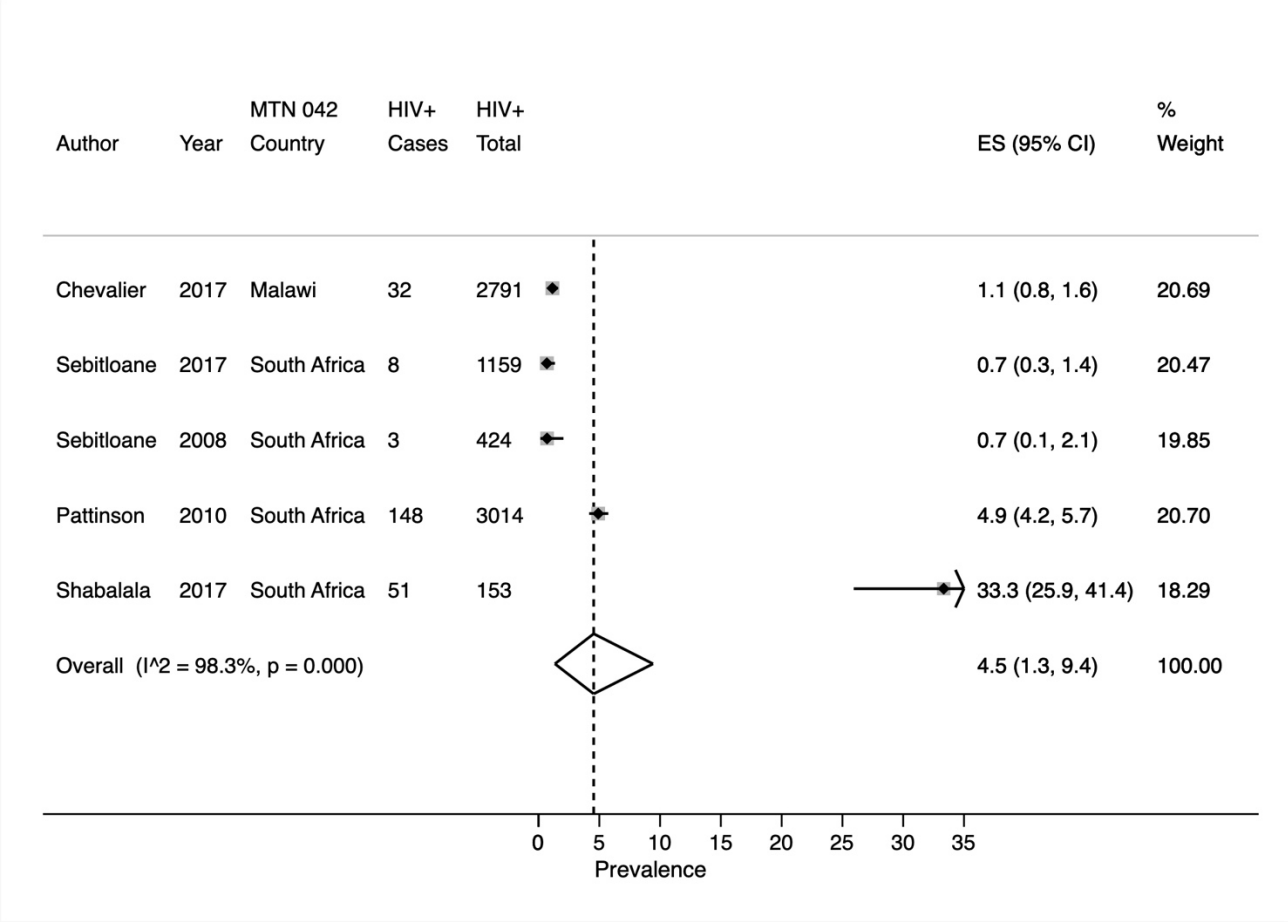


Figure I.3. Forest Plot Summarizing the Pooled Prevalence of Postpartum Hemorrhage Among HIV Negative Women

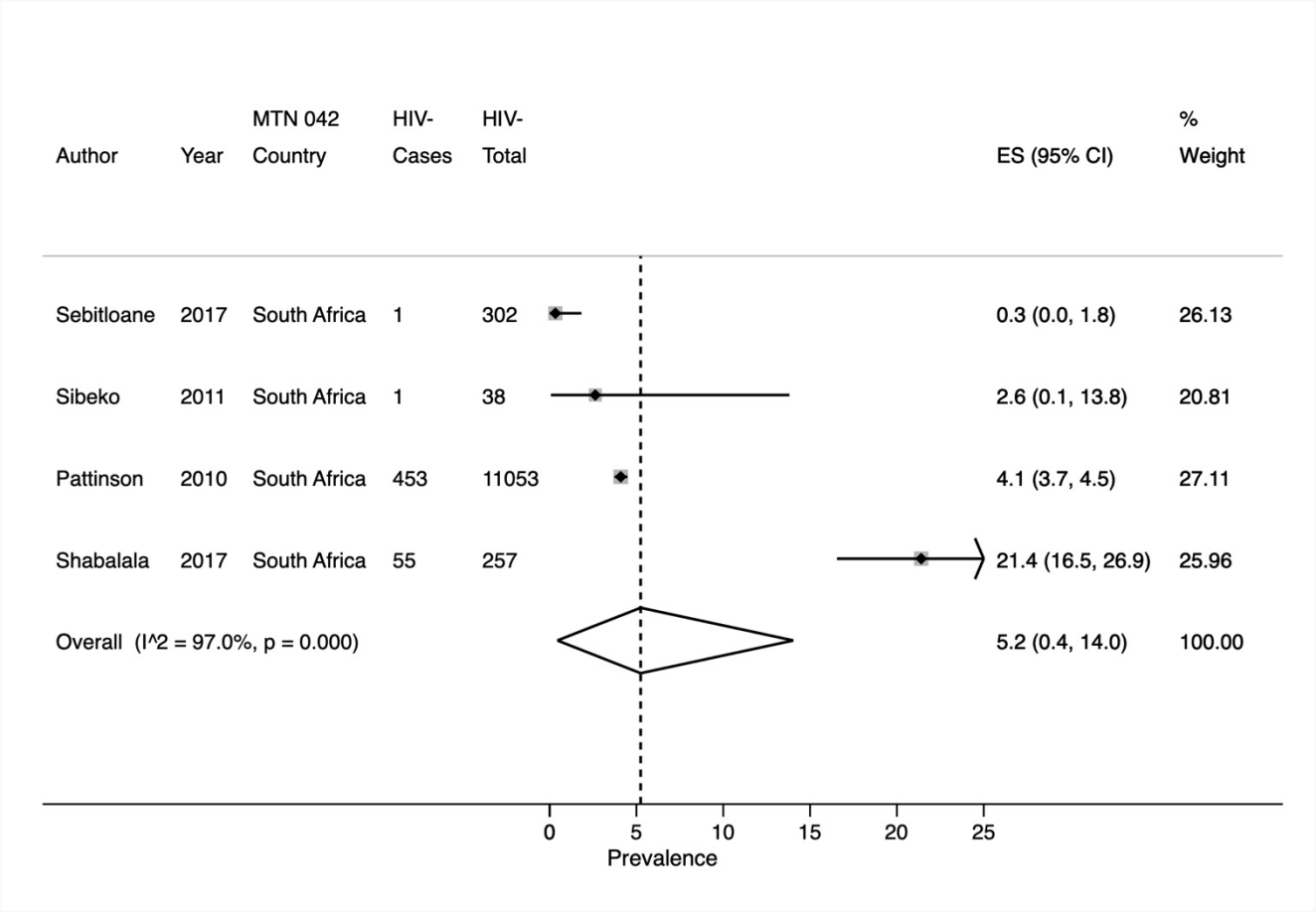
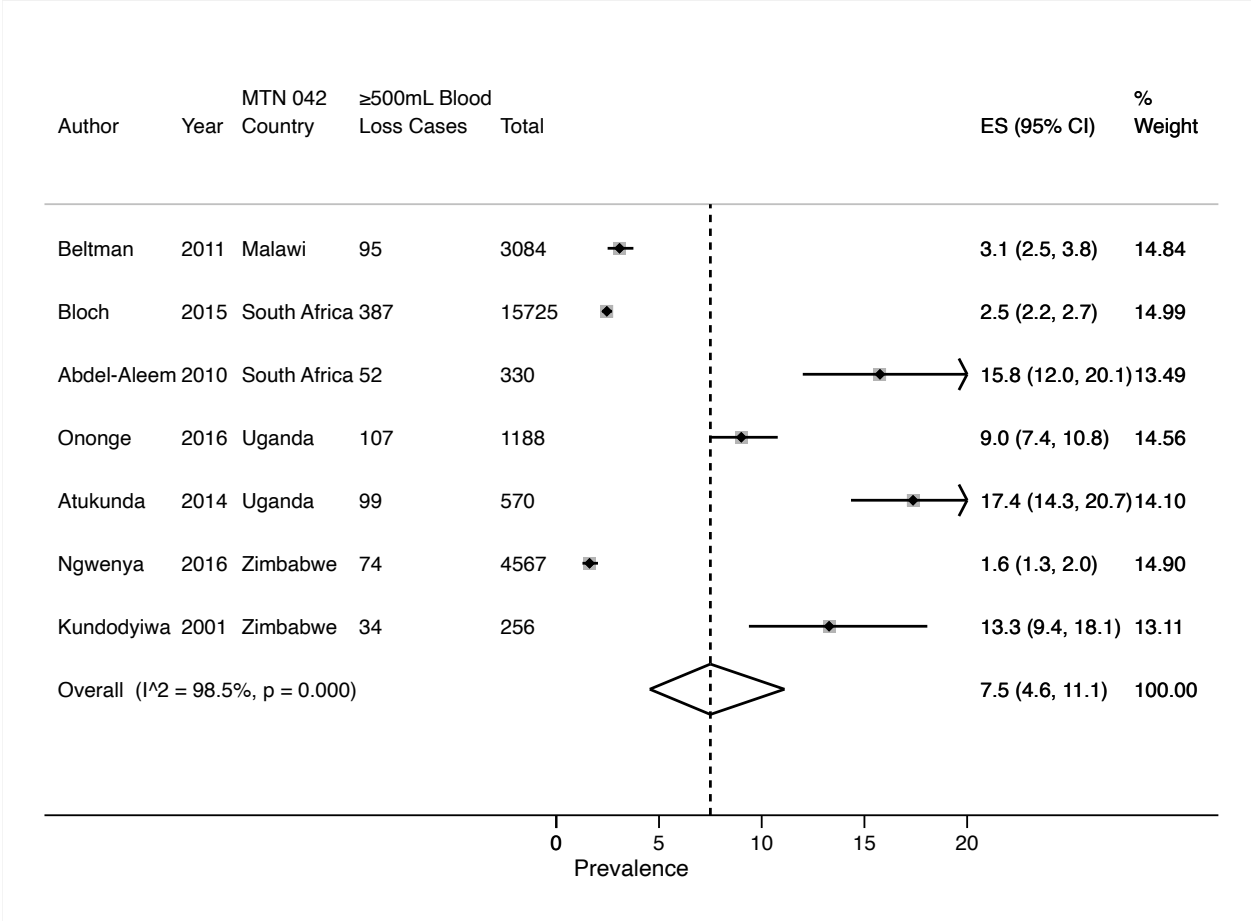


Figure I.4. Forest Plot Summarizing the Pooled Prevalence of Postpartum Hemorrhage - ≥500mL Blood Loss



SECTION J: HYPERTENSIVE DISORDERS OF PREGNANCY

Table J.1. Search Result Flow

	N
Title Review	549
Abstract Review	124
Manuscripts Reviewed*	54
Manuscripts Included From Main Search†	19
Manuscripts Added From Other Searches	33
Total Manuscripts Abstracted	52
Total Manuscripts Included For MTN-042 Analysis	
Gestational Hypertension	14
Pre-eclampsia/Eclampsia	9

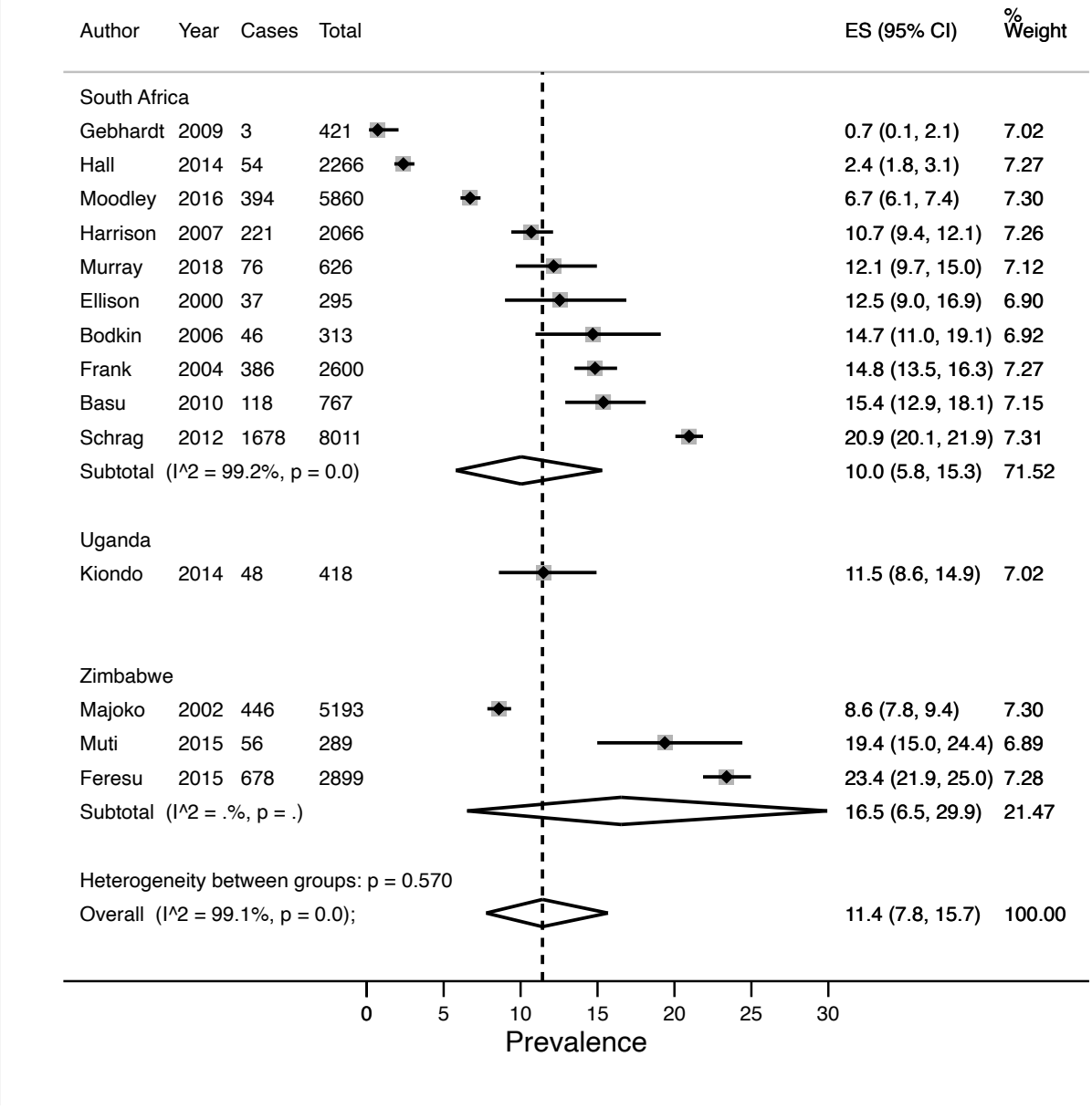
* Does not include number of references from systematic reviews reviewed.
† Includes systematic review reference reviews.

GESTATIONAL HYPERTENSION

Table J.2. Pooled Prevalence of Gestational Hypertension – Overall, by MTN-042 Country and by HIV status

	# of estimates	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95%CI)	I^2 (p-value)
Study Countries	14	32,024	12.4% (0.7, 23.4)	12.3% (8.6, 15.4)	11.4% (7.8, 15.7)	99.1% (<0.01)
Malawi	0	-	-	-	-	-
South Africa	10	23,225	11.1% (0.7, 21.0)	12.3 % (6.7, 14.8)	10.0% (5.8, 15.3)	99.2% (<0.01)
Uganda	1	418	11.5% -	-	11.5% (8.6, 14.9)	-
Zimbabwe	3	8,381	17.1% (8.6, 23.4)	19.4% (8.6, 23.4)	16.5% (6.5, 29.9)	-
Living with HIV	3	3,201	11.1% (1.6, 17.0)	14.8% (1.6, 17.0)	9.6% (1.3, 24.3)	99.0% (<0.01)
HIV Negative	4	2,399	7.1% (0.7, 14.9)	6.5% (1.9, 12.4)	5.8% (0.9, 14.3)	97.7% (<0.01)

Figure J.1. Forest Plot Summarizing the Pooled Prevalence of Gestational Hypertension



*Moodley 2016 refers to Moodley, 2016b (87)

Figure J.2. Forest Plot Summarizing the Pooled prevalence of Gestational Hypertension Among Women Living with HIV

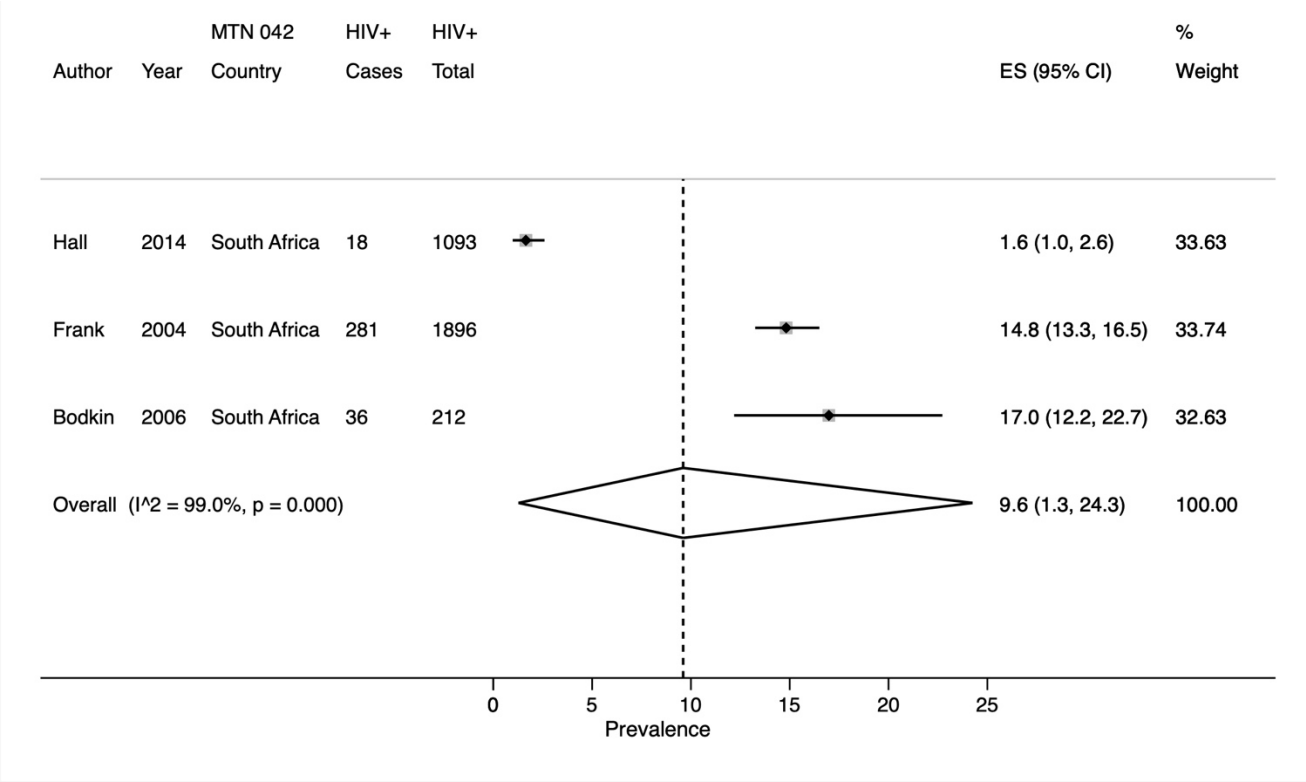
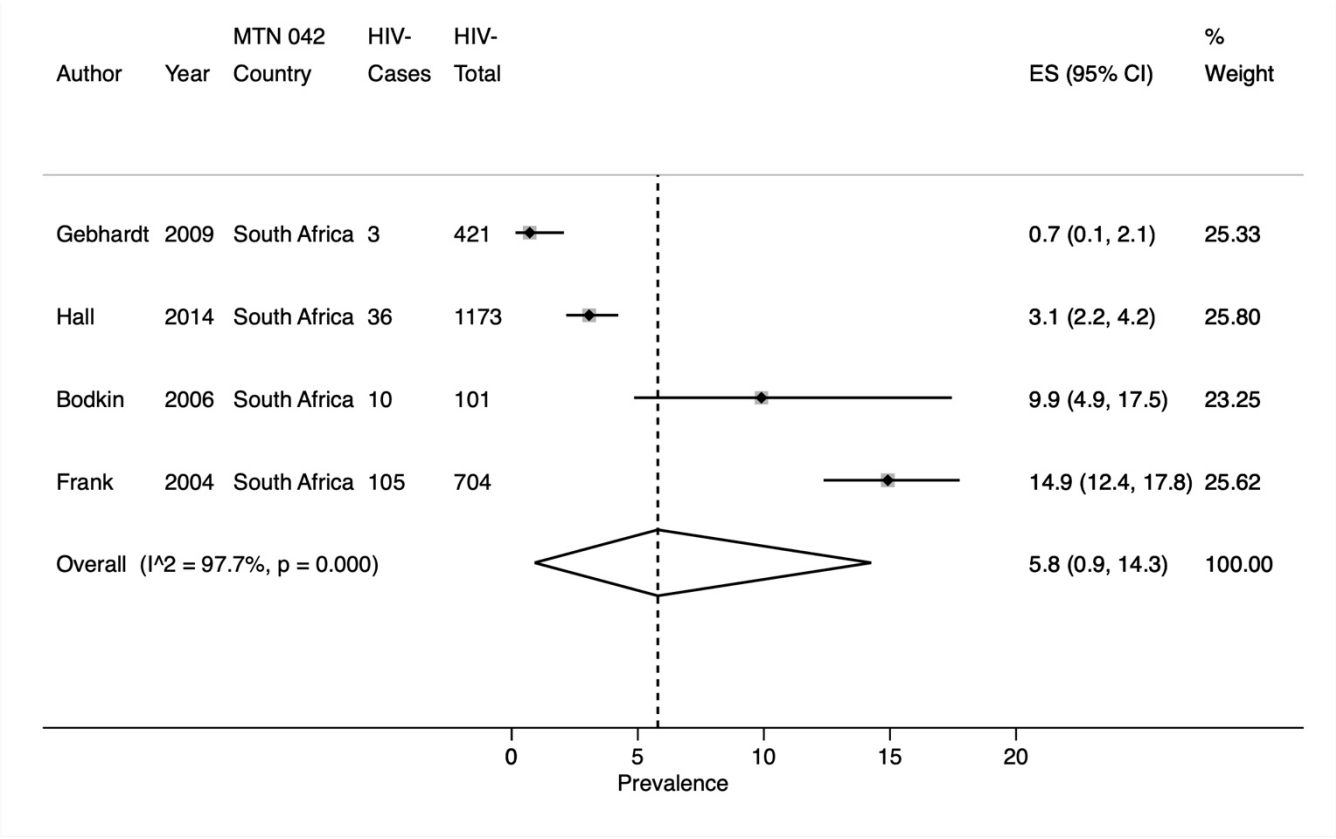


Figure J.3. Forest Plot Summarizing the Pooled prevalence of Gestational Hypertension Among HIV Negative Women



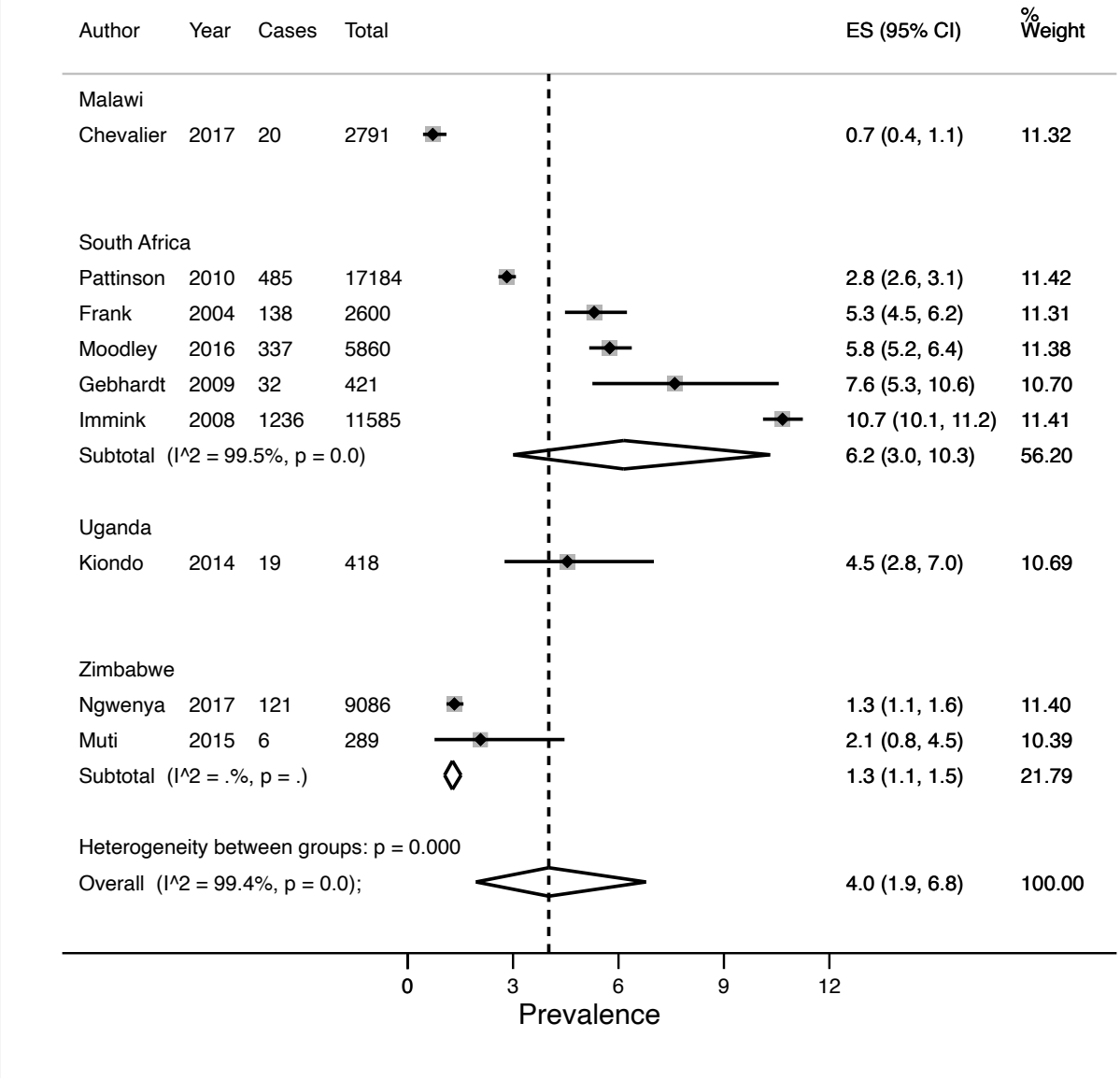
PRE-ECLAMPSIA/ECLAMPSIA

Table J.3. Pooled Prevalence of Preeclampsia/Eclampsia – Overall, by MTN-042 Country, and by HIV status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95%CI)	I^2 (p-value)
Study Countries	9	50,234	4.5% (0.7, 10.7)	4.5% (2.1, 5.8)	4.0% (1.9, 6.8)	99.4% (<0.01)
Malawi	1	2,791	0.7% -	-	0.7% (0.4, 1.1)	-
South Africa	5	37,650	6.4% (2.8, 10.7)	5.8% (5.3, 7.6)	6.2% (3.0, 10.3)	99.5% (<0.01)
Uganda	1	418	4.5% -	-	4.5% (2.8, 7.0)	-
Zimbabwe	2	9,375	1.7 % (1.3, 2.1)	1.7 % (1.3, 2.1)	1.3% (1.1, 1.5)	-
Living with HIV	3	7,701	2.7% (0.7, 5.2)	2.1% (0.7, 5.2)	2.3% (0.6, 5.2)	97.9% (<0.01)
HIV Negative	3	4,910	5.5% (3.1, 7.6)	5.7% (3.1, 7.6)	5.2% (2.6, 8.4)	92.7% (<0.01)

*Only included studies reporting prevalence of both preeclampsia and eclampsia either as a composite or each separately. In addition, no outliers were identified so no sensitivity analysis excluding outliers was conducted.

Figure J.4. Forest Plot Summarizing the Pooled Prevalence of Pre-eclampsia/Eclampsia



*Moodley 2016 refers to Moodley, 2016b (87)

Figure J.5. Forest Plot Summarizing the Pooled Prevalence of Pre-eclampsia/Eclampsia Among Women Living with HIV

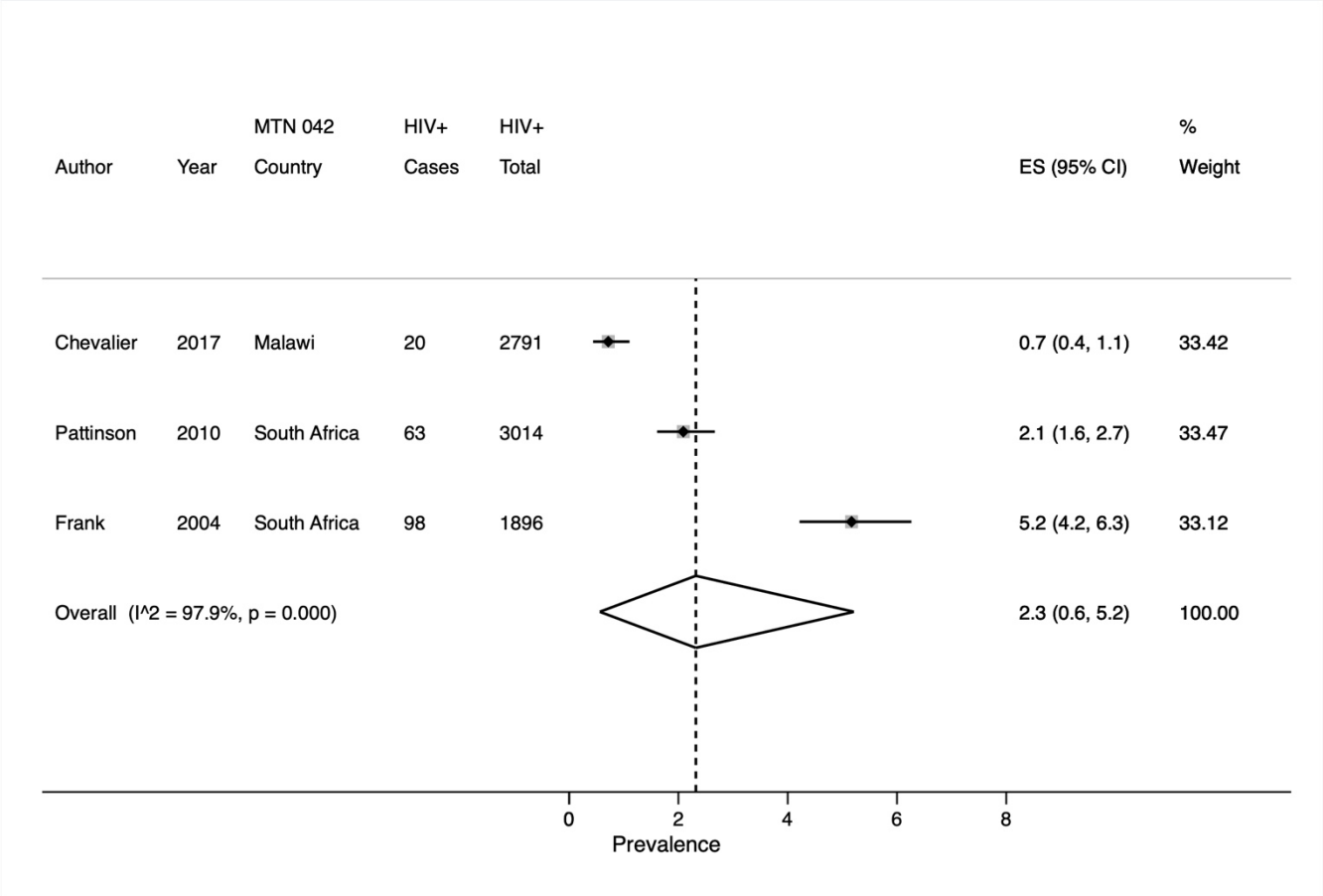
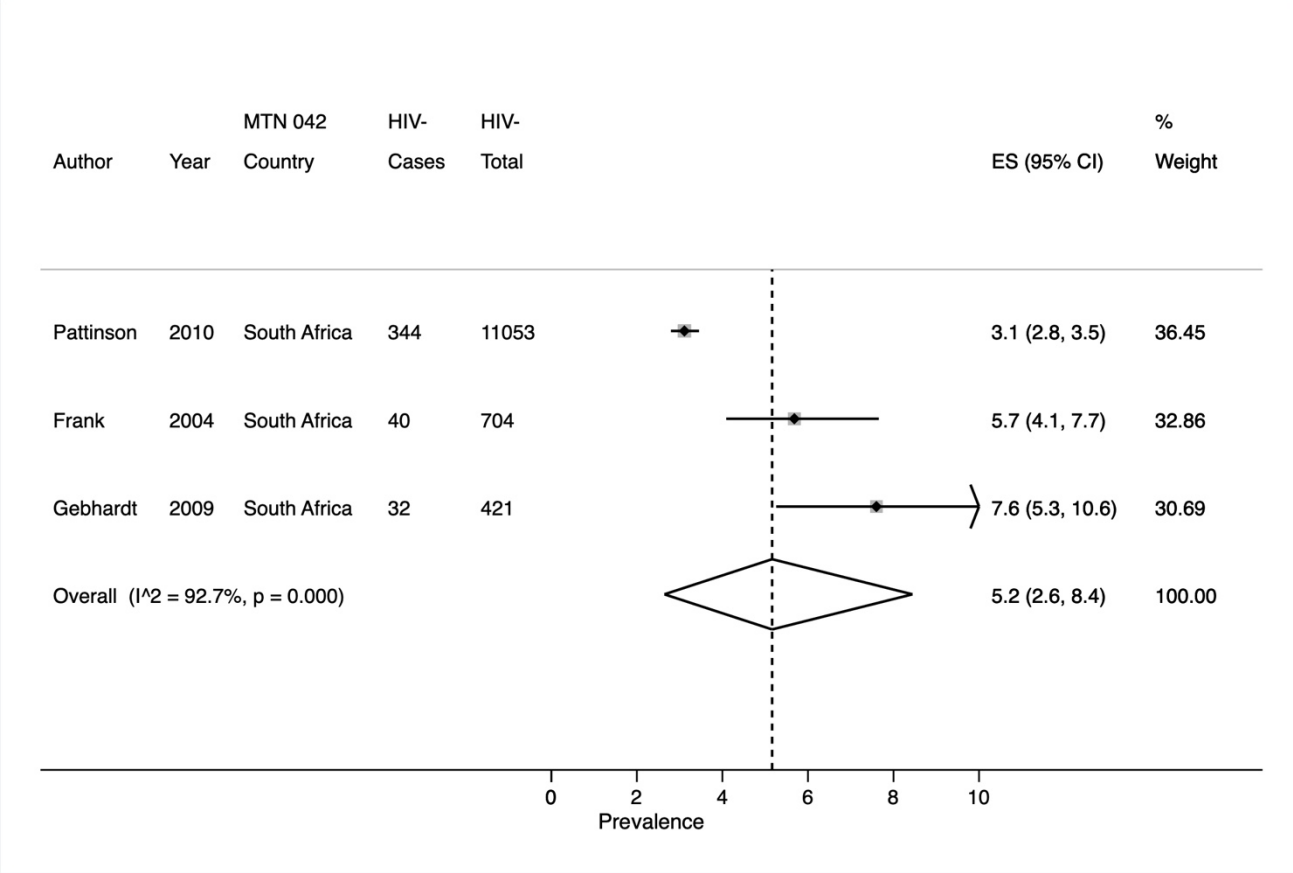


Figure J.6. Forest Plot Summarizing the Pooled Prevalence of Pre-eclampsia/Eclampsia Among HIV Negative Women



SECTION K: PPROM

Table K.1. Search Result Flow

	N
Title Review	50
Abstract Review	23
Manuscripts Reviewed*	9
Manuscripts Included From Main Search†	5
Manuscripts Added From Other Searches	2
Total Manuscripts Abstracted	7
Total Manuscripts Included For Analysis	7

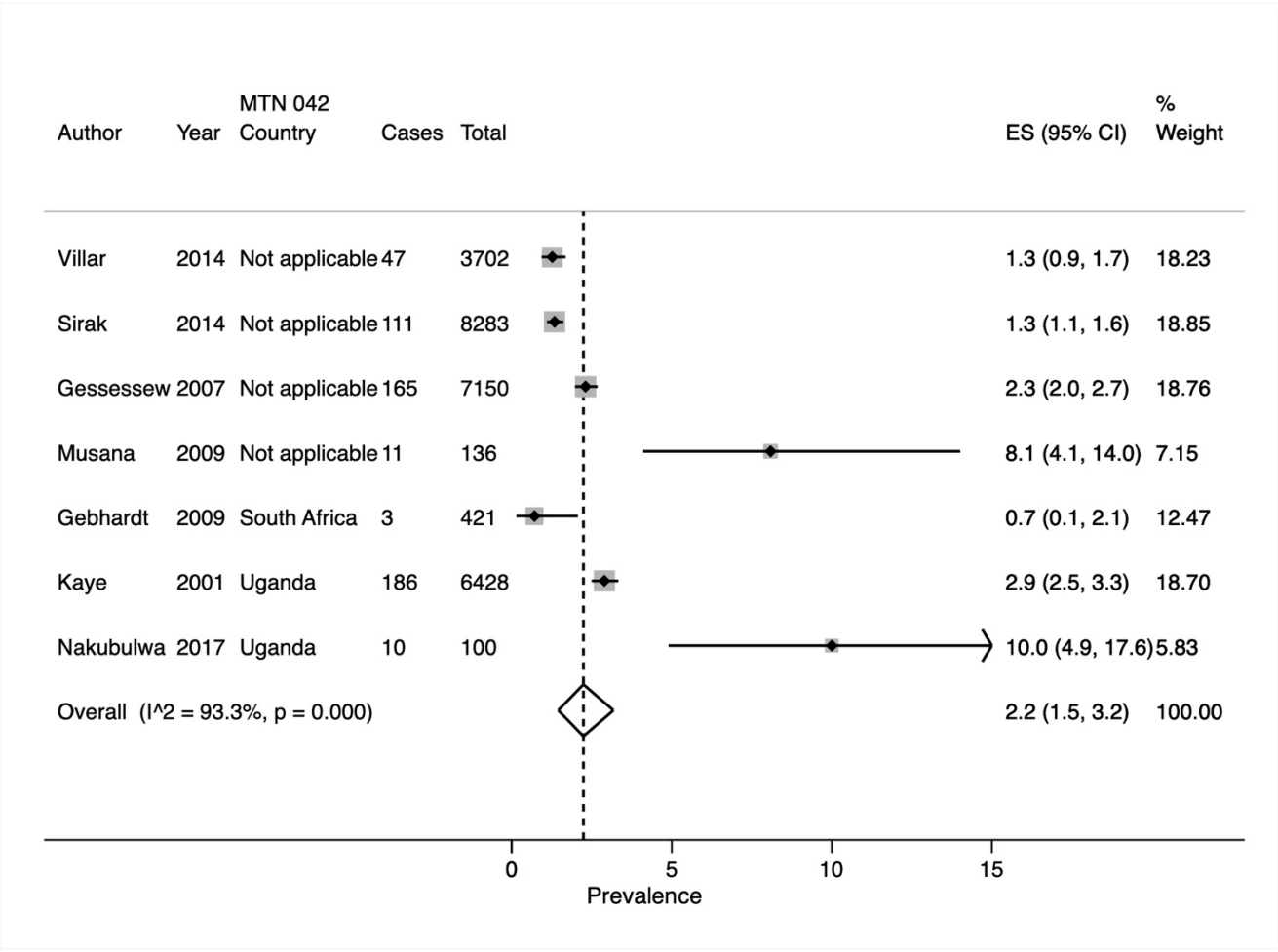
* Does not include number of references from systematic reviews reviewed.
† Includes systematic review reference reviews.

Table K.2. Pooled Prevalence of PPROM– Overall, by MTN-042 Country, and by HIV status

	# of estimates*	# of pregnancies/ infants	Mean (Min, Max)	Median (IQR)	Pooled Prevalence (95%CI)	I^2 (p-value)
Total (all countries)	7	26,220	3.8% (0.7, 10.0)	2.3% (1.3, 8.1)	2.2% (1.5, 3.2)	93.3% (<0.01)
Study Countries	3	6,949	4.5% (0.7, 10.0)	2.9% (0.7, 10.0)	3.1% (0.8, 6.9)	-
Malawi	0	-	-	-	-	-
South Africa	1	421	0.7% -	-	0.7% (0.1, 2.1)	-
Uganda	2	6,528	6.4% (2.9, 10.0)	6.4% (2.9, 10.0)	2.7% (2.3, 3.2)	-
Zimbabwe	0	-	-	-	-	-
Living with HIV	1	68	10.3% -	-	10.3% (4.2, 20.1)	-
HIV negative	2	489	3.3% (0.7, 5.9)	3.3% (0.7, 5.9)	0.9% (0.2, 2.1)	-

* No outliers were identified so no sensitivity analysis excluding outliers was conducted.

Figure K.1. Forest Plot Summarizing the Pooled Prevalence of PPROM



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